POSITIVE SQUARE ENERGY OF MAXIMAL OUTERPLANAR GRAPHS

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Abstract. We study the conjectured inequality $s^+(G) \geq 2|V(G)|$ for finite simple maximal outerplanar graphs of order at least eight. The proof reduces a smallest counterexample to the parent of a deepest branching face. Three rooted parent forms suffice, with two distinct orientations in the single-sibling form. We develop a two-root matrix-ball estimate for the complete graft phase, prove its monotonicity under subtraction of positive lossless rays, and use path continuants to cover unbounded arm lengths. The remaining bounded parameter domains are verified by exact integer and outward rational computations; the infinite-frequency ends are bounded analytically on complete inverse-frequency intervals. We give the finite domains, the arithmetic acceptance inequalities and the accompanying standalone verification sources. The numerical certificates are part of the proof, rather than evidence from sampling.

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1. Introduction and statement

All graphs are finite, simple and undirected, and all adjacency entries corresponding to edges equal one. If $\lambda_1, \dots, \lambda_n$ are the adjacency eigenvalues, counted with multiplicity, write

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \qquad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2.$$

A maximal outerplanar graph on $n \geq 3$ vertices is a triangulation of a polygon with all vertices on its boundary. It has $2n-3$ edges. The following is Conjecture 9.5(ii) of Akbari, Kumar, Mohar, Pragada and Zhang [1].

Theorem 1.1. *For every maximal outerplanar graph G of order $n \geq 8$,*

$$s^+(G) \geq 2n.$$

Equivalently, $s^+(G) - s^-(G) \geq 6$.

The equivalence follows from $s^+ + s^- = 2|E| = 4n-6$. The inequality in the theorem is weak; no assertion about equality cases is needed. The auxiliary graph K_2 is used only as the remainder of a polygon cut, never as a graph satisfying the proposed $2n$ bound.

We distinguish three kinds of arguments. Structural statements below hold for every polygon triangulation by explicit separation or path arguments. Matrix estimates hold on their stated finite-dimensional domains and do not invoke the graph inequality being proved. Finite certificate assertions have explicitly bounded parameter domains and are proved by exact verifiers supplied in the supplement. In particular, hash agreement, an exit status or a stored assertion does not replace any of these arguments.

The published inputs used for the finite basis are the connected-graph bound $s^+(H) \geq |V(H)| - 1$ of Liu, Tang and Zhang [2], the cutvertex inequality of Akbari, Hu and Liu [3, Lemma 2.5], and the outerplanar degree restrictions of Jao and West [4]. Each application of the cutvertex inequality is made before gluing, with both vertex-deleted sides connected. None of these inputs is the maximal-outerplanar $2n$ assertion.

2. Spectral and exact-arithmetic preliminaries

For a real symmetric or Hermitian matrix A let $\mathcal{F}(A) = \operatorname{tr}(A_+^2)$. On a graph adjacency matrix this equals s^+.

Lemma 2.1. *For every Hermitian A,*

$$\mathcal{F}(A) = \max_{X \succeq 0} \{2\operatorname{tr}(AX) - \operatorname{tr}(X^2)\}.$$

If $\mathcal{H} = \bigoplus_j \mathcal{H}_j$ is an orthogonal decomposition, then $\mathcal{F}(A) \geq \sum_j \mathcal{F}(P_j A|_{\mathcal{H}_j})$.

Proof. The squared distance from A to X is minimized on the positive cone by A_+. To verify this directly, write $A = A_+ - A_-$, with $A_+A_- = 0$. For $X \succeq 0$,

$$2\operatorname{tr}(AX) - \operatorname{tr}(X^2) = \operatorname{tr}(A_+^2) - \operatorname{tr}((X - A_+)^2) - 2\operatorname{tr}(A_- X) \leq \operatorname{tr}(A_+^2).$$

The last trace is nonnegative because it is $\operatorname{tr}(A_-^{1/2} X A_-^{1/2})$. Equality holds at $X = A_+$. For the second assertion use the block diagonal matrix consisting of the maximizing positive parts of the compressions. □

Lemma 2.2. *If u is a unit vector, $\alpha = u^*Au$, and R is the compression of A to $u^\perp$, then*

$$\mathcal{F}(R) \geq \mathcal{F}(A) - 2\|Au\|^2 + \alpha^2.$$

Proof. In the decomposition $\mathbb{C}u \oplus u^\perp$ the square trace drops by $2\|Au\|^2 - \alpha^2$. The negative square energy of a compression does not exceed that of the original matrix: this follows either by interlacing, or by applying lemma 2.1 to $-A$ and the subspace $u^\perp$. Subtract this inequality from the trace-square identity. □

Lemma 2.3. *Let L be the disjoint union of two nonempty paths on $d > 2$ vertices, and let J be its join with one new vertex. Put $h = 1 - 2/d$ and*

$$g(d) = 2h^2 + 2h\sqrt{d + h^2} - 6.$$

Then $\mathcal{F}(J) > 2d + g(d)$, and g is strictly increasing on $(2, \infty)$. In particular, $g(16) > 345/128$.

Proof. The path forest has $d-2$ edges, so its positive square energy is $d-2$. Its spectral radius is strictly less than two, since $2I \pm A(P_k)$ are positive definite: their quadratic forms are the sums of consecutive differences or sums squared and the two endpoint squares. The compression of J to the hub and the uniform leaf vector is $\left(\begin{smallmatrix} 0 & \sqrt{d} \\ \sqrt{d} & 2h \end{smallmatrix}\right)$. Its positive eigenvalue is $h + \sqrt{d+h^2}$. Interlacing after removal of the hub therefore yields

$$\mathcal{F}(J) \geq (h + \sqrt{d+h^2})^2 + (d-2) - \rho(L)^2 > 2d + g(d).$$

Both h and $d + h^2$ strictly increase for $d > 2$, and $h > 0$, proving monotonicity. At $d = 16$ the expression is $(7\sqrt{1073} - 143)/32$. The exact inequality $(131/4)^2 < 1073$ gives the stated rational lower bound. □

An integer matrix Y and positive integer s provide the exact certificate $X = YY^{\mathsf{T}}/s^2 \succeq 0$. Its graph score is

$$\frac{4s^2 \sum_{uv \in E} (YY^{\mathsf{T}})_{uv} - \sum_{u,v} (YY^{\mathsf{T}})_{uv}^2}{s^4}. \tag{1}$$

Every entry of Y and the entire squared norm must be included. For negative certificates or upper bounds one may instead use $\mathcal{F}(A) = \min_{X \succeq 0} \|A + X\|_F^2$, proved by the same decomposition.

3. The complete two-root phase estimate

This section proves the analytic bridge used by the local certificates. For a complex matrix M we use the Hermitian parts $\operatorname{Re} M = (M + M^*)/2$ and $\operatorname{Im} M = (M - M^*)/(2i)$; these are not entrywise operations on an arbitrary matrix. The symbol M^* denotes the conjugate transpose. All matrix inequalities are Hermitian inequalities. A matrix is strictly accretive if its Hermitian real part is positive definite. For a finite Hermitian matrix A write $s^+(A) = \sum_{\lambda \in \operatorname{spec}(A),\, \lambda > 0} \lambda^2$, including algebraic multiplicities.

3.1. **The actual boundary-response sector.** Let G be a unit maximal outerplanar graph, and let ab be a boundary edge in a fixed outerplane embedding. In root-first order write

$$A_G = \begin{pmatrix} J & R \\ R^T & C \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathcal{F}_G(t) = R(tI + iC)^{-1} R^T \quad (t > 0).$$

We also allow the auxiliary graph K_2, for which the private block is empty and $\mathcal{F}_{K_2} = 0$. This allowance is a statement about a response; it does not assert the target energy inequality for K_2.

Lemma 3.1 (Two-sided boundary sector)**.** *For every such boundary edge and every $t > 0$,*

$$2 \operatorname{Re} \mathcal{F}_G(t) + t \operatorname{Im} \mathcal{F}_G(t) \succeq 0, \qquad 2 \operatorname{Re} \mathcal{F}_G(t) - t \operatorname{Im} \mathcal{F}_G(t) \succeq 0.$$

Both matrices are positive definite when $|G| \geq 4$.

Proof. Regard the embedding as a triangulated polygon. Its long boundary arc opposite ab is $a, v_1, \ldots, v_{n-2}, b$. Let $S = (N(a) \cup N(b)) \setminus \{a, b\}$. The neighbors of any polygon vertex induce a path: consecutive neighbors bound an incident triangular face, whereas an edge between nonconsecutive neighbors would cross an intermediate spoke. The boundary edge ab has exactly one common neighbor z, its incident face vertex. The two neighbor paths therefore join at z. There is no edge between an exclusive a-neighbor and an exclusive b-neighbor: together with ab it would form an induced four-cycle, since either possible diagonal would contradict exclusivity. A polygon triangulation is chordal (successive ear deletion gives a 2-tree), so this is impossible. It follows that the actual induced graph on S is one path, in long-arc order, beginning at v_1 and ending at v_{n-2}.

Between each consecutive pair s_j, s_{j+1} of S, retain the complete closed polygonal interval and its closing edge $s_j s_{j+1}$. Its strict interior is a pocket U_j. Noncrossing forbids an edge

from U_j to a vertex outside that interval, except its two endpoints. The pockets are disjoint and exhaust the vertices outside $S\cup\{a,b\}$; their closing edges belong to the path on S exactly once. Each nonempty pocket with its two endpoints is a smaller maximal outerplanar graph, rooted at that closing edge. Let P be the adjacency matrix of the path on S, let R_0 be the contacts of a,b with S, and let L_j insert the two pocket-root coordinates into S.

Induct simultaneously on both asserted signs, starting with K_2. For a pocket let $\mathcal{F}_j$ be its actual two-root correction; it is zero for an empty pocket. Schur elimination, with the factors i in the off-diagonal blocks retained, gives

$$Z = tI + iP + \sum_j L_j \mathcal{F}_j L_j^T, \qquad \mathcal{F}_G = R_0 Z^{-1} R_0^T.$$

For a real symmetric private adjacency matrix D, $\operatorname{Re}(tI+iD)^{-1} = t(t^2I+D^2)^{-1} \succ 0$. Consequently $\operatorname{Re} Z \succeq tI$, so Z is invertible. The simultaneous induction hypotheses yield

$$\operatorname{Re}((2+it)Z) = t(2I-P) + \sum_j L_j(2\operatorname{Re}\mathcal{F}_j - t\operatorname{Im}\mathcal{F}_j)L_j^T \succ 0,$$

$$\operatorname{Re}((2-it)Z) = t(2I+P) + \sum_j L_j(2\operatorname{Re}\mathcal{F}_j + t\operatorname{Im}\mathcal{F}_j)L_j^T \succ 0.$$

Here $2I \pm P \succ 0$: their quadratic forms are the sum of the two endpoint squares and the consecutive squared differences or sums. For a one-vertex path the endpoint contribution is $2|x_1|^2$. For any invertible Z and scalar q,

$$Z^{-*}\operatorname{Re}(qZ)Z^{-1} = \operatorname{Re}(\overline{q}\,Z^{-1}).$$

Applying this to $q = 2 \pm it$, then taking real congruence by R_0, proves both signs. If $n \geq 4$, R_0 has row rank two: the rows share z, and at least one root has an exclusive neighbor. If neither did, S would be the singleton z and there would be no pocket, so $n = 3$. Thus the congruences are positive definite for $n \geq 4$. For the remaining triangle, $\mathcal{F}_G = t^{-1}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, which directly verifies the rank-one base case. □

3.2. **The root ball and a sharp determinant-phase lemma.** Fix $t > 0$ and put

$$K = 2I + J \succ 0, \qquad c = t^{-1} - i/2, \qquad \rho = |c| = \sqrt{t^{-2} + 1/4}.$$

Define the enlarged root ball

$$\mathcal{B}_t = \{H : 2\operatorname{Re} H - t\operatorname{Im} H \succeq tH^*KH\}.$$

No symmetry restriction is imposed in this enlarged set. Putting $X = K^{1/2}HK^{1/2}$ and completing the square gives the exact equivalence

$$H \in \mathcal{B}_t \quad\Longleftrightarrow\quad (X-cI)^*(X-cI) \preceq \rho^2 I \quad\Longleftrightarrow\quad X = cI + \rho U, \quad \|U\|_2 \leq 1.$$

Indeed the cross term in the square is $-2\operatorname{Re} X/t + \operatorname{Im} X$. In particular the ball contains $H = 0$ and is compact and convex.

For the physical old graph let $Z_0 = tI + iJ + \mathcal{F}_0$ and $H_0 = Z_0^{-1}$. The spectral theorem gives $\operatorname{Re}\mathcal{F}_0 \succeq 0$, so Z_0 is strictly accretive and invertible. Lemma 3.1 gives $2\operatorname{Re} Z_0 + t\operatorname{Im} Z_0 \succeq tK$. Inverse congruence gives $H_0 \in \mathcal{B}_t$. Notice the sign reversal:

$$H_0^*(2\operatorname{Re} Z_0 + t\operatorname{Im} Z_0)H_0 = 2\operatorname{Re} H_0 - t\operatorname{Im} H_0.$$

Lemma 3.2 (Two-dimensional contraction phase)**.** *If W is any complex 2×2 matrix with $\|W\|_2 < 1$, and its singular values are σ_1, σ_2, then*

$$\max_{\|U\|_2 \leq 1} \operatorname{Im}\operatorname{Tr}\operatorname{Log}(I+UW) = \arcsin\sigma_1 + \arcsin\sigma_2.$$

The matrix logarithm is the accretive branch, equivalently the branch obtained by the path $I + sUW$, $0 \leq s \leq 1$.

Proof. Write $W = P\,\mathrm{diag}(\sigma_1,\sigma_2)Q^*$. Cyclicity of the power-series trace for $\mathrm{Log}(I+UW)$ permits replacing W by $D = \mathrm{diag}(\sigma_1,\sigma_2)$ and U by Q^*UP; the contraction domain is unchanged. For a contraction with fixed singular vectors, replace its two singular values by complex variables in the closed unit disks. The resulting $I+UD$ has real part at least $(1-\|D\|_2)I \succ 0$ throughout those disks. The imaginary trace logarithm is harmonic in each variable. The maximum principle successively moves each variable to the unit circle. The resulting matrix U is unitary.

A unitary 2×2 matrix can be written

$$e^{i\phi}\begin{pmatrix} e^{i\alpha}\cos\theta & e^{i\beta}\sin\theta \\ -e^{-i\beta}\sin\theta & e^{-i\alpha}\cos\theta \end{pmatrix}, \qquad 0\le\theta\le\pi/2.$$

Its determinant expression is

$$1+\sigma_1\sigma_2 e^{2i\phi} + e^{i\phi}\cos\theta(\sigma_1 e^{i\alpha}+\sigma_2 e^{-i\alpha}).$$

For fixed ϕ, the last term describes a filled ellipse. Every point of this ellipse is the determinant of some strictly accretive $I+UD$. The two eigenvalue arguments are each in $(-\pi/2,\pi/2)$, so their sum is in $(-\pi,\pi)$; the ellipse therefore avoids both zero and the negative real axis. The principal argument is harmonic on a neighborhood of this ellipse. Its maximum is on the boundary, obtained with $\cos\theta = 1$. If the ellipse degenerates to a segment, every point is already attained with $\cos\theta = 1$; if $D = 0$ the assertion is immediate.

We have reduced to diagonal U. The two angles $\phi+\alpha$ and $\phi-\alpha$ can be chosen independently. The circle $1+\sigma e^{i\psi}$ has maximal argument $\arcsin\sigma$, attained at $e^{i\psi} = -\sigma + i\sqrt{1-\sigma^2}$; this follows either by tangency from the origin or by direct differentiation. Adding the two independent scalar maxima proves the assertion, including zero and repeated singular values and nonnormal initial W. □

3.3. **A globally lifted algebraic phase bound.** Let F be any complex 2×2 matrix in the convex sector

$$\mathcal{S}_t = \{F : 2\,\mathrm{Re}\,F + t\,\mathrm{Im}\,F \succeq 0\}.$$

Set

$$L = K^{-1/2}FK^{-1/2}, \qquad A = I + cL, \qquad W = \rho LA^{-1}.$$

Both of the following are exact, noncommutative identities:

$$\mathrm{Re}\,A = I + \mathrm{Re}\,L/t + \mathrm{Im}\,L/2 \succeq I,$$
$$A^*A - \rho^2 L^*L = I + 2\,\mathrm{Re}\,L/t + \mathrm{Im}\,L \succeq I.$$

Consequently A is invertible and

$$I - W^*W = A^{-*}(A^*A - \rho^2 L^*L)A^{-1} \succ 0.$$

In particular $\|W\|_2 < 1$. If $H\in\mathcal{B}_t$ corresponds to U, then, by cycling the determinant,

$$\det(I+HF) = \det\big(I+(cI+\rho U)L\big) = \det A\,\det(I+UW) \neq 0.$$

This also holds for sF, $0\le s\le 1$. Define $\psi(H,F)$ as the continuous determinant phase along this radial path, starting at zero. The sector and the ball are convex, and the determinant never vanishes on their product. Thus the continuous phase lift normalized at $F=0$ is single-valued on the whole product, not merely on a chosen ray. The two accretive-factor trace phases in the displayed factorization add to this lift: they are continuous in s, agree modulo 2π, and their difference is zero at $s=0$. Therefore

$$\psi(H,F) = \mathrm{Im}\,\mathrm{Tr}\,\mathrm{Log}\,A + \mathrm{Im}\,\mathrm{Tr}\,\mathrm{Log}(I+UW).$$

Lemma 3.2 now gives the exact enlarged-ball maximum

$$b(F,t) := \max_{H\in\mathcal{B}_t}\psi(H,F) = \mathrm{Im}\,\mathrm{Tr}\,\mathrm{Log}\,A + \arcsin\sigma_1(W) + \arcsin\sigma_2(W). \tag{2}$$

Since $H = 0$ belongs to the ball, $b \geq 0$. Strict accretivity of A puts its trace phase in $(-\pi, \pi)$, while the sum of the two inverse sines is in $[0, \pi)$. Hence $b < 2\pi$.

For a physical cap response $F = T(tI + iB)^{-1}T^T$, the actual old root H_0 is an element of the ball and stays the same across the entire graft. Its actual lifted graft phase is at most $b(F, t)$. Sharpness over the enlarged ball neither asserts that a maximizing root is realizable by a graph nor that the same maximizing root works at different t.

3.4. **Inverse invariance and removal of lossless rays.** If $H \in \mathcal{B}_t$ is invertible, inverse congruence is reversible:

$$2\,\mathrm{Re}(H^{-1}) + t\,\mathrm{Im}(H^{-1}) \succeq tK.$$

Thus, for any $F \in \mathcal{S}_t$,

$$H_F = (H^{-1} + F)^{-1} \in \mathcal{B}_t.$$

Existence of this inverse follows from the nonvanishing determinant factorization above; it is not inferred from accretivity of $H^{-1} + F$, which need not hold on the enlarged domain.

Lemma 3.3 (Lossless positive-ray monotonicity)**.** *For $F \in \mathcal{S}_t$, a real vector $v \in \mathbb{R}^2$, and $\gamma \geq 0$,*

$$b\left(F + \frac{\gamma v v^T}{t + 2i}, t\right) \leq b(F, t).$$

The same assertion holds with any real positive semidefinite matrix Γ in place of $\gamma v v^T$.

Proof. Put $P = vv^T/(t + 2i)$. Direct computation gives $2\,\mathrm{Re}\,P + t\,\mathrm{Im}\,P = 0$. For fixed invertible $H \in \mathcal{B}_t$, all the matrices $H_s = (H^{-1} + F + sP)^{-1}$ exist and lie in the same ball for $s \geq 0$. Differentiating the continuous determinant logarithm gives

$$\frac{d}{ds}\psi(H, F + sP) = \mathrm{Im}\,\mathrm{Tr}(H_s P) = \frac{t\,\mathrm{Im}\,f_s - 2\,\mathrm{Re}\,f_s}{t^2 + 4}, \qquad f_s = v^T H_s v.$$

The ball inequality, tested on the real vector v, gives

$$2\,\mathrm{Re}\,f_s - t\,\mathrm{Im}\,f_s \geq t\,v^T H_s^* K H_s v \geq 0.$$

The derivative is therefore nonpositive. The phase normalization used here is the global lift on the convex sector established above, so no additional 2π term appears when the path starts at F.

Invertible matrices are dense in the root ball: interpolate toward its invertible center cK^{-1} and avoid the finitely many zeros of the resulting determinant polynomial. Nonvanishing of $\det(I + HF)$ and continuity of the normalized lift extend the inequality to singular H. It now holds for every H before taking the maximum, proving the claimed inequality for b. A real spectral decomposition of $\Gamma \succeq 0$ and successive rank-one additions give the last assertion. □

In particular, if a physical response can be written $F = \widetilde{F} + \Gamma/(t + 2i)$ with $\Gamma \succeq 0$, then $\widetilde{F} \in \mathcal{S}_t$ and $b(F, t) \leq b(\widetilde{F}, t)$. Only the phase upper bound has changed. The private graph energy and the original contact count are those of the physical, unnormalized cap.

Lemma 3.4 (Dominance by the linear ball support)**.** *Put $L = K^{-1/2} F K^{-1/2}$ and*

$$E(F, t) = \frac{\mathrm{Im}\,\mathrm{Tr}\,L}{t} - \frac{\mathrm{Re}\,\mathrm{Tr}\,L}{2} + \rho\|L\|_*.$$

For every $F \in \mathcal{S}_t$,

$$0 \leq tb(F, t) \leq \min\{tE(F, t), 2\pi t\}.$$

Here $\|L\|_*$ is the sum of the singular values of the complete complex matrix L.

Proof. The root-ball parameterization and singular-value trace duality give

$$\max_{H\in\mathcal{B}_t} \operatorname{Im}\operatorname{Tr}(HF) = \operatorname{Im}(c\operatorname{Tr}L) + \rho\|L\|_* = E(F,t).$$

For completeness, if $L = PDQ^*$ is a singular-value decomposition, then the diagonal entries of Q^*UP have absolute values at most one, so $\operatorname{Im}\operatorname{Tr}(UL) \le \operatorname{Tr}D$; $U = iQP^*$ attains equality. The support is nonnegative because the ball contains zero. For invertible H in the ball, inverse invariance gives $(H^{-1}+sF)^{-1} \in \mathcal{B}_t$ for $0 \le s \le 1$. Differentiating the globally normalized phase and integrating yields

$$\psi(H,F) = \int_0^1 \operatorname{Im}\operatorname{Tr}((H^{-1}+sF)^{-1}F)\,ds \le E(F,t).$$

Continuity covers singular H, and maximizing gives $b \le E$. The bound $b < 2\pi$ was already proved. □

3.5. **The exact energy identity and complete frequencies.** Let A_H be the actual old adjacency matrix, B the private adjacency, and T the two root-contact rows. Let E insert the two root coordinates in the old vertex space. The whole adjacency matrix is

$$A_G = \begin{pmatrix} A_H & ET \\ T^TE^T & B \end{pmatrix}.$$

The closing old edge is already in A_H and is not inserted a second time. Root and private Schur elimination yield

$$\frac{\det(tI+iA_G)}{\det(tI+iA_H)\det(tI+iB)} = \det(I+H_0F).$$

Define the left phase as the difference of the three accretive matrix trace logarithms. Equivalently, it is

$$\delta(t) = \sum_{\lambda\in\operatorname{spec}(A_G)} \arctan(\lambda/t) - \sum_{\lambda\in\operatorname{spec}(A_H)} \arctan(\lambda/t) - \sum_{\lambda\in\operatorname{spec}(B)} \arctan(\lambda/t).$$

It equals $\psi(H_0,F)$, not an arbitrarily rewrapped scalar argument. For example, scale the actual cross contacts by $\sqrt{s}$; the Schur correction becomes sF, all whole matrices remain Hermitian, and both phase definitions agree continuously from zero at $s = 0$.

For a real $\lambda \neq 0$, direct integration gives

$$\int_0^R t\arctan(\lambda/t)\,dt = \frac{R^2}{2}\arctan(\lambda/R) + \frac{\lambda R}{2} - \frac{\lambda|\lambda|}{2}\arctan(R/|\lambda|) = \lambda R - \frac{\pi}{4}\lambda|\lambda| + o(1).$$

For $\lambda = 0$ the integral is zero. Since $\operatorname{Tr}A_G = \operatorname{Tr}A_H + \operatorname{Tr}B$, the linear terms cancel when summing these expressions. Also

$$\operatorname{Tr}A_G^2 - \operatorname{Tr}A_H^2 - \operatorname{Tr}B^2 = 2\operatorname{Tr}(TT^T).$$

Using $s^+(A) = \frac{1}{2}\operatorname{Tr}A^2 + \frac{1}{2}\sum_\lambda \lambda|\lambda|$ therefore proves the exact identity

$$s^+(A_G) - s^+(A_H) = s^+(B) + \operatorname{Tr}(TT^T) - \frac{2}{\pi}\int_0^\infty t\delta(t)\,dt. \tag{3}$$

The actual finite-spectrum phase integral converges: equal traces cancel the leading term at infinity, and at zero the phase is bounded. Consequently any complete, finite upper bound I on $\int_0^\infty tb(F,t)\,dt$, or on the corresponding lossless-normalized upper bound, gives

$$s^+(A_G) - s^+(A_H) \ge s^+(B) + \operatorname{Tr}(TT^T) - \frac{2}{\pi}I.$$

The sector condition alone does not guarantee that the upper envelope is integrable. For $0 < t < \varepsilon$ the universal $b < 2\pi$ costs at most $2\varepsilon^2$ in this energy inequality. The remaining compact interval and the entire infinite tail require the separately proved outward enclosures in each certificate. The substitution $z = 1/t$ is an exact change of variables, including $z = 0$ for the infinite endpoint; no finite cutoff is a replacement for that tail proof.

3.6. **A consistent exact example requiring a phase lift.** An example must respect the coupling $A = I + cL$ and $W = \rho L A^{-1}$; one cannot specify A and two arbitrary singular values independently. At $t = 1$, keep the actual metric $K = 2I + J$, and set

$$F = 4iK, \qquad H = \frac{1+i}{2} K^{-1}.$$

Then $2\operatorname{Re} F + \operatorname{Im} F = 4K \succ 0$, and $2\operatorname{Re} H - \operatorname{Im} H = H^* K H = \frac{1}{2} K^{-1}$. Thus this is an admissible example in the enlarged algebraic sector and root ball. It is not asserted to be a physical maximal-outerplanar root or cap response. Here $L = 4iI$, $A = (3+4i)I$, and both squared singular values of W are $4/5$. Moreover

$$I + HF = (-1+2i)I, \qquad \det(I + HF) = -3 - 4i.$$

The radial lift has value $2\pi - 2\arctan 2 > \pi$, whereas the principal argument of $-3-4i$ is negative. This is also the sharp ball maximum: each diagonal phase of $I + UW$ is tangent-optimal. All entries of F, H are rational complex numbers; the stated identities and squared singular values admit standard-library rational verification. A scalar A forces scalar L and equal singular values of W, so the pair $(3/5, 4/5)$ cannot be combined with $A = (3+4i)I$ as a coupled root-ball witness.

4. Finite and domination inputs

4.1. **Exact finite basis and order of dependence.** We first prove the following bounded-order assertion. The proof of the domination theorem in Appendix C uses this assertion, but the finite assertion uses only the dominating-*triangle* argument below for orders at least sixty. That triangle argument is independent of every finite $2n$ assertion. This order of dependence is essential.

Theorem 4.1. *Every maximal outerplanar graph of order $8 \le n \le 71$ satisfies $\mathcal{F}(G) \ge 2n$.*

Here and below an exact finite assertion means the stated exhaustive algorithm, applied to all its declared inputs. The integer factors and dynamic-program tables are supplied in the supplement. Their scores, rather than their recorded status strings, are the objects verified.

4.2. **Moments, gap geometry, and the actual hub inequality.** Let $M_2(G) = \sum_v d(v)^2$ and $\mu_j = \operatorname{tr} A^j$. A polygon triangulation has $n-2$ triangles and $n-3$ quadrilateral cycles: a quadrilateral cycle bounds exactly two triangles, since its interior contains no other polygon vertex, and its diagonal is unique. Hence

$$\mu_2 = 4n - 6, \qquad \mu_3 = 6n - 12, \qquad \mu_4 = 2M_2(G) + 4n - 18. \tag{4}$$

Using $X = t(A + rI)^2 \succeq 0$ in lemma 2.1 and optimizing $t \ge 0$ gives

$$\mathcal{F}(A) \ge \frac{(\mu_3 + 2r\mu_2)^2}{\mu_4 + 4r\mu_3 + 6r^2\mu_2 + nr^4} \quad (r > 0). \tag{5}$$

The denominator is $\operatorname{tr}(A + rI)^4 > 0$; the numerator's unsquared quantity is positive. Consequently it suffices that $M_2(G) \le T_r(n)$, where

$$T_r(n) = \frac{1}{2}\left\{ \frac{(6n - 12 + 2r(4n-6))^2}{2n} - \left(4n - 18 + 4r(6n-12) + 6r^2(4n-6) + nr^4\right) \right\}. \tag{6}$$

Choose an actual maximum-degree vertex v, put $d = d(v)$ and $k = n - d - 1$. Its neighbors $x_1, \dots, x_d$, in polygon order, induce a path. Its $d-1$ edges bound independent polygon gaps, of masses $m_i \ge 0$ summing to k; let ℓ be the number of positive masses. Each positive gap of mass $m \ge 1$ induces a triangulation on $m+2$ vertices including its roots; an empty gap contributes zero to S. Every interior vertex sees only those two roots among $N(v)$. Exactly one sees both roots. If the two degrees in that gap block are r, s, the number

$S_i = \sum_{w \text{ in gap}} |N(w) \cap N(v)|^2$ equals $r+s$. Indeed the root incidences number $r+s-2$, and the unique doubly counted vertex contributes two additional units after squaring. Since $r + s \le m + 3$, we obtain

$$S := \sum_{w \notin N[v]} |N(w) \cap N(v)|^2 \le k + 3\ell. \tag{7}$$

The root-degree inequality follows by counting the unique common neighbor of the boundary edge; there are at most $m + 1$ root incidences.

For $k > 0$ define

$$Q(k, \ell) = \min\{k^2, (k-\ell)^2 + 6(k-\ell) + 3\ell - 2\}, \qquad Q(0, 0) = 0.$$

The following three bounds are purely combinatorial:

$$M_2(G) \le U(n, d, k, \ell) := \min\{n^2+7n-18, (2d+8)n-6d-12, d^2+9d-10+Q(k,\ell)+15k+2\ell\}. \tag{8}$$

For the first, delete an ear from the order-n graph. Its two neighbors a, b become a boundary edge of the order-$(n-1)$ triangulation, with one common neighbor. Thus their degrees in that smaller graph sum to at most n. Restoring the ear increases M_2 by $2(d_{\text{old}}(a)+d_{\text{old}}(b))+6 \le 2n + 6$. Starting with the triangle proves the first bound by induction. Summing $(d(w)-2)(d(w)-d) \le 0$ and using $\sum_w d(w) = 4n-6$ proves the second. To prove the third, start with the closed-neighborhood fan, whose degree-square sum is $d^2+9d-10$. Expanding the two root-degree increments of a gap gives at most $m^2+11m+4(r+s)-10 \le m^2+15m+2$; two neighboring nonempty gaps add at most $2m_i m_{i+1}$. Nonneighboring gaps have no common root. Thus the remaining quadratic term is bounded by Q. To check this last claim, the adjacency graph of the positive gaps is a subgraph of a path. Writing $m_i = 1+y_i$ gives a constant at most $3\ell-2$, a linear term at most $6\sum y_i$, and a quadratic term at most $(\sum y_i)^2$; the alternative k^2 bound follows by completing the nonnegative missing cross terms.

Let $R = G - v$, $N = d + k$, $h = (d-1)/d$, and $\rho(R) = \mathcal{F}(R) - (N-1)$. This R is connected: the outer cycle with v deleted is a Hamiltonian path. Let u be the unit vector equal to $1/\sqrt{d}$ on $N(v)$ and zero elsewhere. Then

$$u^* A_R u = 2h, \qquad \|A_R u\|^2 = (4d - 6 + S)/d.$$

The compression on $\operatorname{span}(e_v, u)$ has positive eigenvalue $h + \sqrt{d + h^2}$. Apply lemma 2.2 to R and then orthogonal pinching to G to obtain the exact inequality

$$\mathcal{F}(G) - 2n \ge 2h\sqrt{d + h^2} - k - 5 - 2S/d + 6/d^2 + \rho(R). \tag{9}$$

There is no assertion of monotonicity under edge deletion here. The connected-graph theorem of [2, Theorem 1.2] supplies $\rho(R) \ge 0$. If any proved rational $f \le \rho(R)$ is available, set

$$a = k + 5 + 2(k + 3\ell)/d - 6/d^2 - f. \tag{10}$$

The hub test accepts when $a \le 0$, or when $4h^2(d + h^2) - a^2 \ge 0$. Squaring is used only in the latter branch, where both sides of the unsquared comparison are nonnegative.

A useful independent residual bound is

$$M_2(R) \le 4d - 6 + Q(k, \ell) + 13k. \tag{11}$$

For a gap block the ear bound gives $M_2 \le m^2+11m$. Expanding its root increments against the path degrees, now at most two, changes the extra term to $2(r + s) - 6 \le 2m$. Adjacent gaps have the same cross term as before, proving (11). The residual graph has $d - 1 + 2k$

edges, k triangles, and $k-\ell$ quadrilateral cycles. Thus an upper bound V for $M_2(R)$ gives

$$\mu_2(R) = 2(d-1+2k), \qquad \mu_3(R) = 6k,$$
$$\mu_4(R) \le \overline{\mu}_4 := 2V - 2(d-1+2k) + 8(k-\ell),$$
$$f(V) = \max\left(0, \max_{16\le j\le 64}\left\{\frac{(\mu_3+2r_j\mu_2)^2}{\overline{\mu}_4+4r_j\mu_3+6r_j^2\mu_2+Nr_j^4} - (N-1)\right\}\right), \quad r_j = j/16. \tag{12}$$

Every denominator is checked positive. Replacing the true fourth moment by its upper bound decreases the witness score.

4.3. **The orders through forty-two.** For $n = 8, 9, 10, 11$, enumerate all cyclically labeled polygon triangulations. Their counts are respectively $132, 429, 1430, 4862$. For each, the supplied integer Gram factor has the score (1) strictly greater than $2n$. The verifier checks all edge endpoints, $2n-3$ distinct edges, every boundary edge, no crossing diagonals, all factor entries, distinct edge sets, and these four Catalan counts. A noncrossing polygon graph with all boundary edges and $2n-3$ edges is a triangulation. Every abstract maximal outerplanar graph has a cyclic labeling, so this covers every graph, without assuming that the records represent isomorphism classes.

The same check on all $5+14+42 = 61$ triangulations of orders $5, 6, 7$ gives scores greater than $899/100, 56/5, 27/2$, respectively. At order three, $\mathcal{F}(K_3) = 4$; at order four the graph is $K_4 - e$, with $\mathcal{F} = (9+\sqrt{17})/2 > 6$. Consequently

$$\mathcal{F}(B) \ge 2|B| - 2 \quad (3 \le |B| \le 7). \tag{13}$$

For $d \le 8$, $n \ge 12$, the second bound in (8) gives $M_2 \le 24n-60$, whereas $T_2(n)-(24n-60) = 3(n-12)(5n-9)/n \ge 0$. For $12 \le n \le 33$ and $9 \le d \le n-1$, enumerate exactly $k = n-d-1$, with $\ell = 0$ if $k = 0$ and otherwise $1 \le \ell \le \min(k, d-1)$. All 1949 tuples pass either (8)–(6), using one of the 41 shifts $r = j/16$, $24 \le j \le 64$, or the hub test with $f = 0$. These are finite rational comparisons, including a regenerated full index set, rather than a proposed continuous optimum.

The residual graph R is a chain of the positive gap blocks and intervening bridge paths. It is not a disjoint union of the blocks. Write $c = 8/9$. The gluing lemma of [3, Lemma 2.5, (2.11)] states that two connected graphs of order at least two, identified at vertices whose deletions are both connected, satisfy

$$\rho(H_1 \vee H_2) \ge c(\rho(H_1) + \rho(H_2)) - 2/9. \tag{14}$$

Every split between consecutive nonempty gaps has this property: the cut vertex is an end root on each side of the chain, and deleting it leaves the polygon side attached at the other root connected. A nonempty bridge path has zero reserve and remains connected after its attaching endpoint is deleted. The two possible end paths of one polygon attach at distinct roots. These observations also cover paths consisting of a single edge.

For $34 \le n \le 42$, the cases not yet paid have $k \le 11$; the verifier checks this before using a block reserve. Thus every gap block has order at most thirteen, already covered above or by (13), and $\rho(B) \ge m+1$. Applying (14) up to twice to its two end paths leaves the conservative reserve $f((m)) = (64m+30)/81$. An omitted end path can only improve this lower bound, because $(8x-2)/9 \le x$ for $x \ge 0$. For an ordered mass composition t set

$$f(t) = \max_{1\le j<|t|}\{c[f(t_1,\dots,t_j) + f(t_{j+1},\dots,t_{|t|})] - 2/9\}. \tag{15}$$

Each split is legal on the actual chain. Minimizing this lower bound over all positive ordered compositions of k into ℓ parts gives a uniform reserve. Of the complete 2687 tuples at orders 34–42, 2386 pass the whole-moment test, 130 the zero-reserve hub test, and 171 the composition-reserve hub test. This proves the order-42 basis before it is used below. In particular every gap block of order at most forty-two now has reserve at least $m+1$.

4.4. **A degree-histogram envelope for orders forty-three to seventy-one.** The following exact caps suffice:

n	43–46	47–48	49	50	51–54	55–58	59–64	65–71
$D(n)$	29	28	27	25	26	27	28	29

We give the necessary histogram set used to prove this statement. Let h count degrees at least six, E be their sum of $(d(w)-4)$, and let a, b, c_4, q count degrees $2, 3, 4, 5$. The letter c_4 here is a count, not the constant $8/9$. Jao–West [4, Theorems 3.3–3.4], the degree sum, and nonadjacency of ears give

$$0 \le h \le \lfloor (n-6)/2 \rfloor, \qquad 2h \le E \le \min((D-4)h, n-6),$$

$$0 \le a \le \lfloor n/2 \rfloor, \quad q \ge 0, \quad b = E + q + 6 - 2a \ge 0, \quad c_4 = n - h - q - a - b \ge 0. \tag{16}$$

The case $h = E = 0$ is included. Delete all original ears. They are pairwise nonadjacent, the remaining graph is a polygon triangulation on $m = n - a \ge 22$ vertices, and each surviving high vertex has lost at most two neighbors. Therefore impose also, for $h \ge 1$,

$$E + 4h \le D_{\mathrm{sum}}(m,h) + 2h, \qquad D_{\mathrm{sum}}(m,h) = \begin{cases} m-1, & h = 1, \\ m - 6 + 4h, & 2 \le h < m/2, \\ 2m - 6 + 2h, & h \ge m/2. \end{cases} \tag{17}$$

These are precisely the selected-vertex degree-sum bounds needed here. In particular $E \le n - 6$ follows for $h \ge 2$ from the $h < n/2$ case and for $h = 1$ from $D \le n - 2$.

Put $C = D - 4$. The maximum square sum $S_C(h, E)$ of h integers in $[2, C]$ with sum E is computed by integer knapsack:

$$S_C(0,0) = 0, \qquad S_C(h,E) = \max_{2 \le x \le C} \{S_C(h-1, E-x) + x^2\},$$

with infeasible states absent. Equivalently, if $C > 2$ and $(j, u) = \mathrm{divmod}(E - 2h, C - 2)$, then $S_C = 4h + (C^2 - 4)j + \mathbf{1}_{u>0}((2+u)^2 - 4)$; transferring a unit from a smaller unsaturated coordinate to a larger one proves this convex maximum. At $C = 2$ use $S_2(h, 2h) = 4h$. Since $\sum_w (d(w) - 4) = -6$, every actual graph has

$$M_2(G) \le 16n - 48 + S_C(h,E) + 4a + b + q. \tag{18}$$

The verifier maximizes the right side over every integer in (4.4)–(17). It also independently compares the knapsack values with the displayed packing formula. For every one of the twenty-nine orders the maximum is strictly less than

$$B(n) = 1359n/32 - 729/2 + 441/n = T_{5/2}(n).$$

The smallest rational difference is $1567/2080$, at $n = 65$. This is an optimization over a necessary superset of actual degree histograms; no realization assumption is made about a maximizing histogram. Equation (5) proves the cap table.

4.5. **Completing the high-degree finite tuples.** For $43 \le n \le 71$ enumerate $D(n) < d \le n - 1$, then $k = n - d - 1$ and the full gap-count range used above. There are 11885 tuples. Test the following rules in the stated order: whole moments; hub with zero reserve; hub with the coarse residual moment bound; single-gap partition; balanced block reserve; the exact unmasked degree-state maxima; and the capped zero-mask maxima with their complements. The numbers assigned to these rules are

$$3718, \quad 932, \quad 4049, \quad 527, \quad 2201, \quad 328, \quad 130. \tag{19}$$

Their sum is 11885; the equality of the actual tuple sets, and not only the sum, is checked. Appendix B specifies the last two recurrences and their complete target reconstruction.

For the single-gap rule, the gap block and the hub with its remaining $d - 2$ neighbors are disjoint induced subgraphs. The latter is the join of a hub with at most two paths on $L = d - 2 \ge 21$ vertices. The proof of lemma 2.3, using $h = 1 - 2/L$, gives surplus greater than two provided $4h^2(L + 10) > 100$. Its difference at $L = 21$ is $664/441$, and it increases

thereafter. The block has surplus at least minus two by the already proved order-42 basis. The scan checks its actual order before applying this rule.

For the balanced-block rule every actual gap block has order at most twenty-six, and the scan checks $k+2 \le 42$. Fix a balanced ordered binary gluing tree on the ℓ blocks, with leaf depths t_i equal to $a = \lfloor \log_2 \ell \rfloor$ or $a+1$. There are $2^{a+1} - \ell$ shallow and $2\ell - 2^{a+1}$ deep leaves. This tree is obtained by expanding leaves of the complete depth-a tree, preserving the actual block order. Set $c = 8/9$ and $W = \sum_i c^{t_i}$. Iterating (14) gives

$$\rho(R) \ge \frac{64}{81} \sum_i c^{t_i} m_i + \frac{30}{81} W - \frac{2}{9} \sum_{\text{internal } j} c^{t_j}.$$

For a full binary tree $W = 1 + (2c-1) \sum_{\text{internal}} c^{t_j}$. Also $\sum c^{t_i} m_i \ge W + (k-\ell) \min c^{t_i}$. Consequently the rational uniform reserve is

$$\rho(R) \ge \frac{496}{567} W + \frac{2}{7} + \frac{64}{81} (k-\ell) \min_i c^{t_i}. \tag{20}$$

The construction is legal for each actual ordered composition; there is no averaging over gap orders or split trees.

4.6. **The two complements of the capped zero-mask classes.** We spell out both complements since a maximum over a restricted class alone does not prove a statement about all graphs. The 130 last tuples split into 125 classes with a nonhub cap $C < d$, at orders 63–71, and five classes with $C = d$, at orders 60–63. The zero-mask computation applies only when no triangle containing v dominates the graph and every nonhub degree is at most C.

Lemma 4.2. *A maximal outerplanar graph with a dominating triangle and order $n \ge 60$ satisfies $\mathcal{F}(G) > 2n$.*

Proof. Let the three hubs be h_0, h_1, h_2. Each of their three polygon gaps is either empty or consists of a prefix adjacent to one hub and a suffix adjacent to the other, overlapping in their unique common pivot. To prove this, every gap vertex must see an endpoint of its boundary edge, since an edge to the third hub would cross that edge. Noncrossing makes each endpoint's neighbors consecutive; there is exactly one common neighbor on that side. The resulting spokes and path edges already triangulate the gap, leaving no additional possible diagonal. Assign each nonpivot to its unique hub and each pivot to either endpoint. Each of the three induced blocks is a hub joined to at most two paths, and they partition the vertices.

Let $S(a,b) = K_1 \vee (P_a \sqcup P_b)$, $q = a+b$, and let $q_0, \dots, q_{54}$ be the nondecreasing rational sequence in `three-hub-partition-certificate.json`. Its acceptance conditions are

$$\mathcal{F}(S(a,b)) - 2(q+1) \ge q_q, \qquad q_0 = q_1 = q_2 = -4, \qquad q_{54} = 203/25.$$

At $q \le 2$ this follows by direct spectra. For $3 \le q \le 13$ every $0 \le a \le \lfloor q/2 \rfloor$ has a complete integer Gram factor, checked by (1) with scale 16384. For $14 \le q \le 54$ put $r = (q-2)/q$. Interlacing and hub/uniform compression give surplus strictly greater than $2r^2 + 2r\sqrt{q+r^2} - 8$. The rational check

$$T = q + 8 + q_q > 0, \quad 8 + q_q > 0, \quad 4r^2 T > (8+q_q)^2$$

is equivalent to this last expression exceeding q_q. Both r and the expression increase, so all $q > 54$ have surplus greater than q_{54}. No spectral approximation enters these tests.

If $r_0 = 1, 2, 3$ gaps are nonempty, their pivot edges are respectively one edge, a two-edge path, or all edges of the hub triangle. For $n = 60$, enumerate every ordered nonnegative triple (b_0, b_1, b_2) with sum $57 - r_0$: there are $1653, 1596, 1540$ triples respectively. For every triple try all 2^{r_0} pivot allocations and take the largest sum of the three truncated q-values. The exact minimum over these triples and gap patterns is at least $3/25$. These are the complete loops in `verify_three_hub_partition.py`, with rational comparisons. For

$n > 60$, reduce the three actual nonpivot counts coordinatewise to a triple of sum $57 - r_0$, and use its allocation. Monotonicity of the sequence preserves the bound. If an actual count exceeds 54, its surplus exceeds $203/25 > 8$, while each other surplus is at least -4. In either case the sum is positive. Orthogonal pinching proves the assertion. □

For completeness the finite high-second-degree complement uses the following universal geometry and certificates. If uv is an edge with $h = 1$ or 2 common neighbors, then $G[N[u] \cup N[v]]$ is an induced triangulation of order $d(u)+d(v)-h$. Indeed the two stars and their two neighbor paths force exactly $2|N[u] \cup N[v]| - 3$ distinct edges. The only possible overlap of path edges would join the two common neighbors, producing a forbidden K_4 when $h = 2$. Outerplanarity forbids any further edge. The private paths of either hub may be shortened by deleting their terminal ears independently; this preserves the other degree and the common-neighbor set. Thus all larger degrees contain the smaller certified core as an induced subgraph. If a proper induced triangulation H lies in G, some boundary edge of H spans a nonempty outer interval of G; its triangle on that side supplies an outside vertex. It has precisely two neighbors in H, since a third would exceed the outerplanar edge bound. Therefore G contains an induced one-ear extension of H.

For nonadjacent hubs u, v, their common neighbors lie among the two roots of the u-gap containing v, so there are at most two. If there are two, x, y, they are adjacent: the four-cycle $uxvyu$ requires a diagonal in a chordal polygon triangulation and uv is absent. Counting the two stars and neighbor paths now forces an induced triangulation on $d(u)+d(v)$ vertices, with the common edge xy. The two private path splits describe every such core.

The split-fan inequalities required for the remaining cases are

$$\mathcal{F}(K_1 \vee (P_a \sqcup P_b)) > 2q + 6 \quad (q = a + b \geq 28), \qquad \mathcal{F}(K_1 \vee P_q) > 2q + 8 \quad (q \geq 32). \tag{21}$$

For the first and $q \geq 30$, hub/uniform compression has positive root of $z^2 - cz - q$, where $c \geq 2 - 4/q$. At $q = 30$, $c^2(q+12) \geq 32928/225 > 144$; both factors increase. Thus its square exceeds $q+12$, and path interlacing gives the first claim. At $q = 29$, use weights 107 at the hub and $20 + 2d_L(x)$ on a leaf. The four component types $(0, 29), (1, 28), (2, 27), (\geq 3, \geq 3)$ give respectively $(N, D) = (180152, 27969), (178144, 27793), (177960, 27785), (177952, 27785)$; all satisfy $N^2 > 41D^2$. At $q = 28$ all fifteen unordered splits have scale-1024 Gram certificates with score greater than 62; the smallest margin is $123655042413/1099511627776$. For the full fan, the analogous compression at $q = 33$ gives $c^2(q + 13) - 169 = 4375/1089 > 0$, with $c = 2(q - 1)/q$, and both factors increase. At $q = 32$ weights 99 at the hub, 18 at the endpoints, and 21 on the other thirty leaves give $N = 158958, D = 23679$ and $N^2 - 45D^2 = 36368919 > 0$. This proves (21).

In an adjacent pair, deleting the other hub and assigning the common neighbors leaves disjoint split fans with total leaf count $d(u) + d(v) - 2 - h$. In a nonadjacent pair, the full fan on $N[v]$ and the split fan on $\{u\} \cup (N(u) \setminus N(v))$ are disjoint, with counts $d(v)$ and $d(u) - h$. Consequently the sufficient tests are

$$\begin{aligned} &d(u) + d(v) \geq n - 4 + h && \text{(adjacent, both counts at least 28),} \\ &d(u) + d(v) \geq n - 6 + h && \text{(nonadjacent, both counts at least 28),} \\ &d(u) + d(v) \geq n - 7 + h && \text{(nonadjacent, } d(v) \geq 32,\ d(u) - h \geq 28\text{).} \end{aligned} \tag{22}$$

The adjacent test requires an allocation $a \in \{0, \ldots, h\}$ with leaf counts $q_1 = d(u) - 1 - h + a$ and $q_2 = d(v) - 1 - a$ both at least twenty-eight. The nonadjacent tests allow exchanging the two hubs; the third uses the stronger full-fan reserve. A high nonneighbor in a gap of mass m has degree at most $m+1$. If equality holds, the whole gap is a fan; it and the other hub's remaining $d - 2$ neighbors are disjoint split fans. The reserve-six version pays whenever $d - 2, m + 1 \geq 28$ and $k - m \leq 4$. This is another valid sufficient test; the actual finite complement reader needs only the pair tests and the exceptional-core certificates below.

The exceptional integer-Gram domains, reconstructed as literal graphs, are as follows. An internal core of degrees (d, e) has $(d - 2)(e - 2)$ ordered splits $0 \leq a \leq d - 3$, $1 \leq b \leq e - 2$;

a boundary core has one canonical form. For degrees $(32,32),(33,32)$ the $900,930$ internal forms have scores greater than $138,140$, and the $(32,32)$ boundary core has score greater than 140. At order seventy-one the $961+960$ internal forms of degrees $(33,33),(34,32)$ and the $(33,32)$ boundary core have scores greater than 142. The nonadjacent $(33,32)$ core with two common neighbors has $32\cdot 31 = 992$ forms, all with score greater than 142. Finally each of the 930 internal $(33,32)$ cores has fifty-nine boundary edges avoiding the hubs. Append an ear at each, giving 54870 cases. From its stored Gram rows B_x, B_y choose an appended row with entries $\lfloor (j(B_{xr}+B_{yr})+8)/16 \rfloor$, for some $1 \le j \le 12$. Testing all twelve choices supplies a positive score above 142 in every case. Ears touching a hub use one of the other two order-seventy-one core families. All factors use scale 1024; all graph and score checks are exact. No assertion of energy monotonicity under adding an arbitrary ear is made.

Here is the complete exceptional-core eligibility map after trying the pair tests. Orient degrees with $d \ge e$.

Actual pair conditions	Induced certified object
Adjacent, $h=1$, $e \ge 32$, $n \le 70$	Boundary $(32,32)$
Adjacent, $h=2$, $e \ge 32$, $n \le 69$	Internal $(32,32)$
Adjacent, $h=2$, $d \ge 33, e \ge 32$, $n \le 70$	Internal $(33,32)$
$n=71$, adjacent, $h=1$, $d \ge 33, e \ge 32$	Boundary $(33,32)$
$n=71$, adjacent, $h=2$, $d \ge 34, e \ge 32$	Internal $(34,32)$
$n=71$, adjacent, $h=2$, $d,e \ge 33$	Internal $(33,33)$
$n=71$, adjacent, $h=2$, $(d,e)=(33,32)$	Proper internal core plus one ear
$n=71$, nonadjacent, $(d,e,h)=(33,32,2)$	Two-common-neighbor core

The strict score for each listed object exceeds $2n$ in its indicated range. Core pruning preserves inducedness, so zero extension of its Gram witness pays the whole graph.

The high-pair verifier regenerates all 1150 actual tuples $(n,d,e,h,\text{adjacency})$ required by the 125 capped classes, and requires one of (22) or a listed exceptional core in every case. It verifies all 4760 base Gram cases and all 54870 ears. Thus a graph escaping its nonhub cap is paid. The five classes with $C=d$ have no such complement. The zero-mask classes themselves use the exact maxima and the actual rational hub payments in Appendix B. This completes the case partition in (19) and proves theorem 4.1, with no appeal to a larger-order $2n$ assertion.

4.7. **Small dominating sets.**

Theorem 4.3. *Every maximal outerplanar graph with a dominating set of at most four vertices and order at least eight satisfies $\mathcal{F}(G) \ge 2|G|$.*

The proof, including the finite coupled-compression certificate and its unbounded tail, is given in Appendix C. Its bounded-order input is theorem 4.1, already proved. This theorem will be applied to the whole graph, including cases whose retained remainder is K_2.

5. Local graph increments and unbounded parameters

5.1. **The precise local inputs.** The rooted graph notation $E_{p,q}$, $U(E_{p,q})$ and the two $B(E_{p,q},\mathsf{P}_\ell)$ roles is defined explicitly in Section 6.1. All lengths in the following proposition are positive integers, independently variable. An increment is measured against the same old graph H, with its ordered boundary edge ab retained once. The private vertices have no other outside contacts. The old graph is any maximal outerplanar graph of order at least three; there is no upper bound on its order, degrees or adjacency norm. Coherent reflection also exchanges the ordered roots in H.

Proposition 5.1. *The following increments hold, where N is the private order.*

(i) The first-turn private fan with contacts au, av_1, bu has increment greater than $607/100$ at $N=3$, greater than $83/10$ at $N=4$, and greater than $2N$ at $N=5,6,7$.

(ii) *The literal four-private-vertex Type B defined in Section 6.2 has increment greater than* $807/100$. *The both-private bowtie with two arms of length two has increment greater than* $10 + 1/25$.

(iii) *A mixed branch with external length* $p \geq 2$ *and private-apex length* $q = 2$ *has increment greater than* $2N+1/20$. *For* $q \in \{15, 16, 17, 18\}$ *its increment is greater than* $2N+3/4$.

(iv) *A boundary fan of* $r \geq 19$ *new vertices whose apex has actual old degree at most three has increment greater than* $2r + 1/20$.

(v) *For* $U(E_{p,q})$, $N = p + q + 2$, *the increment is greater than* $2N + 1/20$ *for* $q \in \{1, 2, 15, 16, 17, 18\}$ *and all* $p \geq 1$.

(vi) *For* $B(E_{p,q}, \mathsf{P}_\ell)$, $N = p+q+\ell+2$, *both the external-b and the private-c sibling roles have increment greater than* $2N + 1/50$ *for every* $p, q, \ell \geq 1$.

(vii) *For* $B(E_{p,q}, E_{r,s})$, $N = p+q+r+s+3$, *the four small middle pairs* $(q, r) \in \{1, 2\}^2$ *have increment greater than* $2N + 3/20$, *for all* $p, s \geq 1$. *All actual pairs* $14 \leq q + r \leq 41$ *have increment greater than* $2N + 1/20$, *and all* $q + r \geq 42$ *have increment greater than* $2N + 9/100$.

For other parameters the structural proof uses an ordered two-vertex tip or the degree-three fan deletion, rather than asserting additional universal whole-cap increments. In particular no universal positive increment for a bare $E_{p,q}$ is needed or claimed.

5.2. **A one-sided mode estimate and its finite applications.** We first derive an estimate that uses complete spectral projectors. Diagonalize a real private matrix B. Install every nonpositive mode first, then positive modes in increasing order. For a current root matrix Z and an upper bound $\sigma \geq 2$ on the installed eigenvalues, lemma 3.1 gives

$$\sigma \operatorname{Re} Z + t \operatorname{Im} Z \succeq t(\sigma I + J).$$

Indeed an installed mode of value b_j and contact vector l_j adds $t(\sigma - b_j) l_j l_j^{\mathsf{T}} / (t^2 + b_j^2)$, and the old response contributes its sector inequality plus $(\sigma - 2) \operatorname{Re} F_0 \succeq 0$. Inverse congruence and weighted Cauchy–Schwarz, with $H = Z^{-1}$, imply, for $f = l^{\mathsf{T}} H l = u + iv$ and $\zeta = l^{\mathsf{T}}(\sigma I + J)^{-1} l$,

$$\sigma u - tv \geq t|f|^2/\zeta. \tag{23}$$

The assertion for a zero contact vector is immediate and requires no division by ζ.

Append a positive mode b and compare with the old matrix direct sum $[b]$, so traces agree. Its determinant ratio is $(t+u+i(b+v))/(t+ib)$. If its phase difference Δ is positive, put $p = tv - bu > 0$, $X = tu + bv > 0$. Substitution in (23) gives exactly

$$X^2 + p^2 \leq \zeta(\sigma - b)X - \zeta(t + \sigma b/t)p.$$

If $b \geq \sigma$ this is impossible. If $0 < b < \sigma$, discarding p^2, completing the square in X, and using the positive real part $t^2 + b^2 + X$ of the determinant ratio give

$$\Delta_+ \leq \frac{\zeta(\sigma - b)^2 t}{4(t^2 + \sigma b)(t^2 + b^2)}.$$

The integral of $t^2/[(t^2+\alpha^2)(t^2+\beta^2)]$ over the positive axis is $\pi/[2(\alpha+\beta)]$, by partial fractions, with the repeated case obtained by continuity. The phase-energy identity therefore gives a gain at least $b^2 + \|l\|^2 - \zeta(\sigma - b)^2/[4(\sqrt{\sigma b} + b)]$. Pinching also gives gain at least b^2. Sum within complete eigenspaces and take the better of these two group bounds. For $b < 2$ use $\sigma = 2$; for larger positive modes use $\sigma = b$. Writing P_b for the full projector, this proves

$$\Delta\mathcal{F} \geq \mathcal{F}(B) + \sum_{b>0} \max\{0, g_b - c(b)\zeta_b\}, \tag{24}$$

where

$$g_b = \operatorname{tr}(T P_b T^{\mathsf{T}}), \quad \zeta_b = \operatorname{tr}(K^{-1} T P_b T^{\mathsf{T}}), \quad c(b) = \begin{cases} (2-b)^2/[4(\sqrt{2b} + b)], & b < 2, \\ 0, & b \geq 2. \end{cases}$$

Multiplicity is already included in P_b and in $\mathcal{F}(B)$; it is not applied to g_b a second time. There is no lower private spectral bound, and nonpositive modes are retained in the final matrix.

For a first-turn private fan of orders five, six and seven, its simple Perron root exceeds two. Complete rational Sturm isolation of all eigenvalues, including gcd multiplicity layers, and the Perron residue in (24) give increments greater than $10.138571, 12.468411, 14.876347$, respectively. These decimals denote terminating rational lower bounds, not numerical eigenvalues. The characteristic is reconstructed from integer matrix-power traces by Newton identities. If it is $p(x)$ and the Perron root is b, the contact weight is $N(b)/p'(b)$, where $N(x) = \operatorname{tr}(T \operatorname{adj}(xI-B)T^{\mathsf{T}})$. The expansion of $(xI-B)^{-1}$ in contact moments reconstructs N exactly. The supplied isolating intervals are checked by Sturm counts for every real root, and every positive interval contributes its squared lower endpoint to $\mathcal{F}(B)$.

For the both-private bowtie use vertices c, x_1, x_2, y_1, y_2, hub c adjacent to all four other vertices, edges x_1x_2, y_1y_2, and contact columns $e_c + e_{x_1}, e_c + e_{y_2}$. The complete eigenvalues are $(1 \pm \sqrt{17})/2, 1, -1, -1$. At the Perron root, $g = (85 + 13\sqrt{17})/68$ and $c(b) = 0$; at eigenvalue one, $g = \zeta = 1/2$. Substitution in (24) gives

$$59/8 + (47/68)\sqrt{17} - \sqrt{2}/8 > 10 + 1/25.$$

The strict rational checks use $4123^2 < 17 \cdot 1000^2$ and $1415^2 > 2 \cdot 1000^2$. All five private eigenvalues have been counted.

5.3. **Short turns: variable metrics and the zero-frequency endpoint.** The two short-turn computations use a slightly sharper support estimate than the constant-metric bound. For a complete physical response $F = R + iS$, let $P = 2R/t + S \succeq 0$ and $K_u = K + uP$. At the same old graph, $Z_u = tI + iJ + F_0 + uF$ obeys

$$2 \operatorname{Re} Z_u + t \operatorname{Im} Z_u \succeq tK_u.$$

The root-ball square completion with metric K_u shows that $\operatorname{Im}\operatorname{tr}(Z_u^{-1}F)$ is at most

$$E_{K_u} = \operatorname{Im}\operatorname{tr}(K_u^{-1}F)/t - \operatorname{Re}\operatorname{tr}(K_u^{-1}F)/2 + \frac{\sqrt{t^2+4}}{2t}\left\|K_u^{-1/2}FK_u^{-1/2}\right\|_*. \tag{25}$$

The ball for a larger metric is a subset of that for a smaller metric. Thus its exact support is nonincreasing in u, and a left sum with M equal subintervals is an upper bound for the entire homotopy integral. It is a sum of supports, not sampled actual phase derivatives.

Here is a completely algebraic way to check the scalar integrals. Put $x = t^2$, $Q = T(xI + B^2)^{-1}T^{\mathsf{T}}$, $M_0 = TB(xI+B^2)^{-1}T^{\mathsf{T}}$, so $F = tQ - iM_0$. For the metric $K_u = K + u(2Q - M_0)$ define $q = \operatorname{tr}(K_u^{-1}Q)$, $m = \operatorname{tr}(K_u^{-1}M_0)$, $a = \det Q/\det K_u$, $b = \det M_0/\det K_u$, and $w = (Q_{11}M_{0,22} + Q_{22}M_{0,11} - 2Q_{12}M_{0,12})/\det K_u$. The exact integrand tE_{K_u} is

$$-m - xq/2 + \tfrac{1}{2}\sqrt{(x+4)[x(q^2-2a) + (m^2-2b) + 2\sqrt{(xa-b)^2 + xw^2}]}. \tag{26}$$

Adjugates or contact moments reconstruct all rational functions. The trace norm identity behind this formula is $\|Y\|_*^2 = \operatorname{tr}(Y^*Y) + 2|\det Y|$ in dimension two; no commutation is presumed. On the tail use $z = 1/t$ and the hatted matrices $T(I + z^2B^2)^{-1}T^{\mathsf{T}}$ and $TB(I + z^2B^2)^{-1}T^{\mathsf{T}}$, with the metric $K + uz^2(2\widehat{Q} - \widehat{M_0})$. The signed rationalization below, with this metric, includes $z = 0$ and the full infinite tail.

At private order three the matrix is K_3 with contacts $T = \left[\begin{smallmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right]$, $\mathcal{F}(B) = 4$, and contact mass three. Use $u = 0$ only. The complete directed interval certificate has 2636 cells covering $[0, 4]$ and $z \in [0, 1/4]$; it proves integral less than $29/20$. With $\pi > 157/50$ the increment exceeds $7 - 145/157 = 607/100 + 101/15700$.

For Type B, private vertices are $1, 2, 4, 5$ and the literal edges are those in Section 6.2. Its private matrix is a paw and has $\mathcal{F}(B) > 2403/500$, verified by complete Sturm isolation; the contact mass is four. The two supports $u = 0, 1/2$ have mean integral less than $23/20$.

Their complete scale-72 dyadic cell certificate on $[0,4]$ and $z \in [0,1/4]$ gives increment greater than $4403/500 - 115/157 = 807/100 + 69/19625$.

At private order four the first-turn graph is a diamond, with characteristic $x(x+1)(x^2 - x - 4)$, energy $(9+\sqrt{17})/2 > 164/25$, and contact mass three. Its coupled zero mode must be kept. For the canonical Type D with external roots $0, 3$, start with $01, 02, 03, 12, 23$ and add $15, 24, 25, 45$. The first-turn fan is the same graph after its root-preserving polygon relabeling. The constant-metric support is not integrable at zero, so take $\varepsilon = 1/100$ and use the actual phase bound $\delta < 4\pi$ on $(0, \varepsilon)$. This follows from interlacing for four added vertices, giving a whole-phase change of magnitude less than 2π, and the four private eigenvalues, whose phase has magnitude less than 2π. The resulting energy cost is $4\varepsilon^2 = 1/2500$. The two nested supports at $u = 0, 1/2$ have 10139 directed cells on $[\varepsilon, 4]$ and $z \in [0, 1/4]$. Their mean area plus $2\pi_{\text{upper}}\varepsilon^2$ is less than $39/20$. Consequently the gain exceeds $239/25 - 195/157 = 83/10 + 141/7850$. Every compact and tail divisor and radicand is checked on its entire cell. A positive-frequency cutoff without the stated low cost would not prove this assertion.

5.4. **Path responses and complete infinite length coverage.** For the adjacency of a path on j vertices, define

$$D_0 = 1, \quad D_1 = t, \quad D_j = tD_{j-1} + D_{j-2}, \qquad f_j = D_{j-1}/D_j, \quad e_j = (-i)^{j-1}/D_j,$$
$$g_j = \frac{1 + i(f_j + e_j)}{t + 2i}, \qquad s_j = \frac{j}{t+2i} + \frac{2ig_j}{t+2i}. \tag{27}$$

They are respectively the endpoint diagonal, opposite-endpoint, all-ones-to-endpoint, and all-ones quadratic entries of $(tI + iA(P_j))^{-1}$. The first two follow by the tridiagonal determinant and cofactors. For the latter two apply the inverse to $(tI + iA(P_j))\mathbf{1} = (t+2i)\mathbf{1} - i(e_1 + e_j)$. These formulas also hold at $j = 1$, where the two endpoint vectors refer to the same vertex and their sum counts it twice. Induction on the continuants proves

$$f_j^2 + tf_j - 1 = (-1)^{j+1}/D_j^2 = e_j^2. \tag{28}$$

For an anchor a and all $j = a + 4k$, put $u = D_a/D_j$. Since $D_{j+2} = D_j + tD_{j+1} > D_j$ for $t > 0$, $0 < u \le 1$. Set $d_0 = u/D_a$, $\epsilon_a = (-1)^{a+1}$. The endpoint pair is contained in the full curve

$$f = \frac{2(1 + \epsilon_a d_0^2)}{t + \sqrt{t^2 + 4 + 4\epsilon_a d_0^2}}, \qquad e = (-i)^{a-1} d_0, \qquad 0 \le u \le 1. \tag{29}$$

This is the positive root of (28), rationalized to avoid subtraction; it includes the infinite-length endpoint $u = 0$. Actual u has its own t dependence. When differentiating an enlarged box, t, u are independent coordinates; taking the supremum over the whole box still covers the actual curve and requires no graph realization of every artificial coordinate.

A split of a residue progression always consists of its isolated anchor plus its complete tail starting at $a + 4$. Iterating this rule does not turn the largest anchor into a length cutoff. Fixed actual middle lengths use the actual continuants, not a free endpoint amplitude. Coherent reflection may interchange both old roots and all corresponding arms, but equal arm totals alone never identify responses or contact roles.

5.5. **A signed formula for the entire high-frequency end.** The following calculation supplies the tail used in every large grouped certificate. Let $R = (tI + iB)^{-1}$, $w = t + 2i$, and $V = (B - 2I)T^{\mathsf{T}}$. Since $RT^{\mathsf{T}} = (T^{\mathsf{T}} - iRV)/w$, applying the identity twice gives

$$TRT^{\mathsf{T}} = TT^{\mathsf{T}}/w - iTV/w^2 - V^{\mathsf{T}}RV/w^2. \tag{30}$$

All transposes here are ordinary transposes of complex symmetric matrices. Suppose $TV = -2I$, and subtract only the previously proved nonnegative lossless rays, replacing the leading TT^{T} by a fixed matrix Q. For $z = 1/t$, $x = z^2$, $a = 1 + 4x$, put $C = V^{\mathsf{T}}(I + xB^2)^{-1}V$ and

$D = V^{\mathsf{T}}B(I + xB^2)^{-1}V$. The normalized response is exactly $F_n = zA - ixM$, with real symmetric matrices

$$A = Q/a + 8xI/a^2 - x[(1-4x)C - 4xD]/a^2,$$
$$M = 2Q/a - 2(1-4x)I/a^2 - x[(1-4x)D + 4C]/a^2. \tag{31}$$

For the unary case $TV = -K$ the two constant terms $8xI$ and $2(1-4x)I$ are replaced by $4xK$ and $(1-4x)K$, respectively. This follows directly from (30); both signs of D matter.

Use $K^{-1/2}$ congruence on A, M, and let q, m, u, v, w_0 be their two traces, two determinants, and mixed determinant, respectively. On a tail cell the checker first establishes $A \succ 0$ by its first principal minor and determinant. Thus $q, u > 0$. The generalized eigenvalues of M relative to A are real, so $\Delta = w_0^2 - 4uv \geq 0$, even when M is indefinite. Put

$$Z = \sqrt{(u - xv)^2 + xw_0^2}, \qquad h_0 = \frac{\Delta}{Z + u + xv}, \qquad S = \sqrt{(1+4x)[q^2 + x(m^2 + 2h_0)]},$$
$$J(z) = \frac{(2q - m)^2 + 2(1+4x)h_0}{2(S + q + 2xm)}. \tag{32}$$

These are full two-direction invariants, not individual entry bounds substituted for a singular value. Indeed $Z - u - xv = xh_0$ and $\|A - izM\|_*^2 = q^2 + x(m^2 + 2h_0)$. Rationalizing the exact linear support gives $tE(F_n, t) = xJ(z)$. Therefore

$$\int_T^\infty tb(F_{\text{physical}}, t)\, dt \leq \int_0^{1/T} J(z)\, dz. \tag{33}$$

This uses lossless monotonicity followed by lemma 3.4. It makes no integrability claim based on the sector alone.

The denominator floors used in the implementation are mathematical inequalities: $Z + u + xv \geq 2u$, because $Z \geq |u - xv|$, and

$$S^2 - (q - 2xm)^2 = x[(2q + m)^2 + 2(1+4x)h_0] \geq 0, \qquad 2(S + q + 2xm) \geq 4q.$$

The first identity also proves a finite continuous value at $z = 0$. Each floor is intersected with an outward denominator interval, retaining its upper endpoint; the intersection must be nonempty and the lower bound strictly positive. No arbitrary positive replacement for a failed interval divisor is allowed.

For a whole cell $a_0 \leq z \leq b_0$ and a rational $L \geq \|B\|$, spectral calculus gives

$$\frac{V^{\mathsf{T}}V}{1 + b_0^2 L^2} \preceq C \preceq V^{\mathsf{T}}V, \qquad |D_{ij}| \leq H(a_0, L)\, \|V_i\|\, \|V_j\|,$$

where $H(a_0, L) = L/(1 + a_0^2 L^2)$ if $a_0 L \leq 1$, and $H(a_0, L) = 1/(2a_0)$ otherwise. This is the maximum of $|\lambda|/(1 + a_0^2\lambda^2)$ on $[-L, L]$. If the columns are orthogonal, centering the resolvent at $(1 + (1 + b_0^2 L^2)^{-1})I/2$ improves the off-diagonal bound to

$$|C_{12}| \leq \tfrac{1}{2}(1 - (1 + b_0^2 L^2)^{-1})\, \|V_1\|\, \|V_2\|. \tag{34}$$

These statements hold for every finite arm length on the entire closed cell; no infinite graph operator is introduced.

5.6. **An exact compact-enclosure contract.** For use with the preceding formulas we specify what each finite certificate proves. All rational inputs are enclosed on a fixed integer grid. Addition, multiplication and positive division use integer floor and ceiling in the required direction, with unbounded integers. Square roots use integer square-root brackets. The 72-, 96- and 192-bit grids used by individual supplied checkers are declared separately; a proof at one grid is not inferred from an execution at another. Arctangents use half-angle reduction and alternating series with a retained remainder. A Machin identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ gives rational bounds, including $157/50 < \pi < 22/7$.

Inverse cosine uses the corresponding checked quadrant and radical formulas, not a floating library value.

The function being enclosed is the complete $tb(F_n, t)$ of (2), or the linear support when stated. For a 2×2 response it can be evaluated without symbolic matrix square roots: if $L = K^{-1/2} F_n K^{-1/2}$, $A_0 = I + cL$, $G = I + 2\operatorname{Re} L/t + \operatorname{Im} L$, and $\rho^2 = t^{-2} + 1/4$, then

$$b = \operatorname{Im} \operatorname{tr} \operatorname{Log} A_0 + \arccos \frac{\sqrt{\det G} - \rho^2 |\det L|}{|\det A_0|}. \tag{35}$$

To verify it, let $\theta = \arcsin \sigma_1 + \arcsin \sigma_2$. Then $\cos\theta = \sqrt{\det(I - W^*W)} - \sqrt{\det(W^*W)}$, and use $A_0^* A_0 - \rho^2 L^* L = G$. The two accretive-factor phases fix the lift. For the scalar $\det A_0$, a quadrant must be established on the whole box; using only a principal argument of the determinant product would be invalid.

A natural interval evaluation is intersected, when all analytic prerequisites hold on the box, with the centered mean-value enclosure

$$f(x) \le f(x_0) + \sum_i \sup_{\text{box}} |\partial_i f| \, r_i.$$

The program propagates interval values, center values and all partial derivatives by the exact sum, product, quotient and chain rules. Every divisor, real radical, inverse-trigonometric derivative and quadrant used in this analytic extension is checked on the whole box. A sector clipping used only for a raw bound is never differentiated. If an artificial endpoint box contains nonphysical states, a negative upper principal minor or determinant of the sector may establish that its valid subset is empty. Otherwise the whole actual subset remains covered, using a raw safe bound or $2\pi t$ when the sharper analytic branch is unresolved. Such boxes must be retained or subdivided; inconvenient states may not be dropped.

The recorded tree splits cover both children of every frequency or parameter subdivision. Infinite residue splits cover the singleton and its complete tail as described above; finite progressions are intersected with their exact integer domain. Full tree consumption, all root residues, no extra suffix, positive divisors and complete frequency endpoints are mandatory. For each frequency leaf take the *maximum* bound over all its parameter leaves and multiply by its frequency width, then sum over frequencies. Parameter averaging would not bound every graph and is never an acceptance rule. Floating arithmetic may propose these trees but is not imported by the exact numerical reader. Complete data and the reader, including transitive arithmetic helpers, form the finite proof objects.

5.7. **Mixed branches and the degree-three fan.** For a mixed branch use private order $c, L_1, \dots, L_p, R_1, \dots, R_q$, with a fan at c on the R-path and a pendant L-path at c. The contacts are $a : \{c, L_1, \dots, L_p\}$, $b : \{c, R_1\}$. The private edge count is $p + 2q - 1$, contact mass $p + 3$, and their sum is $2N$, $N = p + q + 1$. With (5.4), eliminate the two paths to get

$$\begin{aligned} d_0 &= t + f_p + s_q, \qquad F_{aa} = s_p + (1 - i g_p)^2 / d_0, \\ F_{ab} &= (1 - i g_p)(1 - i g_q)/d_0, \qquad F_{bb} = f_q + (1 - i g_q)^2 / d_0. \end{aligned} \tag{36}$$

Subtract the nonnegative left bulk $(p - a)/(t + 2i)$, using anchors $a = 2, 3, 4, 5$ for the four residues. Its linear support is exactly zero, so subadditivity increases the upper bound on removing it. This argument applies to the linear support even on artificial curves; it does not assign them graph energy.

For $q = 2$, exact elimination gives

$$\det(tI + iB_p) = D_p[t^3 + 3t + (t^2 + 1) f_p - 2i].$$

The real bracket is positive. The even continuant ratios increase to $\lambda = 2/(t + \sqrt{t^2 + 4})$, and the odd ones decrease to it; this follows by iterating the decreasing map $x \mapsto 1/(t + x)$ twice. Hence $f_p \le f_3$ for all $p \ge 2$. The magnitude of the negative determinant phase is at least its $p = 3$ value. The phase-energy identity implies $\mathcal{F}(B_p) - (p + 3) \ge \mathcal{F}(B_3) - 6 > 17/20$,

the last inequality having a complete scale-integer Gram witness. The four whole compact curves and their signed tails, checked with (5.7), have linear-support integral less than $5/4$ each. The compact interval is $[0,4]$ after the removable endpoint cancellation; the entire tail is $z \in [0,1/4]$. Positive denominators and the positive real response on every tail cell are checked. Therefore the surplus exceeds $17/20 - 125/157 = 169/3140 > 1/20$.

For each $15 \le q \le 18$, a full integer Gram gives $\mathcal{F}(\mathrm{Fan}_{q+1}) > 2q + 23/5$. Pinch B into this fan and its pendant P_p, of energy $p-1$, obtaining reserve greater than $23/5$ above its private edge count, uniformly in p. Use (5.7) and the same four residue curves, retaining the actual fixed right continuants. Each linear-support integral on $[1/100, 4]$ plus $z \in [0,1/4]$ is less than six. The remaining low energy cost is at most $1/5000$ by $b < 2\pi$. Thus the surplus is greater than $23/5 - 600/157 - 1/5000 > 3/4$. The supplied computation also checks the two unused lengths nineteen and twenty; they are not needed here.

The retirement lemma in Appendix D bounds the cost of deleting an actual degree-three vertex by $111/20$. Pinch a grafted apex fan into that fan and the old graph minus its apex. For $r = 19, 20$, full integer Gram scores for the fan exceed $2r + 28/5$; for each $21 \le r \le 27$ they exceed $2r+6$. For all $r \ge 28$ interlacing with its leaf path and hub/uniform compression give

$$\mathcal{F}(\mathrm{Fan}_{r+1}) > 2r + f(r), \qquad f(r) = 2\sqrt{r} - 2/\sqrt{r} - 4 - 2/r + 1/r^2. \tag{37}$$

To obtain this expression, lower the compressed root $h + \sqrt{r + h^2}$, $h = 1 - 1/r$, by $h + \sqrt{r}$, and subtract the path's top squared eigenvalue, strictly less than four. The resulting expansion is precisely (37). The derivative of f is positive for $r \ge 1$ (multiply it by r^3 to see $r^{5/2} + r^{3/2} + 2r - 2 > 0$). Using $\sqrt{28} > 21/4$ gives $f(28) > 6$. Thus every $r \ge 19$ has incremental surplus greater than $28/5 - 111/20 = 1/20$, proving part (iv).

5.8. **Unary parents.** For $U(E_{p,q})$ use private order $L_1, \dots, L_p, d, Q_1, \dots, Q_q, c$. The private graph is the fan at c on $d, Q_1, \dots, Q_q$, with a pendant P_p at d. The root-contact sets are $a : \{L_1, \dots, L_p, d, c\}$ and $b : \{c\}$. It has $p + 2q + 1$ private edges and contact mass $p + 3$. Eliminating both paths leaves

$$Z = \begin{pmatrix} t + f_p + f_q & i + g_q \\ i + g_q & t + s_q \end{pmatrix}, \quad U = \begin{pmatrix} 1 - ig_p & 1 \\ 0 & 1 \end{pmatrix}, \quad F_n = \mathrm{diag}((1 + 2ig_p)/w, 0) + UZ^{-1}U^{\mathsf{T}}. \tag{38}$$

Only the left ray $(p-1)/w$ has been subtracted. For $q = 1, 2$ exact elimination gives

$$\det(tI + iB) = D_p[(C_0 + f_p C_1) - i(M_0 + f_p M_1)],$$

where (C_0, C_1, M_0, M_1) is respectively

$$(t^3 + 3t, t^2 + 1, 2, 0), \qquad (t^4 + 5t^2, t^3 + 3t, 4t, 2).$$

All needed coefficients are nonnegative, and $M_0C_1 - M_1C_0 = 2 + 2t^2$ or $2t^2 + 2t^4$. Thus the negative phase magnitude is minimized by $f_p = 1/t$, the actual $p = 1$ case. Its complete integer Gram gives reserve above the private edge count greater than $4/5$ or $7/5$. For $15 \le q \le 18$, pinch into the actual Fan_{q+2} and P_p. Four integer factors give reserve greater than $23/5$ uniformly in p. A pointwise $p = 1$ comparison is not used in these four cases.

The defect has $TV = -K$ and

$$V^{\mathsf{T}}V = \begin{pmatrix} q + 5 & q + 3 \\ q + 3 & q + 5 \end{pmatrix}, \qquad Q = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}.$$

The bound $\|B\| \le 3$ suffices at $q = 1, 2$; at $q = 15, \dots, 18$ the graph is a star with $q + 1$ leaves plus a path forest, so $\|B\| < \sqrt{q+1} + 2 < 13/2$. These prove the full-cell bounds needed in the unary version of (31). The compact interval is $[1/8, 16]$, starting from ninety-two base cells, with all four infinite p residues. Its exact numerical reader has respectively $9074, 3069, 1060, 1057, 1050, 1044$ length leaves for $q = 1, 2, 15, 16, 17, 18$. All sixty-four tail

cells cover $z \in [0, 1/16]$. The low energy cost is $1/32$. For each actual q, with its own fixed reserve c_q, the acceptance comparison is

$$c_q - 1/32 - 2(I_q + T_q)/\pi_{\text{lower}} > 1/20. \tag{39}$$

All six exact rational inequalities pass. Their weakest surplus exceeds 0.08829; this decimal is descriptive, while (39) is checked rationally. A larger p anchor still starts an infinite progression.

5.9. **Sibling roles: literal responses and private reserves.** In both roles use private order $L_1, \dots, L_p, d, Q_1, \dots, Q_q, c, S_1, \dots, S_\ell$. The external-$b$ role has the core fan at c on $d, Q_1, \dots, Q_q$, a pendant P_p at d, and a pendant P_ℓ at c. Its contacts are $a : \{L_1, \dots, L_p, d, c\}$ and $b : \{c, S_1, \dots, S_\ell\}$. For the private-$c$ role join c to every S_j, and replace the second contact set by $\{c, S_\ell\}$. Both have $N = p + q + \ell + 2$, but different private graphs, edge counts and contact matrices. For $\ell = 1$ they are literally the same graph.

In either role let $U_a = (1 - ig_p, 1)$ and $U_b = (0, 1 - ig_\ell)$ on the heads d, c. For the external role

$$Z = \begin{pmatrix} t + f_p + f_q & i + g_q \\ i + g_q & t + f_\ell + s_q \end{pmatrix}, \quad F_n = \operatorname{diag}((1 + 2ig_p)/w, (1 + 2ig_\ell)/w) + UZ^{-1}U^{\mathsf{T}}. \tag{40}$$

The two outer rays $(p-1)/w, (\ell-1)/w$ are removed. For the private role the formula instead is

$$Z = \begin{pmatrix} t + f_p + f_q & i + g_q \\ i + g_q & t + s_q + s_\ell \end{pmatrix}, \quad F_n = \operatorname{diag}((1 + 2ig_p)/w, f_\ell) + UZ^{-1}U^{\mathsf{T}}, \tag{41}$$

removing only the left ray. These are successive Schur eliminations of disjoint paths, so all private modes and the old root roles are retained.

For the private-energy proof introduce the exact Gaussian polynomials for a path: $D_j = \det(tI + iP_j)$, $X_j = e_j^{\mathsf{T}} \operatorname{adj}(tI + iP_j)\mathbf{1}$, $E_j = \mathbf{1}^{\mathsf{T}} \operatorname{adj}(tI + iP_j)\mathbf{1}$, where here e_j denotes the last *unit coordinate*, not the opposite-endpoint response of (5.4). Starting with $D_0 = 1, E_0 = X_0 = 0, D_1 = t, E_1 = X_1 = 1$, the complete recurrences are

$$D_{j+1} = tD_j + D_{j-1}, \quad X_{j+1} = D_j - iX_j, \quad E_{j+1} = tE_j + E_{j-1} + D_j - 2iX_j. \tag{42}$$

Appending a vertex proves the first two by cofactors. For the third, the additional fraction in a direct Schur expansion is $(D_{j-1}E_j - X_j^2)/D_j = E_{j-1}$, by the Desnanot–Jacobi identity for the path matrix bordered by $\mathbf{1}, \mathbf{1}^{\mathsf{T}}, 0$. Thus these are identities over $\mathbb{Q}[i][t]$, not numerical polynomial fits.

For the external role put $m = q + 1$, $r = f_p$, $s = f_\ell$. After removing the positive factors D_pD_ℓ, its determinant is

$$(t + s)(D_m + rD_{m-1}) + E_m + rE_{m-1} = C - iM. \tag{43}$$

For each $q = 1, \dots, 15$, the exact coefficient tests establish $C, M > 0$ on $t > 0, r, s \geq 0$. Increase s to $1/t$, which decreases M/C; at that value, the coefficient polynomial for the negative r-derivative is nonnegative and nonzero, so increase r to $1/t$. The order of these two comparisons is part of the proof. The continuous determinant phase is $-\arctan(M/C)$, with no wrap, and comparison of its nonnegative magnitude under the full phase-energy integral proves that the private signed energy is at least the literal $p = \ell = 1$ value. At $q = 1$ that value is the bull, with characteristic $x(x^2-x-3)(x^2+x-1)$, giving reserve $c_0 = (\sqrt{13}-\sqrt{5})/2$. At $q = 2$ complete Sturm isolation gives reserve greater than $6/5$. For each $3 \leq q \leq 15$ it gives reserve greater than $17/10$. For every $q \geq 16$, pinching into the fan and the two pendant paths gives reserve greater than $f(q+1) - 1 \geq f(17) - 1 > 153919/59245 > 17/10$. Thus this last reserve is uniform over every $q \geq 3$.

For the private role put $S = q + \ell$. Removing D_p leaves

$$(tD_\ell + E_\ell)(D_{q+1} + rD_q) + D_\ell(E_{q+1} + rE_q) = C - iM, \qquad r = f_p. \tag{44}$$

For the fifty-five ordered pairs $2 \le S \le 11$ all four coefficient polynomials in $C = C_0 + rC_1$, $M = M_0 + rM_1$ are nonnegative, with $C, M > 0$ on $t > 0, r \ge 0$. Thirty-seven pairs have nonnegative nonzero $M_0C_1 - M_1C_0$ and use the $p = 1$ phase comparison; seventeen use pinching into their core and pendant path, with a full-core Sturm bound. One pair, $(q, \ell) = (3, 2)$, requires the two-endpoint envelope described next. These routes prove private reserve greater than $17/10$ at $S \ge 4$, greater than $6/5$ at $S = 3$, and at least c_0 at $S = 2$.

There is a genuine false helper which the proof must avoid. At $(q, \ell) = (3, 2)$, $M_0C_1 - M_1C_0 = -4t^2 + 12t^6 + 10t^8 + 2t^{10}$, negative at $t = 1/4$. The correct replacement uses $t/(t^2 + 1) \le f_p \le 1/t$. Since M/C is fractional-linear with positive denominator, its minimum is at one of these two endpoints. Clear positive denominators. On each of 528 contiguous rational cells of $[1/4, 64]$, evaluate a lower numerator at the left endpoint and an upper positive-coefficient denominator at the right endpoint, then the lower arctangent and lower t. This proves a lower integral

$$\frac{10835742347032107304933190833871 69}{32451855365842672678315602057625 6},$$

sufficient for signed private energy greater than $17/5$. Every omitted frequency has nonnegative integrand, by the all-t coefficient signs, so dropping it is legitimate for this *lower* bound. The stronger private reserve $9/5$, restricted to $\ell \ge 2, S \ge 4$, uses the forty-four pairs at $4 \le S \le 11$: twenty-six valid phase/base routes, sixteen core-pinch routes, and the two endpoint envelopes $(3, 2), (4, 2)$. Their full coefficient and Sturm checks are supplied. A blanket $p = 1$ comparison would be false.

For every $S \ge 12$ pinch into P_p and $c \vee (P_{q+1} \sqcup P_\ell)$. Hub/uniform compression, weakened to the Rayleigh value $\sqrt{r} + 1 - 2/r$, gives the latter energy greater than $2r + g_0(r)$ for $r = S + 1$, where

$$g_0(r) = 2\sqrt{r} - 4/\sqrt{r} - 5 - 4/r + 4/r^2. \tag{45}$$

It is increasing for $r \ge 2$, as direct differentiation shows. Thus the reserve above private edges is greater than $g_0(S + 1) + 1$. At $S = 12$, $\sqrt{13} > 18/5$ gives a value greater than $13726/7605 > 9/5$. At $S = 20$, $\sqrt{21} > 457/100$ gives $41147009/10076850 > 4$. These are whole-integral reserves for the graph's one fixed band, not frequency-dependent allocations.

5.10. **Sibling roles: complete phase certificates and analytic tails.** For the external role, $q = 1, 2$ have all sixteen ordered residue combinations of the two unbounded outer arms, using (40) and fixed actual middle continuants. The compact interval is $[1/4, 8]$, the low energy cost $1/8$, and all sixty-four high-tail cells cover $z \in [0, 1/8]$. For $q = 1$, the reserve is c_0, the compact and tail upper areas are

$$\frac{50115721779771398693611571084123}{81129638414606681695789005144064}, \qquad \frac{74119675461300232313278862288 59}{40564819207303340847894502572032}.$$

With checked radical and Machin bounds, $c_0 - 1/8 - 2(I_c + I_t)/\pi_{\text{lower}} > 1/50$. For $q = 2$, with reserve $6/5$, the areas are

$$\frac{11484411905237813814887040993295}{10141204801825835211973625643008}, \qquad \frac{44369025772678994513909280591 17}{20282409603651670423947251286016},$$

and the analogous surplus exceeds $1/5$. The private degree bounds are respectively three and four; the defect Gram diagonals are $(7, 4)$ and $(8, 5)$. These constants prove the whole-cell spectral tail boxes.

For all $3 \le q \le 45$ use reserve $17/10$, the compact interval $[1/2, 16]$, and the low cost $1/2$. The four middle progressions with anchors $3, 4, 5, 6$, truncated at their largest integer at most forty-five, exhaust the finite middle range. Together with the two outer residues they give sixty-four roots. Retain the actual middle bulk as an independent interval coordinate containing its integer values; this enlarges boxes without deleting any graph. The exact

numerical reading covers 547916 length leaves and 512 frequency leaves. Its compact area is

$$I_c = 337263337104400830158174605686 9/25353012004564588029934064107 52.$$

The defect Gram diagonals are $q+6, q+3$ and the cross entry is $q+2-\mathbf{1}_{\ell=1}$. Removing the c-star leaves a path forest, so $\|B\| < \sqrt{q+2}+2 < 9$. Consequently the complete sixty-four-cell tail on $z \in [0, 1/16]$ uses bounds $C_{11} \le 51, C_{22} \le 48, |C_{12}| < 50$ and $|D_{11}| \le 459, |D_{22}| \le 432, |D_{12}| < 450$. It gives

$$I_t = \frac{651145386437874276590155143303 3}{202824096036516704239472512860 16},$$
$$17/10 - 1/2 - 2(I_c + I_t)/\pi_{\text{lower}} > 1/10.$$

The independent graph degree and contact calculation supplies the unbounded outer-arm hypotheses of this tail.

For the private role, $\ell = 1$ coincides literally with the external role. At $\ell \ge 2, S = 3$ the only pair is $(1, 2)$. Its polynomial cross is $2t^6 + 8t^4 + 10t^2 + 4 > 0$; its $p = 1$ characteristic is $x^6 - 7x^4 - 4x^3 + 9x^2 + 6x - 1$, giving reserve greater than $7/5$ by complete Sturm isolation. The compact interval is $[1/4, 8]$, with low cost $1/8$. Its exact compact and tail areas are

$$\frac{659614880501270079307740647977 7}{507060240091291760598681282150 4}, \qquad \frac{511883158500175951750949443256 1}{202824096036516704239472512860 16},$$

which give surplus greater than $1/4$. The tail uses $TV = -2I$, Gram $\left[\begin{smallmatrix} 8 & 4 \\ 4 & 8 \end{smallmatrix}\right]$ and $\|B\| \le 4$.

For the private bulk $4 \le S \le 49$, $\ell \ge 2$, use one fixed band: reserve $9/5$ for $4 \le S \le 19$, and reserve four for $20 \le S \le 49$. The two bands contain respectively 152 and 975 ordered middle pairs. For each of the sixty-four residue roots $p = 1, \dots, 4$, $q = 1, \dots, 4$, $\ell = 2, \dots, 5$, intersect the finite q, ℓ progressions with the exact band. The total S coordinate may be enlarged to its arithmetic-progression sum interval. Empty leaves require an actually empty integer intersection; a nonphysical analytic midpoint is not a reason to remove a nonempty leaf. The entire compact interval is $[1/2, 16]$, with eighty base cells per band. There are 766777 numerical length leaves and 68224 exact arithmetic-empty leaves in total. The two compact areas are respectively

$$C_s = \frac{441715535923114996886789005822 1}{253530120045645880299340641072 52}, \qquad C_l = \frac{537084929325004063444195859312 5}{126765060022822940149670320537 6}.$$

For every $\ell \ge 2$, direct contact multiplication gives

$$TV = -2I, \quad V^{\mathsf{T}}V = \begin{pmatrix} S+5 & S+1 \\ S+1 & S+5 \end{pmatrix}, \quad Q = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

The star-plus-path-forest norm is less than $\sqrt{S+1}+2$. Use C-entry bound 24, D-entry bound 156 in the small band, and $54,540$ in the large band. Each of the two complete sixty-four cell tails on $z \in [0, 1/16]$ gives respectively

$$T_s = \frac{103649921062628973110353099105 61}{811296384146066816957890051440 64},$$
$$T_l = \frac{321975402136982343769077663723 39}{811296384146066816957890051440 64}.$$

The exact inequalities

$$9/5 - 1/2 - 2(C_s + T_s)/\pi_{\text{lower}} > 1/20, \qquad 4 - 1/2 - 2(C_l + T_l)/\pi_{\text{lower}} > 1/20 \tag{46}$$

hold separately, with approximate lower surpluses 0.10950894 and 0.55008327. Only the graph's actual band reserve is used throughout its whole integral.

For the remaining analytic tails use the bare-branch bound proved in Appendix E, with debt $184/157$, and exactly one degree-four retirement of cost less than $351/50$. In the external role delete only the middle q-path. The retained graph relative to the same old H is $E_{p+1,\ell}$, and its private c has degree four. Pinch the final graph into that retained graph minus c and the fan on c and the q deleted vertices. Equations (37) and the

debt bound give surplus greater than $f(q) - 351/50 - 184/157$. At $q = 40$ this exceeds $179500493/1985736000 > 9/100$, using $\sqrt{40} > 1581/250$; monotonicity covers every $q \geq 40$. In the private role retain only S_ℓ, deleting the q middle vertices and the first $\ell - 1$ siblings. The retained branch is $E_{p+1,1}$, with the same degree-four c. Its disjoint final core is $c \vee (P_q \sqcup P_{\ell-1})$, on $S-1$ leaves. The analogous surplus is $g_0(S-1) - 351/50 - 184/157$. At $S = 50$ it is greater than $2952143/18847850 > 3/20$ and increases thereafter. Every vertex is assigned to one block, with c counted once. These ranges together prove both sibling statements in proposition 5.1.

5.11. **Double parents: both outer arms remain infinite.** Use private order

$$L_1, \dots, L_p, d, Q_1, \dots, Q_q, c, R_1, \dots, R_r, e, S_1, \dots, S_s.$$

It is a path in this order, with c additionally joined to d, e and all middle vertices, keeping repeated edges once. The contact sets are the left prefix through d together with c, and c, e with the right suffix. With $m = q + r$, the private edge count is $p + s + 2m + 2$, the contact mass $p + s + 4$, and their sum $2N$. Elimination leaves three heads:

$$Z = \begin{pmatrix} t + f_p + f_q & i + g_q & 0 \\ i + g_q & t + s_q + s_r & i + g_r \\ 0 & i + g_r & t + f_s + f_r \end{pmatrix},$$

$$U_a = (1 - ig_p, 1, 0), \quad U_b = (0, 1, 1 - ig_s),$$

$$F_n = \operatorname{diag}((1 + 2ig_p)/w, (1 + 2ig_s)/w) + UZ^{-1}U^{\mathsf{T}}. \tag{47}$$

Only the positive outer rays $(p-1)/w, (s-1)/w$ are removed. To reproduce the smaller fixed-middle expression, let $A = t + f_p + f_q$, $E = t + f_s + f_r$, $b = i + g_q$, $f = i + g_r$, $\alpha = 1 - ig_p$, $\beta = 1 - ig_s$, and put

$$z_0 = t + 2i(g_q + g_r)/w - b^2/A - f^2/E, \quad \eta = \frac{w/m}{1 + (w/m)z_0}, \quad v_a = 1 - \alpha b/A, \quad v_b = 1 - \beta f/E.$$

Then its entries are $F_{aa} = (1 + 2ig_p)/w + \alpha^2/A + v_a^2\eta$, $F_{ab} = v_a v_b \eta$, and $F_{bb} = (1 + 2ig_s)/w + \beta^2/E + v_b^2\eta$. This is simply the next two scalar Schur eliminations in (5.11); when an enlarged box cannot certify a divisor, the direct formula or safe fallback is required.

For each of the four ordered $(q, r) \in \{1, 2\}^2$, eliminating the outer paths gives $D_p D_s[C(f_p, f_s) - iM(f_p, f_s)]$, bilinear in the endpoints. The exact coronal recurrence (42) constructs every coefficient. The coefficients of C, M and of the negative derivative cross-polynomials in each endpoint are nonnegative, with $C, M > 0$. Thus their negative phase magnitude is at least that of the actual $p = s = 1$ graph. Complete integer Gram factors at those four bases give respective private reserves $33/25, 187/100, 187/100, 239/100$ above private edges. The compact domain $[1/4, 16]$ uses all sixteen ordered outer residues, with respective numerical length-leaf counts $21729, 17296, 17123, 10037$. The low cost is $1/8$. The all-length defect identities are

$$TV = -2I, \quad V^{\mathsf{T}}V = \begin{pmatrix} m + 6 & m + 3 \\ m + 3 & m + 6 \end{pmatrix}, \qquad Q = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}. \tag{48}$$

For example its first defect column in unit-coordinate notation is

$$-L_1 + 2Q_1 + \sum_{j=2}^{q} Q_j - c + \sum_{j=1}^{r} R_j + e,$$

and the second is $d + \sum_{j=1}^{q} Q_j - c + \sum_{j=1}^{r-1} R_j + 2R_r - S_s$. Squaring and taking their inner product proves (48) for all positive lengths, including one. The star at c has $m + 2$ leaves; removing its edges leaves a path forest. Hence $\|B\| < \sqrt{m + 2} + 2$, uniformly in both outer lengths. At $m = 2$ use the sharper degree bound four; at $m = 3, 4$ use $9/2$. All sixty-four z-cells on $[0, 1/16]$ have their positive real response and every divisor certified. The four exact comparisons $c_{q,r} - 1/8 - 2(I_c + I_t)/\pi_{\text{lower}} > 3/20$ pass.

For the remaining middle $14 \le m \le 41$, pinch the private graph into its two pendant paths and the core $c \vee (P_{q+1} \sqcup P_{r+1})$. By lemma 2.3,

$$\mathcal{F}(B) > |E(B)| + g(m+2). \tag{49}$$

Each actual m retains this one reserve for the whole integral. Coherent reflection interchanges (p,q,r,s) with (s,r,q,p) and both old roots; choose $q \ge r$, giving precisely 378 actual middle pairs. Together with sixteen outer residues these give 6048 fixed-middle roots. No symmetry of H is assumed.

The compact interval $[1/2,5]$ has thirty-six contiguous base slabs. On the first sixteen, $[1/2,2]$, a four-free-endpoint enlargement covers $q \ge 7, r \ge 1, 1/m \in [0,1/14]$ and the two outer residues. Its inverse-m endpoint zero is an auxiliary response limit, to which neither a graph energy nor a finite private norm is assigned. Since reflection gives $q \ge 7$, every actual graph is covered. On the other twenty slabs use the 378 actual finite middle pairs. These are two upper enclosures of the same physical normalized response on disjoint frequency intervals. Both use the full compact contract above; the checked selection targets are $\frac{5}{2}/(1+t_{\mathrm{mid}}^2)$ and $\frac{14}{5}/(1+t_{\mathrm{mid}}^2)$, respectively. No target is accepted without its whole-leaf exact bound. All 284102 numerical length leaves give the exact compact upper

$$C = 291703550006869604049809733907 7/126765060022822940149670320 5376. \tag{50}$$

The low cost is $1/2$ by $b < 2\pi$.

The short tail $t \ge 5$ must preserve both root directions. Let $U = 2^{-1/2}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ and change the defect columns to $R = VU\,\mathrm{diag}(1/\sqrt{3},1)$. This is exactly $K^{-1/2}$ normalization followed by an orthogonal change of coordinates. They are orthogonal and have Gram $\mathrm{diag}((2m+9)/3,3)$. In (31) the leading Q and identity become $\mathrm{diag}(4/3,2)$ and $\mathrm{diag}(1/3,1)$. Use the entire-cell spectral bounds and (34), with a rational outward $L \ge \sqrt{m+2}+2$. For each integer $m = 14,\dots,41$, all 128 equal cells $z \in [0,1/5]$ are computed with the complete signed density (5.5). All 3584 cells pass the positive real-response and divisor conditions. Let their rational area sum be T_m. With the actual lower enclosure $g_{\mathrm{lower}}(m+2)$ the twenty-eight acceptance inequalities are

$$g_{\mathrm{lower}}(m+2) - 1/2 - 2(C+T_m)/\pi_{\mathrm{lower}} > 1/20. \tag{51}$$

Every comparison is strict; the smallest rational surplus exceeds 0.3998686104, at $m = 14$. The entire-cell tail includes $z = 0$. A success on a finite selection of high frequencies would not establish (51).

Finally let $m \ge 42$. Delete only the two middle paths, retaining d, e. Relative to the same old H the retained graph is the bare $E_{p+1,s+1}$, and its private c has actual degree four, with neighbors a, b, d, e. Apply lemma E.3, retire c once with cost less than $351/50$, and pinch the final graph into that graph minus c and $c \vee (P_q \sqcup P_r)$. The surplus is greater than $g(m) - 351/50 - 184/157$. At $m = 42$, $g(42) = (40\sqrt{18922} - 1846)/441$ and $(2751/20)^2 < 18922$, so the surplus exceeds $340213/3461850 > 9/100$. Strict monotonicity of g proves all larger m. This completes the proof of proposition 5.1.

6. Structural reduction and the global conclusion

Write $\mathcal{F}(A) = \sum_{\lambda_i(A)>0} \lambda_i(A)^2$ for a real symmetric matrix A, and write $\mathcal{F}(G)$ when A is the unit adjacency matrix of G. The symbol P_k denotes the ordinary path on k vertices; a rooted fan cap is written P_ℓ below. Here a maximal outerplanar graph of order at least three is supplied with its outerplane embedding. Its outer boundary is a Hamiltonian cycle and its bounded faces are triangles. The auxiliary graph K_2 will occur as a remainder, but is not included by extending any incremental energy lemma.

6.1. **Definitions, orientations, and vertex accounting.** A cap rooted at an ordered edge (a, b) is a triangulated polygon whose outer boundary consists of the edge ba and an a–b arc. Its private vertices are all its vertices other than a, b. Grafting such a cap onto a boundary edge of an old graph keeps ab once; every other new edge has at least one private endpoint. Different grafts use disjoint private sets.

For $\ell \geq 1$ the straight fan cap P_ℓ has boundary arc $u, z_1, \ldots, z_\ell, v$. An apex choice is part of its description. With apex u its edges are the consecutive boundary edges and all uz_j; with apex v they are the consecutive boundary edges and all vz_j. Thus it has ℓ triangular faces and ℓ private vertices. When $\ell = 1$ both apex descriptions give the same triangle. When $\ell \geq 2$ its apex among its two roots is unique.

For rooted caps C, D, the ordered operation $B(C, D)$ has root edge (a, b), a fresh vertex c, root triangle abc, a copy of C on (a, c), and a copy of D on (c, b). The operation $U(C)$ is the same construction with only C on (a, c) and no cap on (c, b). Empty and one-triangle children are different operations; no identity $\mathsf{P}_0 = \mathsf{P}_1$ is used.

The cap $E_{p,q}$, $p, q \geq 1$, is $B(\mathsf{P}_p, \mathsf{P}_q)$ with the left fan apex a and right fan apex b. In particular *the cap $E_{p,q}$ itself*, rather than either one of its children, consists of the root triangle and two straight fans. Its private order is $p + q + 1$. Consequently

$$N(U(E_{p,q})) = p + q + 2, \qquad N(B(E_{p,q}, \mathsf{P}_\ell)) = p + q + \ell + 2,$$

and

$$N(B(E_{p,q}, E_{r,s})) = p + q + r + s + 3.$$

Each such cap adds exactly $2N$ unit edges. For $B(E_{p,q}, \mathsf{P}_\ell)$ the right fan apex is either c or b; these are distinct rooted contact models when $\ell \geq 2$ and must both be covered.

A coherent reflection reverses the entire cap, exchanges the ordered old roots, and exchanges every associated arm role. It need not extend to an automorphism of the old graph. Equal arm sums do not identify rooted graphs or their response matrices.

Lemma 6.1 (Chord separation)**.** *Deleting the private vertices on one closed side of a polygon chord leaves an induced maximal outerplanar graph, or leaves exactly the chord K_2. The deleted vertices have no contacts with the remainder except through the chord endpoints. The same statement holds for the cap consisting of all faces, with a boundary edge designated as its root.*

Proof. In the cyclic boundary order, the chord endpoints split the other vertices into two disjoint open arcs. An edge joining the two open arcs would cross the chord, so no such edge exists. Each closed side is triangulated by exactly the faces on that side. The surviving arc and the chord therefore bound a triangulated polygon; if the surviving open arc is empty, only the chord survives. In the all-face case both surviving vertices are precisely the designated boundary-edge endpoints. This also proves the contact assertion and inducedness. □

6.2. **The precise energy inputs used in this section.** The argument below uses the following statements, proved in Sections 4–5 and Appendices A and C. They are listed with their full domain restrictions to distinguish structural reasoning from numerical input.

(1) **Finite basis and whole domination class.** By Theorem 4.1, every unit maximal outerplanar graph of order $8 \leq n \leq 71$ satisfies $\mathcal{F}(G) \geq 2n$. Every such graph of order $n \geq 8$ with domination number at most four also satisfies $\mathcal{F}(G) \geq 2n$ by Theorem 4.3. The latter result is obtained from the independent selected-hub decomposition and its finite certificates, not from the extraction argument in this section.

(2) **Short first turns.** For every old unit maximal outerplanar graph H of order at least three and every ordered boundary edge (a, b), append private vertices $u, v_1, \ldots, v_{L-1}$ with all uv_j, the path edges $v_j v_{j+1}$, and precisely the root contacts au, av_1, bu. For $L = 3$ the increment exceeds 6.07; for $L = 4$ it exceeds 8.3; for each

$L \in \{5, 6, 7\}$ it exceeds $2L$. All coherent reflections are included and no upper old degree, order, norm or spectrum bound is imposed.

(3) **The ordered C_0 inputs.** For an arbitrary finite Hermitian matrix C and real R with $RR^t = \left[\begin{smallmatrix} r & 1 \\ 1 & 1 \end{smallmatrix}\right]$, $3 \le r \le 14$, append two private adjacent vertices at the root block $J = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ with root contact matrix $T = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right]$. The increment of $\mathcal{F}$ is greater than 4.09 by Lemma A.1. The two root rows are ordered.

(4) **The four-private-vertex Type B.** Starting with roots $0, 3$, take base edges

$$01, 02, 03, 12, 23,$$

then add vertices $4, 5$ and edges $24, 25, 35, 45$. At every ordered boundary of every old unit maximal outerplanar graph of order at least three, its increment is greater than 8.07, including coherent reflection.

(5) **The complete positive-mode filter.** At every ordered boundary of an old unit maximal outerplanar graph of order at least three, a real private matrix B and real root-contact matrix T obey

$$\Delta\mathcal{F} \ge \mathcal{F}(B) + \sum_{b>0} \max\{0, g_b - c(b)\zeta_b\},$$

where the sum is over distinct positive eigenvalues, P_b is the complete spectral projector, $g_b = \operatorname{Tr}(TP_bT^t)$, $\zeta_b = \operatorname{Tr}((2I + J)^{-1}TP_bT^t)$, and $c(b) = (2 - b)^2/[4(\sqrt{2b} + b)]$ for $b < 2$, and $c(b) = 0$ otherwise.

(6) **Mixed and degree-three fan inputs.** For a bare branch with an outward fan of $p \ge 2$ faces and an inward fan of $q = 2$ faces, the full branch increment is greater than twice its private order. The same holds for $p \ge 2$ and each $q \in \{15, 16, 17, 18\}$. These statements hold at every ordered boundary of every old unit maximal outerplanar graph of order at least three. Separately, appending a fan of $r \ge 19$ private vertices at an actual boundary edge of an old unit maximal outerplanar graph whose chosen apex has actual old degree at most three has increment greater than $2r + 1/20$. No other root degree, old order, norm or spectral upper bound occurs.

6.3. **Small old remainders and the minimality rule.**

Lemma 6.2 (Anchored domination of a small remainder)**.** *Let H be a unit maximal outerplanar graph of order $3 \le h \le 7$ with boundary edge ab, or let $H = K_2$ with edge ab. There is a dominating set of H contained in $\{a, b, v\}$ for some old vertex v. If a graft's private vertices are dominated by $\{a, b, c\}$ for one private vertex c, the whole graph has domination number at most four.*

Proof. The K_2 and triangle cases use $\{a, b\}$. Otherwise write the outer cycle as $a = w_0, w_1, \dots, w_{h-2}, w_{h-1} = b, a$. The two roots already dominate $w_0, w_1, w_{h-2}, w_{h-1}$. The cycle interval $w_2, \dots, w_{h-3}$ has at most three vertices. Choose its midpoint, if nonempty; it dominates the interval by the cycle edges. This choice is made on that *cycle interval*, regardless of additional chords. The union with c dominates the whole graft and has size at most four. □

Lemma 6.3 (Two induced MOP blocks)**.** *In a smallest counterexample to $\mathcal{F}(G) \ge 2|V(G)|$ in the original class $|V(G)| \ge 8$, the vertex set cannot be partitioned into two induced maximal outerplanar graphs each of order at least eight.*

Proof. For any real symmetric A, $\mathcal{F}(A) = \max_{X \succeq 0}\{2\operatorname{Tr}(AX) - \operatorname{Tr}(X^2)\}$. A block diagonal choice of X proves superadditivity over disjoint principal blocks. Both proper blocks satisfy the original inequality by minimality. Their two inequalities sum to $\mathcal{F}(G) \ge 2|V(G)|$, a contradiction. No edge-addition monotonicity is used. □

6.4. **Straight terminal paths.**

Lemma 6.4 (First-turn signature)**.** *Let a closed terminal cap have a rooted weak dual which is a unary path. If it is not a straight external-apex fan, it contains a closed subcap with private adjacency $K_1 \vee P_{L-1}$ for some $L \geq 3$ and with precisely the contacts au, av_1, bu from input 2 above.*

Proof. A one-face cap is a triangle. A two-face cap is a quadrilateral with one diagonal, and its root edge contains exactly one common apex of its two faces. Start at the leaf end and choose the straight-fan apex of this two-face prefix; this avoids treating the nonunique one-face description as a genuine turn. Suppose the current prefix has faces $uv_0v_1, uv_1v_2, \ldots, uv_{\ell-1}v_\ell$ and closing edge uv_0. Here $\ell \geq 2$ until a genuine first turn is found. The next face across that edge is uv_0x. Since the dual is a path, one of ux, v_0x is the next closing edge; its other edge has no further face inside this cap. If the closing edge is ux, the enlarged prefix is still a straight fan with apex u. If it is v_0x, the new roots are $a = v_0, b = x$ and the private set is $\{u, v_1, \ldots, v_\ell\}$. Its edges are exactly all uv_j and the path $v_1 \cdots v_\ell$. The contacts are exactly au, av_1, bu: the triangulated fan contains no further chords and chord separation excludes all additional outside contacts. Thus $L = \ell + 1 \geq 3$. If the entire cap were never to turn, all its faces would have the same external apex, contrary to the hypothesis. □

Lemma 6.5 (Terminal fan normalization)**.** *In a smallest original counterexample every closed terminal unary cap is a straight fan with an external apex. If the entire weak dual is a path and is rooted at a leaf, the whole graph is a straight fan.*

Proof. The finite basis forces $n \geq 72$. Apply Lemma 6.4 to a purported first turn. For $L = 3, 4, 5, 6, 7$ the retained old graph has order $n - L \geq 65 \geq 8$, and input 2 together with minimality excludes it. For $L \geq 8$, if the retained graph has order at least eight, its vertex set and the private fan form the two induced MOP blocks of Lemma 6.3. If the retained order is $2, \ldots, 7$, the private hub dominates all private vertices and Lemma 6.2 gives domination number at most four for the whole graph. Input 1 excludes this case. Hence no first turn exists. In the all-face path case the same leaf-to-root induction reaches the root with one common apex; the whole fan has domination number one. □

6.5. **The four child-apex patterns at a deepest branch.** Let a deepest branching face be abc, rooted at ab. Its terminal child caps on ac and cb have face counts $p, q \geq 1$. Their apex choices are respectively a or c and c or b. We now exclude the patterns with one or both private apex choices, except where a one-face child admits an identical external description.

Lemma 6.6 (Bounded ordered C_0 tips)**.** *In the both-private pattern, if $d = p + q \leq 15$ and $\max(p, q) \geq 3$, a closed two-vertex tip has whole root degrees $(d + 2, 3)$ and retained degrees $(d, 2)$. In the mixed pattern with outward $p \geq 2$ and inward $3 \leq q \leq 14$, the corresponding degrees are $(q + 3, 3)$ and $(q + 1, 2)$. Input 3 excludes both occurrences in a smallest original counterexample.*

Proof. In the both-private case the private graph is $c \vee (P_p \dot{\cup} P_q)$. Choose a path with at least three private vertices and delete the two at its free end, i.e. the end next to its apex in the outer boundary arc. Write the retained predecessor as w. Vertex c sees both deleted vertices, w sees exactly the first, and the two deleted vertices are adjacent. They have no other outside contacts. The private vertex c has precisely $p + q + 2$ neighbors, namely both roots and all path vertices; w has its two path neighbors and c. Deletion lowers their degrees by two and one. The same argument in the mixed case gives $q + 3$ neighbors at c: the q inward private vertices, a, b, and the last outward private vertex. Again w has degree three.

The retained closing edge cw is a boundary edge of a triangulated polygon. It has exactly one common old neighbor. After removing the edge itself from the root rows the old contact Gram is therefore $\left[\begin{smallmatrix} r & 1 \\ 1 & 1 \end{smallmatrix}\right]$ with $r = d-1$ or $r = q$, respectively. In the first case $4 \le d \le 15$; hence $3 \le r \le 14$ in both cases. Order the deleted vertices so that the one adjacent to w is first. The actual contact matrix is exactly input 3. Deleting them leaves an induced MOP of order $n - 2 \ge 70 \ge 8$. Its original inequality and the gain > 4.09 contradict $\mathcal{F}(G) < 2n$. □

Lemma 6.7 (Both-private branch exclusion)**.** *The both-private pattern is impossible in a smallest original counterexample unless $(p, q) = (1, 1)$, in which case both children can be described with external apices without changing any edge.*

Proof. First suppose $d = p + q \ge 16$ and put $h = 1 - 2/d$. Delete the hub from the private graph $B = c \vee (P_p \dot\cup P_q)$. The resulting path forest L has d vertices, $d-2$ edges, $\mathcal{F}(L) = d - 2$, and spectral radius less than two. Interlacing gives $\mathcal{F}(B) \ge \rho(B)^2 + \mathcal{F}(L) - \rho(L)^2$. Compression onto the hub and the uniform unit leaf vector gives $\rho(B) \ge h + \sqrt{d + h^2}$. Consequently

$$\mathcal{F}(B) > 2d + G(d), \qquad G(d) = 2h^2 + 2h\sqrt{d + h^2} - 6.$$

Every positive factor in this expression increases with $d > 2$. Moreover $G(16) = (7\sqrt{1073} - 143)/32 > 345/128$, since $(131/4)^2 < 1073$. Therefore $\mathcal{F}(B) > 2(d+1) + 89/128$. If the old remainder has at least eight vertices, minimality and disjoint principal block superadditivity exclude the whole graph. Otherwise Lemma 6.2 applies, because c dominates every private vertex.

For $d \le 15$ and a child length at least three, use Lemma 6.6. The remaining ordered pairs are $(1,1), (1,2), (2,1), (2,2)$. The pairs $(1,2), (2,1)$ are input 4 and its coherent reflection: for $(1,2)$ send its old labels $(0, 3, 2, 1, 4, 5)$ to (a, b, c, L_1, R_1, R_2). The old remainder has order $n - 4 \ge 68$, so the gain > 8.07 excludes them.

For $(2,2)$ the private bowtie has eigenvalues $\lambda_+ = (1 + \sqrt{17})/2$, 1, -1, -1, $\lambda_- = (1 - \sqrt{17})/2$, so $\mathcal{F}(B) = (11 + \sqrt{17})/2$. For contacts $e_c + e_{L_1}$ and $e_c + e_{R_2}$ the Perron vector $(\lambda_+ - 1, 1, 1, 1, 1)$ gives $g_{\lambda_+} = (85 + 13\sqrt{17})/68$. The unit eigenvector $(0, 1/2, 1/2, -1/2, -1/2)$ gives $g_1 = \zeta_1 = 1/2$. Input 5 therefore yields

$$\Delta\mathcal{F} \ge \frac{59}{8} + \frac{47\sqrt{17}}{68} - \frac{\sqrt{2}}{8} > \frac{59}{8} + \frac{47 \cdot 4123}{68000} - \frac{1415}{8000} > 10 + \frac{1}{25}.$$

The radical comparisons follow from $4123^2 < 17 \cdot 1000^2$ and $1415^2 > 2 \cdot 1000^2$. This excludes five private vertices. Finally, when $(p, q) = (1, 1)$ each child is a triangle and either of its root vertices is an apex. Choose a, b as the two external apices. □

Lemma 6.8 (Mixed branch exclusion)**.** *Every deepest branching cap in a smallest original counterexample admits the external/external description $E_{p,q}$.*

Proof. The external/external case already has this description. The both-private case is handled by Lemma 6.7. Coherent reflection reduces either mixed case to left apex a and right apex c. If $p = 1$, reassign that child triangle to apex c and apply the both-private analysis. If $q = 1$, reassign its triangle to b, obtaining an external/external description of the same graph. Thus assume $p, q \ge 2$.

If the old remainder has order at most seven, the private vertices are dominated by a, b, c and the whole graph is covered by Lemma 6.2. Otherwise its order is at least eight. For $q = 2$ and $q = 15, 16, 17, 18$, input 6 gives a strict full-branch payment. For $3 \le q \le 14$ use Lemma 6.6. For $q \ge 19$ delete only the q inward private fan vertices. The retained private apex c has exactly the neighbors a, b, L_p, hence actual degree three, independently of p. It is an induced retained MOP containing the original old remainder, and therefore has order at least eight. Input 6's fan theorem, applied to the actual retained boundary edge cb with apex c, gives a gain greater than $2q$. Minimality excludes the occurrence. All positive lengths have now been covered. □

6.6. **The all-depth parent trichotomy.**

Theorem 6.9 (Deepest-parent extraction)**.** *Assuming the energy inputs stated above, any smallest original counterexample of order $n \geq 8$ contains a closed parent cap of exactly one of the following structural forms (descriptions need not be unique):*

$$U(E_{p,q}), \qquad B(E_{p,q}, \mathsf{P}_\ell), \qquad B(E_{p,q}, E_{r,s}),$$

where every displayed length is a positive integer, with no upper bound. The middle form must allow both possible sibling apices. If its old remainder has order at most seven, the whole graph is already excluded by domination number at most four.

Proof. Root the weak dual at a leaf face, choosing one of that face's boundary edges as the graph's root edge. The root then has at most one child. If there is no branching node, Lemma 6.5 makes the whole graph a fan, contrary to input 1. Choose a branching node X of greatest depth d. It is not the root and has a parent Y. Both child subtrees of X are unary paths, since any descendant branch would have greater depth. Lemma 6.8 gives *the whole closed cap at* X an $E_{p,q}$ representation.

If Y has one child its cap is $U(E_{p,q})$. Otherwise let Z be the sibling of X. If Z has no branching descendant, its cap is a straight fan P_ℓ by Lemma 6.5; its apex on the parent's right edge is either c or b. This gives both roles of $B(E_{p,q}, \mathsf{P}_\ell)$. If the subtree at Z contains a branch, its root Z must itself branch: any branch strictly below it would have depth greater than d. Its children again have no branches, and its whole cap admits an $E_{r,s}$ representation. This gives $B(E_{p,q}, E_{r,s})$. If X was the other child of Y, apply coherent whole reflection.

Chord separation proves closedness, inducedness of the old remainder, and that the new parent vertex c is counted exactly once. In all three forms every new vertex either is c or is adjacent to at least one of a, b, c: in each E child, its center sees its two roots and each fan vertex sees its external apex; each of those roots belongs to a, b, c. The sibling fan has apex b or c. Thus Lemma 6.2 handles all old remainders of order $2, \ldots, 7$, including K_2 as a whole-graph argument. No local increment theorem is extended to an old K_2. □

Lemma 6.10 (Ordered tips in the grouped parents)**.** *For $U(E_{p,q})$ with $3 \leq q \leq 14$ there is a closed C_0 tip with whole root degrees $(q+3, 3)$ and retained degrees $(q+1, 2)$. For $B(E_{p,q}, E_{r,s})$ with $m = q + r \leq 13$ and $\max(q, r) \geq 3$ there is one with whole degrees $(m+4, 3)$ and retained degrees $(m+2, 2)$. Both are excluded by input 3 in a smallest original counterexample.*

Proof. In the unary form let the left child E have center d. Its inner fan is rooted at (d, c) with apex c, and has outer arc $d, Q_1, \ldots, Q_q, c$. Vertex c has exactly the neighbors a, b, d and all Q_j. Delete Q_{q-1}, Q_q and retain $w = Q_{q-2}$. Vertex w has degree three, including the neighbor d when $q = 3$. The tip contacts are exactly those in Lemma 6.6.

In the double form the two child centers are d, e. Vertex c sees exactly a, b, d, e and all $q + r$ inner fan vertices. If $q \geq 3$ use the same left inner tip as above. Otherwise $r \geq 3$; its right inner fan is rooted at (c, e) with apex c and arc $c, R_1, \ldots, R_r, e$. Delete R_1, R_2 and retain $w = R_3$, whose degree is three, including neighbor e when $r = 3$. Each deletion lowers the apex degree by two and the other root degree by one. The retained closing edge has exactly one common old neighbor, so the ordered old contact Gram index is $g = q$ in the unary case, or $g = m + 1$ in the double case. For the latter case $4 \leq m \leq 13$, hence $5 \leq g \leq 14$; the unary case has $3 \leq g \leq 14$. The retained graph has order $n - 2 \geq 70$. Chord separation, input 3, and minimality give the claimed exclusion. □

The remaining energy task is exactly to exclude the three displayed parent forms at their actual contacts when the retained old MOP has order at least eight. One may use full increments or a smaller closed removal paid against minimality. A bare-branch theorem for every old parent, a universal good ear, and a convention obtained by setting $\ell = 0$ in a theorem for $\ell \geq 1$ are not required.

6.7. **Exhaustive parent payments and the original conclusion.**

Proof of Theorem 1.1. Assume a counterexample exists and choose one of minimum order n. Theorem 4.1 forces $n \geq 72$. Apply Theorem 6.9. If the retained old remainder has order at most seven, Lemma 6.2 and Theorem 4.3 already contradict the counterexample. We may therefore suppose that the actual retained MOP H has order at least eight. Every proper induced retained MOP of order at least eight satisfies the original bound by minimality.

For $U(E_{p,q})$, if $q \in \{1, 2, 15, 16, 17, 18\}$, Proposition 5.1 gives the actual increment $\Delta\mathcal{F} > 2(p+q+2)+1/20$. If $3 \leq q \leq 14$, use Lemma 6.10. If $q \geq 19$, remove only the q inner fan vertices on the actual child edge dc, with apex c. The retained vertex c has exactly the neighbors a, b, d, independently of p. The retained graph contains H and is an induced MOP of order at least eight. The degree-three fan statement in the inputs gives gain $> 2q+1/20$, so minimality again excludes the occurrence. These ranges exhaust every positive q without asserting a stronger all-q full increment relative to every arbitrary old H.

For $B(E_{p,q}, \mathsf{P}_\ell)$, both apex roles are covered by Proposition 5.1: for every independent $p, q, \ell \geq 1$ the actual increment is greater than $2(p+q+\ell+2)+1/50$. The one-triangle sibling equivalence is an equality of the actual rooted graph; it is not a substitution $\ell = 0$ and it does not create a second energy credit.

Finally consider $B(E_{p,q}, E_{r,s})$ and put $m = q+r$. If $(q, r) \in \{1, 2\}^2$, Proposition 5.1 gives gain $> 2(p+q+r+s+3)+3/20$ for every independent $p, s \geq 1$. If $m \leq 13$ and at least one of q, r is at least three, use Lemma 6.10. The two other ranges are $14 \leq m \leq 41$ and $m \geq 42$; Proposition 5.1 gives gains $> 2(p+q+r+s+3)+1/20$ and $> 2(p+q+r+s+3)+9/100$, respectively. Thus every positive pair (q, r) is covered, keeping both independent outer lengths and both ordered root roles.

Each full increment $> 2N$ contradicts the counterexample after adding the inequality $\mathcal{F}(H) \geq 2|H|$. A C_0 removal leaves order $n-2 \geq 70$ and has gain $> 4.09 > 4$; a fan removal leaves a proper MOP of order at least eight and gains more than twice its deleted order. These also contradict $\mathcal{F}(G) < 2n$. The three parent forms exhaust all possibilities, so no counterexample exists. This proves exactly the original statement for every finite simple unit maximal outerplanar graph of order at least eight, with equality allowed. □

7. Exact verification and reproducibility

The supplementary archive is an explicit collection of original mathematical sources, finite witnesses, interval partitions, and independent replay drivers. It is not an export of the author's database. The file `REPRO_MEMBERS.json` lists each mathematical source or replay-code pathname, uncompressed byte length, and SHA–256 digest. `ARCHIVE_MEMBERS.json` indexes the complete archive except that self-index. No earlier full ZIP archive or database is an archive member. Historical observation files, when included, retain their original scope; a historical `PASS` string is not a mathematical axiom. The integrity command below checks byte identity only.

Environment and basic commands. The independent executions used CPython 3.12.14 and NumPy 2.3.5 on Windows. The finite Gram reader uses NumPy solely for matrix products whose integer overflow bounds are checked explicitly; final squares and sums use Python integers. The rational and interval computations use integers and `fractions.Fraction`. A second installed CPython 3.12.3/NumPy 1.26.4 environment was available, but this is not a claim that every command was replayed in both environments. No database, account, network connection, SciPy, or symbolic algebra system is required by the commands below. Assertions must remain enabled: do not invoke Python with `-O`. The new drivers enforce this with an explicit exception outside any `assert` statement; six optimization-attempt controls failed closed. The unchanged original validators require the stated normal interpreter mode. Extract the archive into a short path on Windows, preserving its directory structure.

All commands are run from its root, with a fresh output directory for each independently materialized family tree.

```
python -B repro_check_integrity.py
python -B repro_verify_finite.py quick --output finite_results
python -B repro_verify_finite.py positive --output finite_results
python -B repro_verify_finite.py masked --output finite_results
python -B repro_verify_finite.py assemble --output finite_results
python -B repro_verify_finite.py charge190 --output charge_results
python -B extraction_gamma3_verify.py
python -B extraction_gamma_tail_checks.py
python -B verify_coupled_bridge_244.py
python -B verify_removed_runs_244.py
python -B repro_verify_core.py c0 --output c0_results
```

Alternatively, the single command

```
python -B repro_replay_b71.py --output fresh_B71
```

executes the complete finite quick/positive/masked/assembly sequence in a new, nonexistent output directory, checking all source hashes before and afterwards. Use this command or the individual finite stages above; both are not needed. The assembly stage alone is conditional on the five mathematical result files and does not infer fresh execution from a JSON name or status field. In this review the stages were actually run and observed separately; the newly provided combined launcher was not used to repeat them a second time.

The first finite command reconstructs all 29 degree envelopes and verifies all 6,853 labelled polygon triangulations of orders 8–11 with their integer Gram witnesses, all 61 small-gap witnesses of orders 5–7, all 4,636 parameter rows of orders 12–42, all 4,760 high-pair kernels, all 54,870 specified exterior-ear extensions, and all 1,150 degree-cap complements. The positive and masked commands recompute integer Catalan dynamic programming maxima over their complete domains; stored extrema in target tables are compared with freshly computed maxima. They do not read native acceptance receipts. The final assembly evaluates all 11,885 high-degree rows of orders 43–71 using those new results and the independently computed 29 low-degree envelopes. Its five required mathematical inputs are the new positive, masked, degree, early-cover, and high-pair/class result files. Omitting an input causes failure before a complete finite conclusion can be emitted. The reduction from these explicit domains to the finite graph theorem is the written proof above; a parameter computation alone does not establish that reduction.

The `charge190` command runs all four original exact finite readers for the tighter and nonlinear split-fan charges. They verify respectively 4,951 and 4,950 parameter cells, including complete exceptional forest Gram coverage. The aggregate explicitly compares the freshly discovered residual triples (q, p, s) with the initial-screen obligation sets; their cardinalities are 21 and 62. This connects the parameter and forest enumerations without treating an old status field as a substitute for coverage. The gamma-three driver checks all 3,795,253 allocations required by the finite bridge. The tail driver checks the six and 33 rational constant comparisons for the analytic charge tails and the gamma-four threshold. The all-order recurrence and structural implications remain the mathematical arguments given in the corresponding appendices.

Universal parent families. The following command runs the original serial-turn, triangle-wing, singular-turn, first-turn, mixed-$q = 2$, mixed-$q = 15, \ldots, 20$, degree-three fan, and quantitative trace readers in an explicitly copied tree:

```
python -B repro_verify_families.py fast --output family_results
python -B repro_verify_families.py normalized-sign --output sign_results
```

The normalized-sign reader and its 144 matrix jets, 924 coefficient PSD checks, all-index recurrence proof, and rational path controls are separately included under `work/mop-normalized-sign-20261105`. Its proof supplies the imaginary-part sign needed to improve the general two-sector phase bound from $4\arctan(2/t)$ to $2\arctan(2/t)$ in the bare-branch argument. The quantitative trace input includes the actual numerator/denominator polynomials, 288 initial jets, 256 full scalar tail intervals, and their separate Newton/moment reader. The qualitative sector estimate alone would not justify the stated 184/157 branch debt.

For the slow compact parent families, the independent drivers perform actual numerical interval evaluation at every accepted leaf:

```
python -B repro_verify_families.py unary --output unary_results
python -B repro_verify_families.py small-double --output small_results
python -B repro_verify_families.py external --output external_results
python -B repro_verify_core.py private --output private_results
python -B repro_verify_core.py double-compact --output double_results
python -B repro_verify_core.py double-tail \
  --compact double_results/double_compact.json --output double_results
```

The backslash above denotes a line continuation in POSIX shells; on Windows the last command may be written on one line. Compact and signed-tail stages must be separate Python processes. The compact kernels install the 96-bit rational-grid backend, whereas the selected double-family tail uses the original default 192-bit interval backend. Sharing an imported backend across these stages would change the arithmetic program. The private-family tail uses its stated 96-bit backend. The double compact driver includes the numerically evaluated slab 23; it does not substitute the earlier preflight area. The full private driver reads all 160 raw frequency objects for the two bands, and the full double compact driver reads all 36 selected raw slab objects. Neither driver accepts a saved area or an execution-session receipt as numerical proof.

The explicit smoke commands are materially smaller:

```
python -B repro_verify_core.py private --smoke --output smoke
python -B repro_verify_core.py double-compact --smoke --output smoke
python -B repro_verify_core.py double-tail --output tail_results
```

The private smoke evaluates only the first frequency leaf and first ordered residue root in each band: 24 and 60 numerical leaves. The double smoke evaluates only the first root of slab 23: eight numerical leaves. Neither is a family theorem. Without a new full compact result, the tail command uses the explicit target compact budget and labels its conclusion `CONDITIONAL_TARGET_BUDGET_ONLY_NO_COMPACT_PROOF`; it does recompute all 3,584 signed tail intervals and all 28 resulting conditional inequalities.

What was actually replayed in this review. The C0 driver completed all 2,304 exact node envelopes and reproduced every mathematical output field of the source result. The finite quick checks completed in approximately 9.33 seconds (degree envelopes), 2.06 seconds (early cover), 31.61 seconds (high pairs and ears), and 0.77 seconds (class payments) on the review machine. All four finite charge readers completed, as did their explicit residual-set linkage. The full positive 328-target DP completed in 370.36 seconds. The normalized-sign reader completed all 144 jets and 924 PSD coefficient checks in 1.72 seconds. The full coupled gamma-four reader freshly checked all 286,769 profiles of orders 72–159, including its independently rebuilt integer tables and small split-fan Gram witnesses, in 17.44 seconds. The private and double compact smoke tests and all 3,584 double tail cells completed. The complete eight-family command also completed: serial-turn certificates in 205.56 seconds, triangle wing in 4.31 seconds, singular turns in 23.30 seconds, first turns in 1.16 seconds, mixed $q = 2$ in 129.56 seconds, mixed $q = 15, \dots, 20$ in 65.38 seconds, fan retirement in 0.19 seconds, and quantitative trace in 0.52 seconds. The seven saved family result objects agree

with their original mathematical fields after excluding only runtime and source-location fields. The removed-run graph-domain calibration for orders 4–9 completed in 21.17 seconds; its finite calibrations supplement the written all-order proof.

The complete masked DP then finished all 130 targets in 29 groups in 2,331.39 seconds, below its 5,400-second limit; even the original $(31, 31)$ preflight group was computed anew. The new standalone finite assembly completed in approximately 4.98 seconds, checking all 11,885 rows from the new mathematical inputs. All five single-input omission controls, and an altered actual-maximum control, were rejected. Thus the complete finite arithmetic recipe was freshly executed in this review. This is still not a fresh full global numerical replay of the slow compact families.

The executed pre-guard masked and bounded-runner source variants and their exact differences from the packaged guarded versions are preserved. The running masked script remained byte-identical to its preserved source until its terminal observation; the guarded successor was copied only afterwards. The changed assertion-mode check is a no-op under normal Python, and all executed masked/payment/helper function ASTs agree. No original DP kernel or target file changed, and their hashes are recorded. A launch-time hash of the entire earlier positive orchestration file was not captured before unrelated modes were added; whole-file identity is not asserted for that launcher. Its mathematical positive/payment functions and the original DP kernels remained unchanged. This distinction is recorded rather than silently attributing earlier execution to a later source variant.

The original private compact result records 766,777 numerical leaves and about 26,494 seconds of runtime. The selected double compact composition retains 284,102 source-bound numerical leaf subproofs; the later selected reader rechecks their grammar and provenance and recomputes its signed tails, while explicitly reusing their numerical areas. Those original sources and data are retained unchanged. This review did not repeat the multi-hour private and double compact computations. The new full commands above make such a repeat possible; smoke success is not reported as its completion. Runtime is hardware dependent, and the source-bound historical completions and fresh executions are separate evidence.

A separate bounded counterobject attempt. An independent approximate flip search used 522 starts in total: 517 at the selected large orders $72, 73, 74, 80, 88, 96, 112, 128$, plus single-start controls at orders $8, 9, 10, 12, 16$. It performed 220,931 floating eigenvalue solves and 138,860 floating flip evaluations in about 285.05 seconds. These are operation counts, not distinct-graph counts or an exhaustive search. No floating candidate below $2n$ was found. The retained best large candidate has order 72, and an independent integer Gram verification gives

$$\mathcal{F}(G) - 144 \;\geq\; \frac{423972407978449381}{36028797018963968} > 0.$$

The full graph, factor, search code, environment, actual terminal records, and guarded exact reader are included in the archive. Its exact verifier can be run independently:

```
python -B review_results/countersearch/phase_countersearch_verify_best.py
```

The approximate search is additional adversarial evidence, and is not used as a proof of any unsearched graph class.

Acceptance boundary. The mathematical conclusion requires the written polygon/dual extraction, reflection and retained-degree arguments; the published graph inequalities at their stated scopes; the lossless-root Neumann and phase-branch derivations; the Gram and interval finite algorithms; all unbounded residue and recurrence coverage; and the final parent-role case partition. Content hashes certify identities, and process receipts certify observations. Neither certifies those mathematical implications. The supplementary drivers therefore report their exact scopes, including partial smoke and conditional-budget modes, and do not automatically promote a successful receipt to the global theorem.

Appendix A. Bounded serial matrix extension

The following bounded-contact matrix lemma is stronger than the graph statement required by the reduction. It places no norm, degree, order, trace, or spectrum bound on the far block.

Lemma A.1 (The actually used serial kernel)**.** *Let C be any finite Hermitian matrix and R a real matrix satisfying*

$$RR^T = G = \begin{pmatrix} r & 1 \\ 1 & 1 \end{pmatrix}, \qquad r \in \{3, 4, \dots, 14\}.$$

Put $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and form

$$A_0 = \begin{pmatrix} J & R \\ R^T & C \end{pmatrix}, \qquad A_1 = \begin{pmatrix} J & R & T \\ R^T & C & 0 \\ T^T & 0 & J \end{pmatrix}.$$

Then $s^+(A_1) - s^+(A_0) > 409/100$.

The two contact roles are ordered: the first root sees both new private vertices and the second sees only the first. The proof does not identify $G(r, 1)$ and $G(1, r)$ by swapping their diagonal entries alone.

A.1. **A uniform phase envelope for the same far matrix.** Fix $t > 0$. To avoid confusion with the whole adjacency matrices above, write the two root Schur matrices as

$$S_0 = tI + iJ, \quad Q = T(tI + iJ)^{-1}T^T = \frac{1}{t^2+1}\begin{pmatrix} 2t - 2i & t - i \\ t - i & t \end{pmatrix}, \qquad S_1 = S_0 + Q.$$

Their real parts are positive definite, so $\|S_j^{-1}\|_2 \le 1/t$. Root-first Schur elimination leaves

$$L_j = tI + R^T S_j^{-1} R = U_j + iV_j, \qquad U_j \succeq tI \succ 0.$$

Here U_j, V_j are Hermitian, and in fact real symmetric. Since $\det S_0 = t^2 + 1$ and $\det(tI + iJ) = t^2 + 1$ are positive, the actual lifted phase difference for the extension is

$$\delta(t) = \operatorname{Im} \operatorname{Tr} \operatorname{Log} S_1 + \operatorname{Im} \operatorname{Tr} \operatorname{Log}(L_1 + iC) - \operatorname{Im} \operatorname{Tr} \operatorname{Log}(L_0 + iC).$$

All three matrix logarithms use the accretive branch; no determinant product is reset to a principal scalar argument.

Let

$$X = U_0^{-1/2}(C + V_0)U_0^{-1/2} = X^*, \qquad D = U_0^{-1/2}\{U_1 + i(V_1 - V_0)\}U_0^{-1/2}.$$

Then $\operatorname{Re} D \succ 0$. Congruence, with its positive real determinant factors, shows that the far phase difference is the phase of $(D + iX)(I + iX)^{-1}$ with its continuous normalization. Set

$$U = (I - iX)(I + iX)^{-1}, \qquad Z = (I + D)^{-1}(D - I), \qquad B = (I + D)/2.$$

Because X is Hermitian, U is unitary. Direct multiplication on the right by $I + iX$ gives

$$(D + iX)(I + iX)^{-1} = B(I + ZU).$$

Also

$$I - Z^*Z = 4(D + I)^{-*}(\operatorname{Re} D)(D + I)^{-1} \succ 0,$$

so $\|Z\|_2 < 1$.

The space $W = \operatorname{range} R^T$ has dimension two since $G \succ 0$. Each $U_j - tI$ and V_j is supported on W and vanishes on its orthogonal complement. Thus $D = I$ and $Z = 0$ on that complement. The far matrix C need not preserve W. This causes no loss of rank-two control: the compression U_{WW} of its unitary Cayley transform is a contraction, and Sylvester's determinant identity gives

$$\det(I + ZU) = \det_W(I + Z_W U_{WW}).$$

The normalized logarithmic phases agree as well, by scaling Z from zero and using $\|Z\|_2 < 1$. Lemma 3.2 applied on W therefore bounds this phase by the sum of the inverse sines of the two nonzero singular values of Z.

There is no unaccounted multiple of 2π in the factorization. To see this without an argument-product convention, let $X = sI$ and $s \to +\infty$. The original far phase difference tends to zero, $U \to -I$, and $I - Z = B^{-1}$. Both B and B^{-1} are accretive and their trace phases cancel exactly. The difference of the two phase definitions is a continuous integer multiple of 2π on the connected space of all Hermitian X, since no factor is singular. It is zero at this limit, and therefore identically zero. For the SAME actual C at each t we have established

$$\delta(t) \le \operatorname{Im}\operatorname{Tr}\operatorname{Log} S_1 + \operatorname{Im}\operatorname{Tr}\operatorname{Log} B + \arcsin\sigma_1(Z) + \arcsin\sigma_2(Z). \tag{52}$$

No common maximizing far matrix at different frequencies is asserted or needed: the displayed bound holds for each fixed matrix C throughout.

A.2. **Exact intrinsic computation on the two-dimensional contact space.** Use coordinates $z \mapsto R^T z$ on W. Their Gram metric is G, and the intrinsic matrices are

$$\widehat{U}_j = tI + \operatorname{Re}(S_j^{-1})G, \qquad \widehat{V}_j = \operatorname{Im}(S_j^{-1})G.$$

These are G-self-adjoint; an ordinary transpose or Euclidean adjoint in these nonorthonormal coordinates would be incorrect. Define

$$B_0 = \widehat{U}_0 + \widehat{U}_1 + i(\widehat{V}_1 - \widehat{V}_0), \quad E_0 = \widehat{U}_1 - \widehat{U}_0 + i(\widehat{V}_1 - \widehat{V}_0), \quad V = B_0^{-1}E_0, \qquad V^\sharp = G^{-1}V^*G.$$

In an orthonormal realization of W, the Cayley matrix is similar to $U_0^{1/2}VU_0^{-1/2}$, and

$$M = \widehat{U}_0 V \widehat{U}_0^{-1} V^\sharp$$

has the same eigenvalues as ZZ^*. Thus, with

$$a = \operatorname{Tr} M, \quad b = \tfrac{1}{2}\{(\operatorname{Tr} M)^2 - \operatorname{Tr}(M^2)\}, \qquad d = a^2 - 4b,$$

the two squared singular values are $(a \pm \sqrt{d})/2$. The center phase in (52) is $\arg\det B_0$: the omitted determinant factors are positive, and the accretive two-dimensional trace phase lies in $(-\pi, \pi)$. This center phase is added separately to $\arg\det S_1$, without resetting their sum.

At rational t all these matrices and invariants are in $\mathbb{Q}(i)$. Positive definiteness of $G\widehat{U}_j$ is verified by its two principal minors. The arithmetic also verifies $b \ge 0$, $d \ge 0$, $a < 2$ and $1 - a + b > 0$, and proves every singular-value upper endpoint below one. These are rational comparisons, not floating spectral estimates.

Here is the complete outward node algorithm. If $x \ge 0$ is rational and $D_* = 10^{14}$, put

$$k = \lfloor D_*\sqrt{x} \rfloor = \operatorname{isqrt}\left(\left\lfloor D_*^2 x \right\rfloor\right).$$

Then $k/D_* \le \sqrt{x} \le (k+1)/D_*$, with equality used when $(k/D_*)^2 = x$. Enclose $\sqrt{d}$ by $[d_-, d_+]$ and use the two upper squared-singular-value bounds $\bar{\lambda}_+ = (a + d_+)/2$ and $\bar{\lambda}_- = (a - d_-)/2$. The smaller squared singular value therefore uses the LOWER discriminant endpoint. For either bound, enclose

$$\arcsin\sqrt{\lambda} = \arctan\sqrt{\frac{\lambda}{1-\lambda}} \le \arctan\frac{\operatorname{sqrtUpper}(\bar{\lambda})}{\operatorname{sqrtLower}(1-\bar{\lambda})}.$$

The denominator must be strictly positive.

The arctangent routine is entirely rational. For $|y| \le 1/2$ use

$$\arctan y = \sum_{j=0}^{11} \frac{(-1)^j y^{2j+1}}{2j+1} + \varepsilon_y, \qquad |\varepsilon_y| \le |y|^{25}/25.$$

The remaining positive arguments use

$$\arctan x = \begin{cases} \pi/4 + \arctan((x-1)/(x+1)), & 1/2 < x \le 2, \\ \pi/2 - \arctan(1/x), & x > 2. \end{cases}$$

Oddness handles negative arguments. Machin's identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239),$$

with eighteen alternating terms and their rational remainder bounds, supplies the outward interval for π. The sign of the real and imaginary parts determines the quadrant for each nonzero complex determinant. A negative real determinant is excluded by strict accretivity in dimension two. Finally the upper value of (52) is rounded upward to a multiple of 10^{-9}. All roundings are outward; no floating value enters any acceptance condition.

A.3. **A quadrature bound for the actual common-source phase.** This step bounds the smooth ACTUAL phase, not the potentially nonsmooth upper envelope in (52). Since A_0 is a principal submatrix of A_1 of codimension two, interlacing implies that the spectral counting difference

$$\xi(x) = \#\{\lambda(A_1) > x\} - \#\{\lambda(A_0) > x\}$$

lies in $[0, 2]$. Integration by parts gives

$$\delta(t) = -\pi + \int_{\mathbb{R}} \frac{t\xi(x)}{t^2 + x^2}\, dx.$$

The two private eigenvalues $1, -1$ have zero combined phase, so this is the same phase as in (52). Since ξ is a bounded step function, differentiating under this integrable kernel is justified on every compact positive t interval. Writing $k_t(x) = t/(t^2 + x^2)$,

$$\partial_t k_t = \frac{x^2 - t^2}{(t^2 + x^2)^2}, \qquad \partial_t^2 k_t = \frac{2t(t^2 - 3x^2)}{(t^2 + x^2)^3}.$$

Both derivatives have integral zero. Their positive-part integrals are $1/t$ and $3\sqrt{3}/(4t^2)$, respectively, by integration at the zeros $|x| = t$ and $|x| = t/\sqrt{3}$. Therefore

$$|\delta(t)| < \pi, \qquad |\delta'(t)| \le 2/t, \qquad |\delta''(t)| \le 3\sqrt{3}/(2t^2).$$

For $f(t) = t\delta(t)$, this gives

$$|f''(t)| \le \frac{4 + 3\sqrt{3}/2}{t} < \frac{53}{8t},$$

using $\sqrt{3} < 7/4$.

On $[1/10, 20]$ use $N = 192$ equal cells of width $h = 199/1920$ and their midpoints $t_j = 1/10 + (j + 1/2)h$. The midpoint error on a cell starting at u is at most $53h^3/(192u)$. Since $1/t$ decreases,

$$\sum_{j=0}^{N-1} \frac{1}{1/10 + jh} \le 10 + \frac{\log 200}{h} < 10 + \frac{6}{h}.$$

The last inequality follows, for example, from $e^6 > \sum_{j=0}^{6} 6^j/j! > 200$. If $q_{r,j}$ is the rational upper node value, the ENTIRE compact integral for every fixed C is bounded by

$$\int_{1/10}^{20} t\delta(t)\, dt \le h \sum_{j=0}^{191} t_j q_{r,j} + \frac{53}{192}(10h^3 + 6h^2). \tag{53}$$

The same C is used on both sides. Independent maximization of the node upper bounds does not assume that maximizing matrices agree; the quadrature error is uniform over every actual fixed matrix. The full low interval contributes at most $\int_0^{1/10} \pi t\, dt = \pi/200 < 11/700$.

A.4. **A signed bound for the entire infinite tail.** Let $g = r+1 = \operatorname{Tr} G \le 15$ and $t \ge 10$. The root/private four-vertex diamond has characteristic polynomial $x(x+1)(x^2-x-4)$, second trace10, third trace12 and norm less than 8/3. Its phase, equal to the phase of S_1, has the signed expansion

$$\operatorname{Im}\operatorname{Tr}\operatorname{Log} S_1 \le -4/t^3 + 41/t^5.$$

Indeed the cubic term is $-\operatorname{Tr}(A^3_{\text{diamond}})/(3t^3)$; the arctangent remainder is at most $\sum|\lambda|^5/(5t^5) \le 10(8/3)^3/(5t^5) < 41/t^5$.

The far correction remains uniform without a bound on C. The accretive inverse bounds and $\|T\|_2^2 \le 3$ give

$$\|Q\|_2 \le 3/t, \quad \|S_1^{-1} - S_0^{-1}\|_2 \le 3/t^3, \quad \|D-I\|_2 \le \frac{3g}{t^4} =: \varepsilon.$$

Here the last step uses $U_0 \succeq tI$ and $\|R\|_2^2 \le g$. The rank-two center phase is at most $2\arcsin(\varepsilon/2)$, and $\|Z\|_2 \le \varepsilon/(2-\varepsilon)$ bounds the two Cayley angles. Using $\arcsin x \le x/(1-x)$ for $0 \le x < 1$ yields a total far-phase upper bound

$$\frac{\varepsilon}{1-\varepsilon/2} + \frac{\varepsilon}{1-\varepsilon} \le \frac{2\varepsilon}{1-g/t^2},$$

since $\varepsilon = 3g/t^4 \le g/t^2 < 1$. Consequently

$$\delta(t) \le -\frac{4}{t^3} + \frac{6g}{t^4(1-g/t^2)} + \frac{41}{t^5}.$$

Integration from $T = 20$ to infinity, retaining the negative leading term, gives

$$\begin{aligned}\int_T^\infty t\delta(t)\,dt &\le -\frac{4}{T} - 3\log(1-g/T^2) + \frac{41}{3T^3} \\ &\le -\frac{4}{T} + \frac{3g}{T^2-g} + \frac{41}{3T^3}. \end{aligned} \tag{54}$$

The last inequality is $-\log(1-x) \le x/(1-x)$ for $0 \le x < 1$. Thus the tail is complete, signed, and independent of the far dimension.

A.5. **The finite rational conclusion and graph transfer.** The supplemental standard-library checker computes all $12\cdot 192 = 2304$ node upper values by the exact algorithm above. For each ordered r it forms the rational upper bound

$$I_r = h\sum_j t_j q_{r,j} + \frac{53}{192}(10h^3+6h^2) + \frac{11}{700} - \frac{4}{20} + \frac{3(r+1)}{400-(r+1)} + \frac{41}{24000}.$$

It verifies $I_r < 0$ and $4 - (7/11)I_r > 409/100$ for EACH $r = 3,\dots,14$. The four new unit edges give the constant4 in the exact energy identity (3). Since $\pi < 22/7$ and $I_r < 0$, that identity and (53)–(54) imply

$$s^+(A_1) - s^+(A_0) \ge 4 - \frac{2}{\pi}I_r > 4 - \frac{7}{11}I_r > \frac{409}{100}.$$

For transparency, the exact minimum checker lower bound is

$$\frac{131605851646082517349}{32115916800000000000} > \frac{409}{100},$$

attained at the ordered input $r = 14$. These rational comparisons and all2304 node endpoints are retained in the supplement, and are regenerated by the checker rather than inferred from printed decimals. The arithmetic kernel uses only Python integers and `Fraction`. Its Gaussian rational arithmetic, metric adjoints, outward square-root and arctangent bounds are specified above; the whole numerical obligation is finite and has no unstated far-block discretization.

For the original graph application, a boundary edge in a MOP of order at least4 has one common neighbor. Thus its actual old contact Gram is

$$RR^T = \begin{pmatrix} d_H(a) - 1 & 1 \\ 1 & d_H(b) - 1 \end{pmatrix}.$$

Deleting the stated serial tip reduces the apex degree by2 and the other root degree by1. In the unary-parent case $q = 3, \dots, 14$, the whole degrees are $(q+3, 3)$ and the retained degrees $(q + 1, 2)$, giving $r = q$. For a double parent with $m = q + r \le 13$ and a long inner arm, the whole degrees are $(m + 4, 3)$ and retained degrees $(m + 2, 2)$, giving the ordered kernel parameter $m+1 \in \{5, \dots, 14\}$. For the both-private branch with total $d \le 15$ and an arm at least3, the analogous degrees are $(d + 2, 3)$ and $(d, 2)$, giving parameter $d - 1 \in \{3, \dots, 14\}$. Each tip has exactly the stated contacts and no additional private contacts. In a smallest original counter of order $n \ge 72$, its deletion leaves an actual MOP of order $n-2 \ge 70$, paid by minimality. The gain greater than4.09 then excludes the tip. No unrestricted all-root serial increment, no auxiliary K_2 induction step, and no old degree selector are asserted by this bounded-contact lemma.

Appendix B. Finite dynamic programs

B.1. **Polygon tables and their exact interface.** Fix an integer cap C. For a polygon with p vertices and an ordered boundary edge define $T_p(r, s)$ to be the largest sum of squared degrees of its *internal* vertices (all vertices other than the two boundary roots) among triangulations with endpoint degrees r, s, with every internal degree at most C. An infeasible state is absent, not assigned zero. The base is $T_2(1, 1) = 0$. For $p \ge 3$, the unique triangle incident with the boundary edge splits the polygon into children of sizes a and $p - a + 1$, $2 \le a \le p - 1$. If their ordered states are (r, u) and (w, s), the shared vertex has degree $u + w$ and the outer degrees are $r + 1, s + 1$. Hence the update is

$$T_p(r+1, s+1) \leftarrow \max\{T_p(r+1, s+1), T_a(r, u)+T_{p-a+1}(w, s)+(u+w)^2\}, \qquad u+w \le C. \tag{55}$$

This is an exact recurrence. Every triangulation has exactly that incident triangle and two children. Conversely every child pair glues to a triangulation, with no additional cross edge. Every internal vertex is scored when it first becomes the shared vertex, and the two external endpoints remain unscored. Endpoint degrees are retained jointly, since their separate maxima are not a legal interface. Translation invariance permits storing sizes, rather than all translated copies of a polygon interval.

B.2. **The unmasked residual optimization.** Choose an actual maximum-degree hub v of degree d. Its neighbor path has d vertices and $d - 1$ gap positions. A zero-mass gap has state $(m, r, s, t) = (0, 1, 1, 0)$. A positive-mass gap uses (m, r, s, t) with $t = T_{m+2}(r, s)$. It adds $r - 1, s - 1$ to the degrees already present in the residual path and adds t to the score of its internal vertices. The completed residual degree of every neighbor is at most $d - 1$, because its omitted edge to v restores its actual degree. Gap-interior vertices retain cap d.

For each declared mass bound K, first build all polygon tables for $2 \le p \le K + 2$ with the chosen interior degree cap. Then run exactly the $d - 1$ gap positions, using every feasible (r, s) and $0 \le m \le K - M$ at each retained mass M. For clarity the following is full pseudocode for a layer of the residual dynamic program; maxima over missing entries remain absent. The index i counts the current gap, starting with $i = 1$.

1. Initialize $D_0(0, 0, 0) = 0$. Its state is (M, L, z): used gap mass, number of nonempty gaps, and the pending degree increment at the left endpoint of the next gap.
2. At gap i, for every retained state and every polygon choice (m, r, s, t), put $b_i = 1$ if $i = 1$ and $b_i = 2$ otherwise. The degree of the now completed path vertex is $b_i + z + r - 1$. Reject it if it exceeds $d - 1$. Update the state

$$(M', L', z') = (M + m, L + \mathbf{1}_{m>0}, s - 1)$$

with score $D_{i-1}(M,L,z)+t+(b_i+z+r-1)^2$. Retain the largest score at each identical new state.

3. Reject only $M'>K$ or $L'>L_{\max}$, the declared all-mass and all-gap upper bounds. After the last gap add $(1+z')^2$ for the final path endpoint, rejecting degree greater than $d-1$. Maximize over its pending increment for each final (M',L').

The future feasible choices depend exactly on the layer and retained state. Two histories with the same state have the same possible continuations and additive future scores, so discarding the smaller score is valid. Histories are identified only for this optimization; this does not identify their graphs, their individual vertex degrees, or root contacts for other operations. Mass and gap count are nondecreasing, so the all-size caps cannot remove a history needed by any retained final target. The zero-mass result is the path square-degree sum $4d-6$, an immediate independent calibration.

The precise positive target list contains 328 distinct tuples (n,d,k,ℓ), with $n=d+k+1$, arising as the tuples not paid by the preceding finite rules. There are ninety (d,k) pairs and thirteen degrees $28\le d\le 40$, at orders $56\le n\le 71$. The supplement regenerates the list from the original complete finite index, rather than accepting only a supplied list of favorable rows. For each d choose K and $L_{\max}$ as the largest required mass and gap count. At each target the resulting exact maximum $V=M_2(R)$ is inserted in (4.2) and (10). Every exact rational payment must pass. A hash or a match to an old maximum is a consistency check; the recurrence, full domain, and payment are the mathematical proof.

B.3. **Exact candidate masks and nonhub caps.** Suppose every nonhub degree is at most $C\le d$. Use cap C for polygon interiors in (55) and cap $C-1$ for completed residual neighbors of v. There is one candidate dominating triangle through v for each gap index $j=1,\dots,d-1$, namely $\{v,x_j,x_{j+1}\}$. Retain a bit mask of the candidates that still dominate all interiors processed so far. Initially all bits are present.

For a positive gap i of mass m and root degrees r,s, its interior vertices see no other vertex of the neighbor path. Thus the only possible surviving bits and their exact conditions are

(56)

j	i-gap coverage condition
$i-1$	$r=m+1$
i	$r+s=m+3$
$i+1$	$s=m+1.$

Omit indices outside $1,\dots,d-1$. The first condition says that every interior vertex sees the left root; the third is its reflection. For the middle condition, there is exactly one common neighbor of the boundary roots. Hence the number of distinct interior vertices seen by at least one root is $r+s-3$, proving the claimed equivalence. An empty gap imposes no condition. Intersect the old mask with (56) at every positive gap. The final mask is therefore *exactly* the set of triangles through v that dominate the whole graph: all path vertices already see v.

Add this full mask to the state (M,L,z). Maxima may be merged only for identical masks. A single Boolean indicating whether the current mask is empty would lose which candidates a later gap could remove, and is not used. The same induction and future-state argument proves exactness of the masked recurrence.

The full target set consists of 125 capped tuples (n,d,C,k,ℓ) with $C<d$, plus five unrestricted tuples with $C=d$, in twenty-nine (d,C) groups. For each group compute all masses and gap counts up to its declared maxima, and read the zero-mask maximum at every target. The source target domains, cap, order equation and uniqueness are checked. All 130 targets are nonvacuous in the completed computation; an absent state would have to be proved absent by the full recurrence before declaring its conditional class empty. Insert each maximum in the same residual moment and hub formulas. A nonzero mask is paid by

lemma 4.2; for $C < d$, an escaping nonhub degree is paid by the complete pair partition in Section 4.

The order-sixty target needs one domain distinction. Its historical upper bound 1216 applied to the broader class with no adjacent shared-fan pair. Zero mask is a subclass. A fresh maximum at most 1216 is sufficient; equality is not logically required. The complete recurrence in fact gives 1216. For every other target, the stored and recomputed domains agree. This distinction prevents a stricter-domain improvement from being misreported as a counterexample to an older maximum.

The independent small calibration enumerates all 625 polygon triangulations of orders three through nine and every actual maximum-degree root. It compares residual degrees, every candidate's actual closed-neighborhood union, and all nonhub caps from two to d. The masked diagnostic has 896 root choices, 1627 root/cap presentations, 144 parameter/mask maxima and six empty domains. These checks diagnose an implementation mistake in the recurrence; they do not extrapolate an all-size statement from finite samples.

B.4. **Literal exceptional high-pair graph constructors.** We record the constructors used for the finite Gram part so that the data can be decoded without a graph library or a database. All edge collections are sets of unordered pairs. Every constructor is additionally checked to have a simple outer Hamiltonian boundary, no crossing diagonals in that boundary order, the declared hub degrees and exactly $2n - 3$ edges.

For an internal (d, e) core put $p = d - 3$, $q = e - 1$, hub $u = 0$, hub $v = p + 1$, and $f = p + 2$. For $0 \le a \le p$, $1 \le b < q$, take the stars

$$0j \ (1 \le j \le p), \qquad vj \ (f \le j < f + q), \qquad 0v, \ 0f, \ 0(f + b).$$

Add consecutive path edges in the four intervals $[1, a]$, $[a+1, p]$, $[f, f+b-1]$, $[f+b, f+q-1]$; an interval of size zero or one has no such edge. Add $1f$ if $a > 0$, and $(a+1)(f+b)$ if $a < p$. This graph has $d + e - 2$ vertices. The two path splits exhaust the private segments on the two sides of the internal edge, by the neighbor-path and induced-core counting argument.

For a boundary core put $p = d - 2$, $q = e - 1$, $v = p + 1$, $c = p + 2$. Take stars $0j$ for $1 \le j \le p$ and vj for $c \le j < c + q$, consecutive edges in both intervals, and edges $0v, 0c, pc$. It has $d + e - 1$ vertices. Reversing a private path does not create a second boundary-core type.

For the nonadjacent $(33, 32)$ two-common-neighbor core, use vertices $0, \dots, 64$, with hubs $0, 34$ and common roots $32, 33$. Take $0j$ for $1 \le j \le 33$, $34j$ for $35 \le j \le 64$, and $32\,34$, $33\,34$, $32\,33$. For $0 \le a \le 31$ and $0 \le b \le 30$, add consecutive edges inside intervals of lengths $a, 31 - a, b, 30 - b$ starting at $1, a + 1, 35, 35 + b$, respectively. Join their endpoints to the common roots in this order: the last vertex of the first interval to 32, the first of the second to 33, the last of the third to 33, and the first of the fourth to 32, omitting an empty interval. The opposite orientations reflect that the two hub triangles lie on opposite sides of the shared edge.

For each scale-s Gram factor the verifier may calculate $\left\|YY^{\mathsf{T}}\right\|_F^2$ as $\left\|Y^{\mathsf{T}}Y\right\|_F^2$, an exact trace identity. If fixed-width matrix multiplication is used, it first checks

$$\max(n, r) \max_{i,j} |Y_{ij}|^2 < 2^{63},$$

so every entry product and sum fits in signed sixty-four-bit arithmetic. All final squares and score sums use unbounded integers. The ear-row update has its own product bound before multiplication. A missing certificate, invalid factor, absent domain member, extra duplicate, nonpositive margin, or unproved overflow bound must reject the entire corresponding finite assertion.

Appendix C. Graphs with small dominating sets

This appendix proves Theorem 4.3. All graphs and all adjacency matrices in it are finite, simple and unweighted. A path forest includes isolated vertices. If it has $q > 0$ vertices and p components, write

$$J(L) = K_1 \vee L, \qquad m = q - p.$$

The graph $J(L)$ need not itself be maximal outerplanar. Its role is an *induced vertex block* of the actual maximal outerplanar graph. Consequently every use of additivity below is orthogonal pinching, which follows from the positive-semidefinite variational principle. The proof never requires monotonicity of $\mathcal{F}$ under adding edges.

C.1. **Spectral estimates for the induced blocks.** For any real symmetric matrix A,

$$\mathcal{F}(A) = \max_{X \succeq 0} \{2\,\mathrm{tr}(AX) - \mathrm{tr}(X^2)\}. \tag{57}$$

Indeed, in an eigenbasis of A, the expression is $\mathrm{tr}(A^2) - \|A - X\|_F^2$, whose maximum is attained at the positive part A_+. Restricting X to orthogonal blocks proves $\mathcal{F}(A) \geq \sum_i \mathcal{F}(A|_{W_i})$ for mutually orthogonal subspaces W_i. Here $A|_W$ means orthogonal compression to W, not restriction to an invariant subspace.

Lemma C.1 (Residual loss)**.** *Let B be symmetric, u a unit vector, $\alpha = u^t B u$, and $\sigma^2 = \|Bu\|^2 - \alpha^2$. If $R = B|_{u^\perp}$, then*

$$\mathcal{F}(R) \geq \mathcal{F}(B) - \alpha^2 - 2\sigma^2.$$

Proof. Cauchy interlacing bounds the sum of squared negative eigenvalues of R by that of B. In the decomposition $\mathbb{R}u \oplus u^\perp$,

$$\mathrm{tr}(R^2) = \mathrm{tr}(B^2) - 2\,\|Bu\|^2 + \alpha^2.$$

Subtract the negative-square-energy inequality from this identity. □

For a path on ℓ vertices the squared spectral radius is $4\cos^2(\pi/(\ell+1))$. The following rational upper bounds will be used:

$$C(1) = 0, \quad C(2) = 1, \quad C(3) = 2, \quad C(4) = 8/3, \quad C(5) = 3,$$
$$C(\ell) = 4 - \frac{16}{(\ell+1)^2} \quad (\ell \geq 6). \tag{58}$$

The last inequality follows from $\sin x \geq 2x/\pi$ for $0 \leq x \leq \pi/2$; the cases 1 through 5 follow directly from the displayed exact radius formula. If $m > 0$ and s components of L are nontrivial, then

$$1 \leq s \leq \min(p, m), \qquad \ell_{\max} \leq m - s + 2, \qquad D_2 := \sum_{v \in L} d_L(v)^2 = 4m - 2s. \tag{59}$$

Each nontrivial component has two degree-one endpoints and all its other vertices have degree two, proving these identities.

Lemma C.2 (Uniform compression)**.** *For $q \geq p \geq 1$ and $m = q - p > 0$, put $\ell = m - s + 2$ and*

$$U(q,p,s) = q + m - \min\left\{C(\ell), \frac{8m - 4s}{q} - \frac{4m^2}{q^2}\right\} + \frac{2m^2}{q^2} + \frac{2m}{q^2}\sqrt{q^3 + m^2}.$$

Then

$$\mathcal{F}(J(L)) \geq \max\left\{q, \min_{1 \leq s \leq \min(p,m)} U(q,p,s)\right\}. \tag{60}$$

For $m = 0$, $\mathcal{F}(J(L)) = q$.

Proof. Let $u = q^{-1/2}\mathbf{1}$ on the leaves. Compress $J(L)$ to its hub and u. This gives the matrix $\left(\begin{smallmatrix} 0 & \sqrt{q} \\ \sqrt{q} & 2m/q \end{smallmatrix}\right)$, whose positive square energy is

$$q + \frac{2m^2}{q^2} + \frac{2m}{q^2}\sqrt{q^3 + m^2}.$$

The leaf adjacency is bipartite and has positive square energy m. Its compression to $u^\perp$ has energy at least $m - 2D_2/q + 4m^2/q^2$ by Lemma C.1. Positive-eigenvalue interlacing gives the alternative lower bound $m - \rho(L)^2 \geq m - C(\ell)$. Pinch the full matrix into this residual space and the two-dimensional compression. Their sum gives U. Taking the minimum over all possible s removes dependence on shape. For the alternative q, the Rayleigh vector with hub coordinate $\sqrt{q}$ and all leaf coordinates 1 gives $\rho(J(L)) \geq \sqrt{q} + m/q \geq \sqrt{q}$. For a star the eigenvalues are $\sqrt{q}, -\sqrt{q}$, and zeros. □

We require two stronger uniform charges:

$$\mathcal{F}(J(L)) > 2q - \frac{11p}{7} - \frac{7}{4} + 2\sqrt{q}, \qquad (q \geq 0), \tag{61}$$

$$\mathcal{F}(J(L)) \geq 2q - p + 2\sqrt{q} - 2\sqrt{p} - \frac{q-p}{4q}, \qquad (q \geq p \geq 1). \tag{62}$$

For $q = 0$ set $p = 0$ in (61). Equality in (62) is allowed for stars; the inequality is strict for every nonstar. Here is a proof with its finite part stated as explicit exact arithmetic obligations.

Besides (60), two shape-independent certificates suffice for the finite part. For rational $t > 0$ and $R > 0$, take leaf weights $1 + td_L(v)$ and hub weight $b = (q + 2tm)/R$. Their squared norm is

$$b^2 + q + 4tm + t^2 D_2.$$

Their adjacency quadratic form is at least

$$2b(q + 2tm) + 2m + 2tD_2 + 2t^2 D_3, \qquad D_3 = 4m - 4s + \max(0, 2s - m).$$

To check the last bound, consider the sum

$$\sum_{xy \in E(L)} d_L(x) d_L(y).$$

A nontrivial path with e edges contributes $4e - 4$, except that a one-edge path contributes 1 instead of 0. At least $\max(0, 2s - m)$ paths have one edge. If the quadratic form is greater than R times the squared norm, the Rayleigh principle and leaf interlacing give

$$\mathcal{F}(J(L)) > R^2 + m - C(\ell). \tag{63}$$

For a second certificate put $N = m + s$, $n_0 = p - s$, and $a = 2m/N$. Compress to the hub, the uniform vector on the N nonisolated leaves, and, if $n_0 > 0$, the uniform vector on isolated leaves. The matrix is

$$\begin{pmatrix} 0 & \sqrt{N} & \sqrt{n_0} \\ \sqrt{N} & a & 0 \\ \sqrt{n_0} & 0 & 0 \end{pmatrix}.$$

Its negative eigenvalue has magnitude $r > 0$, the unique positive root of $g(r) = r^3 + ar^2 - qr - an_0$; uniqueness follows from the single sign change of its coefficients. Thus $g(R) > 0$ implies $r < R$. The leaf residual bound, now using the uniform vector on N leaves, yields

$$\mathcal{F}(J(L)) > 2q + a^2 - R^2 + m - \min\{C(\ell), 2D_2/N - a^2\}. \tag{64}$$

When $n_0 = 0$, omit its zero direction; the same formula holds.

Finally, if a particular forest is specified by its sorted component orders, form its exact $(q+1)$-vertex unit adjacency matrix A. An integer rectangular matrix Z and positive integer D certify

$$\mathcal{F}(A) \geq \frac{2D^2 \operatorname{tr}(AZZ^t) - \operatorname{tr}((ZZ^t)^2)}{D^4}, \tag{65}$$

since ZZ^t/D^2 is positive semidefinite. Every matrix entry is used; there is no numerical eigenvalue premise in this test.

For clarity, the finite obligations establishing (61) and (62) are the following. Loop through every q, p, s with $0 \leq q \leq 99$, $p = 0$ if $q = 0$, $1 \leq p \leq q$ otherwise, and $s = 0$ if $m = 0$, $1 \leq s \leq \min(p, m)$ otherwise. This gives 82 175 shapes for (61); omit $q = 0$ for the 82 174 shapes of (62). Stars are exact; $(q, p) = (2, 1)$ is a triangle of energy 4. For every other shape, certify the desired strict inequality by (60), (63), (64), or (65). The rational trial values for t are

$$\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{16}, \tfrac{1}{32}, 1, 2, \tfrac{1}{64}, \tfrac{3}{4}, \tfrac{3}{2}.$$

All square roots are enclosed by integer square roots at scale 10^6: for $x \geq 0$ rational, use $\lfloor 10^6\sqrt{x} \rfloor / 10^6$ below and either that exact value, if its square equals x, or that value plus 10^{-6} above. Positive target radicals are enclosed above and negative target radicals below; energy radicals are enclosed below.

The executable obligations are `verify_tighter_split_fan_charge_190.py`, `verify_tighter_charge_small_190.py`, `verify_nonlinear_split_fan_190.py`, and `verify_nonlinear_small_190.py` with their accompanying `*-190-small-certificate.json` and `*-190-initial-screen.json` files. The first and third readers recompute all shape bounds. Their sets of residual (q, p, s) shapes must equal exactly the corresponding initial-screen failure sets; this equality is part of the coverage obligation, not an assumption about a stored status word. The second and fourth readers enumerate all ordered positive compositions $(e_1, \ldots, e_s)$ of m, convert each into the sorted forest orders $(e_1 + 1, \ldots, e_s + 1, 1, \ldots, 1)$, and require their set to equal the set of certified matrices, with no duplicates. There are 21 residual shapes and 22 actual forests for (61), and 62 residual shapes and 62 forests for (62). Each Gram score is evaluated by (65) in integers, and a zeroed factor must be rejected. These are exhaustive local parameter certificates, not samples of graphs.

To complete the infinite range, (60) gives, for $q \geq 100$,

$$\mathcal{F}(J(L)) \geq 2q - p - 4 + \frac{2(q-p)}{\sqrt{q}} + \frac{2(q-p)^2}{q^2}.$$

Its excess over the right side of (61) is at least

$$p(4/7 - 2/\sqrt{q}) - 9/4 + 2(1 - p/q)^2.$$

The coefficient of p is at least $13/35$. For $p \geq 7$ the first two terms alone are at least $7/20$. For $1 \leq p \leq 6$, the lower bounds $13p/35 - 9/4 + 2(1 - p/100)^2$ have minimum $2857/35000 > 0$. This proves (61) at every order.

For (62) stars are exact. Otherwise put $t = m/q > 0$ and $u = \sqrt{1 - t}$. Keeping the better loss term in (60) gives the following lower bound for the excess over (62):

$$-\min(4, 8t - 4t^2) + 2t^2 + t/4 + 2\sqrt{q}(u - u^2). \tag{66}$$

If $t \leq 1/2$, then $u > 7/10$ and $u/(1 + u) > 7/17$, so (66) is greater than $33t/68 + 6t^2$. If $t \geq 1/2$ and $p \geq 34$, its polynomial part is at least $-27/8$ and its radical part is at least $2\sqrt{34}(1 - 1/\sqrt{2}) > 27/8 + 4/625$; the last strict inequality follows from $\sqrt{34} > 583/100$ and $1/\sqrt{2} < 71/100$. For $1 \leq p \leq 33$, all its terms increase with q at fixed p, so at $q \geq 100$ it is at least

$$-4 + 2(1 - p/100)^2 + (1 - p/100)/4 + 2\sqrt{p} - p/5.$$

For completeness these 33 finite constants may be checked without floating arithmetic by substituting $\lfloor 10^6\sqrt{p}\rfloor/10^6$; the minimum of the resulting rational bounds is $77/10000$ at $p=1$. All three ranges are strict, proving (62).

C.2. **A normalized minimum dominating set.** Fix a minimum dominating set H, first maximizing $e(G[H])$, then maximizing $\sum_{v\in H} d_G(v)$. All uses below have $n \geq 72$. Every selected vertex has degree at least four. Indeed, a degree-two vertex has its closed neighborhood contained in that of either neighbor. Replacing it by such a neighbor preserves domination and all its adjacencies within H, and either increases selected edges or increases the degree sum. The neighbor is not already selected, as otherwise the old vertex was redundant, and has degree at least three when $n \geq 4$. If a selected degree-three vertex has neighbor path x, y, z, its closed neighborhood is contained in $N[y]$. The middle neighbor y has degree at least four for $n \geq 5$: if both degrees were three, the four outer edges of their two incident triangles would already form the whole outer boundary, forcing $n = 4$. The same replacement argument applies.

Each vertex of a minimum dominating set has a private closed neighbor, meaning a vertex dominated by it and by no other member of H. A nonisolated selected vertex cannot be its own private neighbor, and therefore has a private neighbor outside H. If an isolated selected vertex i had no such neighbor, choose any neighbor w; all its neighbors are shared with another selected vertex. Replacing i by w preserves domination and adds a selected edge, a contradiction. If its only outside private neighbor were x, choose a vertex w adjacent to x along the induced path $G[N(i)]$. This w is shared and is outside H, because i is isolated in $G[H]$. Replacing i by w still dominates i, x and all vertices formerly dominated only by i, while adding a selected edge. This too is impossible. Consequently, writing k_i for the number of outside private neighbors,

$$k_i \geq 1, \qquad k_i \geq 2 \text{ if } i \text{ is isolated in } G[H]. \tag{67}$$

Put $h = |H|$, $b = c(G[H])$, $e = e(G[H])$, and let t count its triangles. An induced subgraph of a maximal outerplanar graph is chordal and outerplanar. In each of its nontrivial blocks all bounded faces are triangles. Euler's formula, including isolated vertices, therefore gives

$$e = h - b + t. \tag{68}$$

Every vertex $v \notin H$ has the nonempty type $T = N(v) \cap H$. The graph $G[H[T]]$ is a path forest because it lies in $G[N(v)]$, which is an induced path. In addition, its intersection with each component of $G[H]$ that it meets is connected. Otherwise choose a shortest path inside that selected component between two separated neighbor portions, shortening between consecutive occurrences of neighbors of v. Together with v this is an induced cycle of length at least four, contrary to chordality. Write $r_T = c(G[H[T]])$, and set $\sigma = \sum_{v\notin H}(r_T - 1)$.

Lemma C.3 (The component and path budgets)**.** *For $h \geq 2$,*

$$\sigma \leq 2b - 2, \qquad \sum_{v\notin H} (|T| - 1) \leq 2h - 2 - t. \tag{69}$$

There is an assignment of every outside vertex to one of its selected neighbors for which its assigned leaf sets induce path forests with $q_i \geq p_i \geq 1$ and

$$\sum_i q_i = n - h, \qquad P := \sum_i p_i \leq 3h - 2 - t. \tag{70}$$

The assignment can be any one satisfying the following rule: if $G[H[T]]$ has an isolated vertex, choose such a vertex as owner; otherwise choose any member of T.

Proof. Contract each connected selected component, delete outside–outside edges, and merge parallel edges. The resulting simple bipartite outerplanar graph has b marked selected-component vertices, and one vertex for each original outside vertex. Its degree

at the latter vertex is r_T, by the preceding connectedness fact. Let κ be its number of components. Every component contains a marked vertex. If f denotes its number of bounded faces, Euler's formula says $\sigma = b - \kappa + f$. For a two-connected bipartite outerplane block with B marked vertices the outer cycle has $2B$ vertices. All bounded faces have length at least four. Counting face incidences and applying Euler gives at most $B - 1$ bounded faces. A bridge gives the same bound with $B = 1$. Summing over blocks gives $f \le b - \kappa$: repetition of a marked cut vertex is offset by the block count, while repetition of an outside cut vertex can only reduce this upper bound. More explicitly, in each connected component $\sum_{\text{blocks}}(B-1) = b_{\text{comp}} - 1 - c_O$, where $c_O \ge 0$ is the sum over its outside cut vertices of their block multiplicity minus one. Hence $\sigma \le 2b - 2\kappa \le 2b - 2$.

Let t_2 count triangular faces with exactly two selected vertices and let e_∂ count selected edges on the outer boundary. Every triangle is a face in a maximal outerplane embedding. Counting faces at selected edges gives $t_2 = 2e - e_\partial - 3t$. On the other hand $\sum_v |E(H[T])| = t_2$ and $|T| - 1 = |E(H[T])| + r_T - 1$. Thus

$$\sum_v (|T| - 1) = t_2 + \sigma \le 2e - 3t + 2b - 2 = 2h - 2 - t.$$

For each selected i let $a_i = d_{G[H]}(i)$ and let τ_i count selected triangles containing i. Removing selected neighbors from the neighbor path of i leaves at most $\phi_i = 1 + a_i - \tau_i$ outside intervals: the removed neighbor set has a_i vertices and τ_i edges and hence $a_i - \tau_i$ runs. Its sum is $\sum_i \phi_i = h + 2e - 3t = 3h - 2b - t$.

An outside vertex v whose occurrence at i has an adjacent selected neighbor in $H[T]$ is an endpoint of one of these original outside intervals. Indeed, the edge between those two selected vertices and v is a triangular face. Removing this occurrence cannot split an outside interval. Only isolated vertices of $H[T]$ can give interior cuts. Under the stated priority rule the number of lost interior occurrences is at most $r_T - 1$: if there are singleton components one is retained by its owner, and if there are none no interior cut is possible. Simultaneous removal of several vertices increases the component count by at most the number of their original interior occurrences. Therefore $P \le \sum_i \phi_i + \sigma \le 3h - 2 - t$. Every assigned leaf graph is an induced subgraph of a neighbor path. Private vertices are necessarily retained by their owner, so (67) gives $q_i \ge p_i \ge 1$. The assigned blocks partition $V(G)$, proving the leaf-count identity. □

C.3. **Domination numbers one, two, and three.** Orders 8 through 71 are covered by Theorem 4.1; assume $n \ge 72$. A dominating vertex makes G a hub joined to its full neighbor path. Formula (61) with $q = n - 1, p = 1$ immediately implies $\mathcal{F}(G) > 2n$.

For two dominating vertices u, v, partition the vertices into $S = \{u\} \cup (N(u) \setminus \{v\})$ and $T = V(G) \setminus S$. The induced blocks are hubs joined to at most two paths, with leaf counts $q_S + q_T = n - 2$; zero leaves are allowed. For S, delete v from the neighbor path of u if it occurs. For T, remove u and the common neighbors from the neighbor path of v. If uv is an edge, the removed vertices are a contiguous run at u. If uv is not an edge, there are at most two common neighbors, since three would give a $K_{2,3}$ subgraph; when there are two they are adjacent by chordality of their four-cycle with u, v. They again form a contiguous run. The remaining graph has at most two paths in every case.

For any hub joined to at most two paths on q leaves,

$$\mathcal{F}(J(L)) \ge 2q - 2 \quad (q \ge 0), \qquad \mathcal{F}(J(L)) > 2q + 6 \quad (q \ge 30). \tag{71}$$

Here are details that avoid a finite graph input. Cases $q = 0, 1, 2$ are immediate from the empty graph, edge, two-edge path or triangle. For $q = 3, 4, 5$, the vector with hub coordinate 2 and all leaf coordinates 1 has Rayleigh quotient at least $(6q - 4)/(q + 4)$. Its squares are respectively $4, 25/4, 676/81$, each at least $2q - 2$. For $q \ge 6$, put $h_0 = 1 - 2/q$ and use $m \ge q - 2$ in the uniform compression and the strict path radius bound $\rho(L)^2 < 4$.

This gives

$$\mathcal{F}(J(L)) > 2q + 2h_0^2 + 2h_0\sqrt{q + h_0^2} - 6.$$

The sum $2h_0^2 + 2h_0\sqrt{q + h_0^2}$ increases with q and at $q = 6$ equals $(8 + 4\sqrt{58})/9 > 4$, proving the first inequality. For the second, the positive eigenvalue of the hub/uniform compression solves $z^2 - cz - q = 0$ with $c = 2m/q \geq 2 - 4/q$. At $x = \sqrt{q + 12}$ the polynomial is negative, since $(2 - 4/q)^2(q + 12) > 144$ for $q \geq 30$ (already strict at 30, and increasing). Thus $\rho(J(L))^2 > q + 12$. Leaf interlacing adds at least $m - \rho(L)^2 > q - 6$, proving the claim. At $n \geq 72$ at least one block has at least 30 leaves, so pinching and (71) give $\mathcal{F}(G) > 2(q_S + q_T) + 4 = 2n$.

For $h = 3$, Lemma C.3 gives $P \leq 7$ and $\sum q_i = n - 3$. At $72 \leq n \leq 104$ the following independent integer enumeration proves the required scalar inequality. Precompute (60) for $1 \leq q \leq 99$, $1 \leq p \leq \min(q, 5)$, enclosing its positive radical below at scale 10^5, and round the resulting maximum with q down at scale 10^{10}; call the integer value $E_3(q, p)$. For each $n = 72, \ldots, 104$ enumerate

$$p_i \geq 1, \quad \sum_i p_i \leq 7, \quad q_i \geq p_i, \quad \sum_i q_i = n - 3.$$

Check $\sum_i E_3(q_i, p_i) > 2n\,10^{10}$. The file `extraction_gamma3_verify.py` performs this loop using only integer arithmetic and rational fractions. Its independent count for order n is

$$\sum_{r=3}^{7} \binom{r-1}{2} \binom{n-1-r}{2},$$

obtained by first choosing the three positive p_i of sum r and then the three nonnegative $q_i - p_i$ of sum $n - 3 - r$. The total is 3 795 253. The smallest certified excess is $200606059/5000000000 > 0$, at $n = 72$, $q = (1, 1, 67)$, $p = (1, 1, 5)$. This is complete coverage of all allowed allocations.

For all larger orders sum (61). Positive q_i with fixed sum Q satisfy $\sum_i \sqrt{q_i} \geq (h - 1) + \sqrt{Q - h + 1}$: repeatedly move mass from a smaller coordinate greater than one to a larger coordinate; concavity of the square root can only decrease the sum. With $h = 3$ and $P \leq 7$ this gives

$$\mathcal{F}(G) - 2n > 2\sqrt{n - 5} - 73/4 > 0 \qquad (n \geq 105).$$

This completes every case with domination number at most three.

C.4. **The four-hub structural catalogue.** Assume now $\gamma(G) = 4$ and use the normalized set H above. The induced graph on four labeled vertices is chordal, outerplanar, and K_4-free. Enumerating all 64 edge subsets and all 24 permutations gives exactly nine isomorphism types. For each outside vertex the shared type is a subset $T \subseteq H$ of size at least two; private vertices have the counts k_i already defined. The catalogue stores shared-type multisets, not individual graphs. Lemma C.3 says

$$\sum_{\text{shared}} (|T| - 1) \leq 6 - t, \qquad \sum_{\text{shared}} (r_T - 1) \leq 2b - 2, \qquad P \leq 10 - t. \tag{72}$$

In particular there are at most six shared vertices.

For any selected pair i, j the number of outside common neighbors is at most $2 - |N_H(i) \cap N_H(j)|$, because a $K_{2,3}$ is forbidden. If ij is an edge there must be at least $\max(0, 1 - |N_H(i) \cap N_H(j)|)$ outside common neighbors: every edge belongs to at least one triangular face. If ij is not an edge and has two outside common neighbors, the latter must be adjacent, by chordality of the four-cycle. Retain these forced outside–outside edges in the partial core and require that core to be outerplanar. Do not make this last inference for an adjacent selected pair: its two triangular faces form a diamond and their opposite vertices need not be adjacent.

For each shared vertex choose an owner obeying the priority rule in Lemma C.3. Every actual graph realizes every such choice. Let s_i count shared vertices owned by i, X_i count shared neighbors of i owned elsewhere, and

$$J_i = \sum_{\substack{v \text{ shared} \\ \text{owner}(v)=i}} d_{H[T(v)]}(i), \qquad r_i = a_i - \tau_i, \qquad q_i = k_i + s_i.$$

The following are necessary individual path-count bounds:

$$p_i \geq \max\{1, J_i - r_i + 1, J_i - s_i + 1, \lceil J_i/2 \rceil\}, \tag{73}$$

$$p_i \leq X_i + 1 - r_i + J_i, \tag{74}$$

$$p_i \leq 1 + r_i + c_i, \tag{75}$$

where c_i is the number of distinct owners in other selected components receiving a lost shared neighbor at which i is isolated in $H[T]$. To justify (73) and (74), retain in the neighbor path of i its selected neighbors and its assigned leaves. This induced forest has $r_i + p_i - J_i$ components, so its component count is between 1 and $X_i + 1$. Also $J_i \leq 2r_i$ and $J_i \leq 2s_i$, as all degrees in this retained path are at most two. Since at least one private vertex is present, the assigned forest has at most $k_i - 1$ private–private edges and at least $s_i - p_i + 1$ other edges. At most $2s_i - J_i$ incidences remain at its shared leaves, so $s_i - p_i + 1 \leq 2s_i - J_i$, giving $p_i \geq J_i - s_i + 1$. Each assigned leaf-path component is a consecutive interval in the full neighbor path and can contact selected neighbors only at its two ends. Thus the assigned forest has at most $2p_i$ selected-neighbor contacts, giving $J_i \leq 2p_i$. This proves the last lower bound as well.

For (75), begin with at most $1 + r_i$ outside intervals. A lost shared occurrence with a selected neighbor in $H[T]$ is an endpoint of an original interval. A lost occurrence with i isolated has, by connectedness within each selected component, its owner in another selected component. For any fixed such owner j, the common neighbors of i, j form either a singleton or an adjacent pair, since ij is absent and three common neighbors are forbidden. The assigned-away subset for that owner is therefore a single interval in the neighbor path. Removing it creates at most one additional interval. Counting distinct owners proves the bound.

Two further upper bounds retain the forced edges. The removed subset of $N(i)$ has $R_i = a_i + X_i$ vertices. If at least e_i^0 edges among them are known, it has at most $R_i - e_i^0$ runs, and

$$p_i \leq 1 + R_i - e_i^0. \tag{76}$$

Use for e_i^0 the τ_i selected–selected edges, every selected–lost-shared edge specified by a type, and every forced outside–outside edge whose endpoints are both lost. Each is an edge of the actual neighbor path; no unrecorded edge is assumed. If f_i forced outside–outside edges have both endpoints owned by i, the leaf forest has at least f_i edges, so also

$$p_i \leq q_i - f_i. \tag{77}$$

Finally $p_i \leq q_i$ and (72) give $p_i \leq 7 - t$. The degree and private requirements give

$$k_i \geq \max\{1 + \mathbf{1}_{a_i=0},\ 4 - a_i - \nu_i\}, \qquad \nu_i = \#\{v \text{ shared} : i \in T(v)\}. \tag{78}$$

Every domain restriction just proved is necessary. Conversely no claim is made that every admitted abstract profile has a realization. An overestimate of this domain is safe for exhaustive lower bounds.

C.5. **Coupling induced blocks with rational vectors.**

Lemma C.4 (A positive semidefinite common core)**.** *Partition the actual adjacency matrix A into induced vertex blocks B_i. In some of them choose nonnegative unit vectors u_i, with*

$\alpha_i = u_i^t B_i u_i \geq 0$ and $\sigma_i^2 = \|B_i u_i\|^2 - \alpha_i^2$. Let $K_{ij} = u_i^t A_{ij} u_j$ be their actual cross-contacts. Suppose $d_i \leq \alpha_i$, $0 \leq L_{ij} \leq K_{ij}$, $L_{ii} = 0$, and $\operatorname{diag}(d_i) + L \succeq 0$. Then

$$\mathcal{F}(A) \geq \sum_i \mathcal{F}(B_i) + 2\sum_{i<j} L_{ij}^2 - 2\sum_{i \text{ selected}} \sigma_i^2. \tag{79}$$

Several disjoint selected cores may be used, paying the variance of each selected block once.

Proof. The compression to the chosen vectors is $C = \operatorname{diag}(\alpha_i) + K$. The matrix $X = \operatorname{diag}(\alpha_i) + L$ is positive semidefinite because it is obtained from the assumed positive semidefinite matrix by adding a nonnegative diagonal. Evaluating (57) at X gives

$$\mathcal{F}(C) \geq \sum_i \alpha_i^2 + \|L\|_F^2 + 2\operatorname{tr}((K-L)L) \geq \sum_i \alpha_i^2 + \|L\|_F^2 .$$

The final inequality uses the entrywise nonnegativity hypotheses. Pinch the original space into this common span, the residual spaces inside each selected block, and the unselected whole blocks. Apply Lemma C.1 to the residuals; the α_i^2 cancel, yielding (79). The same argument with several orthogonal common spans proves the last assertion. □

Here are the exact rational quantities used in the finite bridge. Set $S = 256$ and $S_2 = S^2 = 65536$. For $1 \leq q \leq 155$, $1 \leq p \leq \min(q, 10)$, define an energy lower bound by taking the maximum of (60), (61), (62), and the small table below, using square-root lower or upper enclosures at scale 10^6 in the appropriate directions, and flooring at scale S_2. Write the resulting integer as $E(q,p)$. Thus $E(q,p)/S_2 \leq \mathcal{F}(J(L))$ for every forest with these q, p. Using a strict lower bound as a non-strict bound is harmless.

The small table has all 78 cells $1 \leq q \leq 12$, $1 \leq p_{\max} \leq q$. In a cell it certifies the energy of every forest on q vertices with *at most* $p_{\max}$ paths. Its files are `small-splitfans-243-certificate.json`, `small-splitfans-243-witnesses.npz`, and `verify_small_splitfans_243.py`. The reader reconstructs the exact unit matrix for each of the 271 sorted path forests, checks its full integer Gram score (65), and takes the minimum over all forest orders in each cell. A separate enumeration of all 4095 ordered positive compositions for $q = 1, \ldots, 12$ proves that this sorted list omits no forest. For a block with exactly p paths the at-most-p cell is a valid lower bound.

For a nonnegative rational trial vector use the integer coordinates

$$a = \left\lfloor S\sqrt{q^3 + m^2}/q \right\rfloor - \lfloor Sm/q \rfloor > 0,$$
$$v = (a, S, \ldots, S), \quad M = a^2 + qS^2, \quad T = 2aSq + 2S^2 m,$$
$$V_0 = q^2S^2 + qa^2 + 4aSm + S^2\bar{D}_2, \qquad \bar{D}_2 = \begin{cases} 4m-2, & m > 0, \\ 0, & m = 0. \end{cases}$$

The first floor is implemented as $\operatorname{isqrt}(\lfloor S^2(q^3+m^2)/q^2 \rfloor)$. In fact $a \geq S$: since $m \leq q-1$, $q^2 \geq q + 2m$ for integer $q \geq 1$, so $\sqrt{q^3+m^2} \geq q + m$ and the two quantities being floored differ by at least S. The normalized vector $u = v/\sqrt{M}$ has exact Rayleigh quotient T/M, exact hub coordinate $a/\sqrt{M}$, and common leaf coordinate $S/\sqrt{M}$. Formula (59) proves $\|Bv\|^2 \leq V_0$, so $\sigma^2 \leq V_0/M - T^2/M^2$. Define integers

$$D(q,p) = \lfloor ST/M \rfloor, \qquad Z(q,p) = \left\lfloor Sa/\sqrt{M} \right\rfloor,$$
$$W(q,p) = \left\lfloor S^2/\sqrt{M} \right\rfloor, \qquad V(q,p) = \left\lceil 2S^2(V_0/M - T^2/M^2) \right\rceil. \tag{80}$$

Coordinate floors use integer square roots of $\lfloor S^2a^2/M \rfloor$ and $\lfloor S^4/M \rfloor$. All four bounds hold for the same actual vector of a given block.

If a block has only a path interval $l \leq p \leq u$ known, define E, D, Z, W on (q, l, u) as their minima over the interval and V as its maximum. These are simultaneous bounds valid for

every actual p in the interval. It is unnecessary that all extrema be attained at one common p. In the exact-p fallback below, a single p is used for every quantity throughout.

For an ownership assignment let HH_{ij} be the indicator of the selected edge ij, HO_{ij} the number of shared vertices owned by j adjacent to hub i, and OO_{ij} the number of distinct forced outside edges crossing the assigned blocks i, j. In particular a forced edge inferred from more than one selected pair is counted only once. For the chosen interval quantities define the symmetric integer matrix

$$C_{ii} = D_i, \qquad C_{ij} = \left\lfloor \frac{HH_{ij}Z_iZ_j + HO_{ij}Z_iW_j + HO_{ji}Z_jW_i + OO_{ij}W_iW_j}{S} \right\rfloor. \tag{81}$$

These satisfy $C_{ii}/S \le \alpha_i$ and $0 \le C_{ij}/S \le K_{ij}$. Every recorded edge is an actual edge; unrecorded nonnegative contacts can only improve the variational evaluation used in Lemma C.4. If all principal minors of $C[U]$ are nonnegative, it is positive semidefinite and the following integer is a certified lower bound for $S_2\mathcal{F}(G)$:

$$\sum_i E_i + 2\sum_{i<j\in U} C_{ij}^2 - \sum_{i\in U} V_i. \tag{82}$$

The principal-minor criterion is for every principal minor, not only leading ones. Also consider each of the three pair matchings on four vertices. Each pair is used only when its 2 by 2 core is positive semidefinite and its bonus is positive. Two disjoint such bonuses may be added. Keeping no core gives the valid block bound $\sum E_i$.

C.6. **The whole-hub fallback.** We use the established connected-graph theorem

$$\mathcal{F}(R) \ge |V(R)| - 1 \qquad \text{for every connected graph } R, \tag{83}$$

which is Theorem 1.2 of Liu, Tang and Zhang [2]. This external input is a theorem about connected graphs and does not presuppose the maximal-outerplanar $2n$ conclusion proved here.

Let v have degree $d \ge 4$, put $k = n - d - 1$ and $h_0 = 1 - 1/d$. The graph $R = G - v$ is connected, since the outer cycle with v removed is a spanning path. Let u be uniform on $N(v)$ and zero elsewhere in R. Its Rayleigh quotient in R is $\alpha = 2h_0$, since the induced neighbor graph is a path with $d-1$ edges. Write $S_0 = \sum_{x\notin N[v]} |N(x)\cap N(v)|^2$. Then

$$\|Ru\|^2 = (4d - 6 + S_0)/d, \qquad S_0 \le 4k.$$

The equality counts squared neighbor-path degrees and squared nonneighbor contacts. The inequality follows because a nonadjacent pair v, x has at most two common neighbors, or it contains a $K_{2,3}$. Compressing G to v, u gives positive square energy $d + 2h_0^2 + 2h_0\sqrt{d + h_0^2}$. Lemma C.1 applied to R and u, followed by (83), gives

$$\begin{aligned} \mathcal{F}(G) - 2n &\ge 2h_0\sqrt{d + h_0^2} - 5 - k - 2S_0/d + 6/d^2 \\ &\ge 2h_0\sqrt{d + h_0^2} - 5 - k - 8k/d + 6/d^2. \end{aligned} \tag{84}$$

Thus a profile is paid whenever the last scalar expression has a nonnegative certified lower bound. For $72 \le n \le 159$ and $4 \le d < n$, the reader computes it with its square root enclosed below at scale 10^6. No hypothesis about any smaller MOP's $2n$ bound occurs in this calculation.

C.7. **Complete finite coverage for four hubs.** Here is a specification of the finite certificate sufficient to reproduce its mathematical meaning independently of the producer. The parameter domain is $72 \le n \le 159$, with all four selected vertices and every shared vertex explicitly retained. The following steps describe its complete generation and readback.

(1) Generate the 64 edge subsets on $\{0, 1, 2, 3\}$, discard K_4 and the chordless 4-cycle, and identify the rest under all 24 permutations. Sort canonical edge lists by edge count and then lexicographically. This produces the nine rows listed below.

(2) For a core, enumerate every count vector of shared types $T \subseteq \{0,1,2,3\}$ of size at least two. The bounds (72) make this finite. Require that $H[T]$ is a path forest connected within each selected component it meets; require the pair upper and edge lower bounds proved above. Insert the forced outside edges for nonadjacent selected pairs, remove repeated edges, and test outerplanarity of this partial graph. Represent a multiset by its sorted list of bit masks $\sum_{i\in T} 2^i$. Reduce it only by permutations preserving the selected core. For each raw list retain its canonical list and the permutation giving it. Require equality of the entire raw-to-orbit map, not merely equality of orbit counts.

The independent outerplane test first splits disconnected components and then cut vertices. A two-connected outerplanar graph has a Hamiltonian outer boundary. Enumerate all edge-respecting Hamiltonian cycles in such a block, fixing the first vertex and orientation to remove duplicate cycles. A cycle works precisely when no two nonboundary edges have alternating endpoints on it. Finding such a cycle gives an explicit outerplane embedding; if no cycle works the block is not outerplanar. This proves completeness of that test.

(3) Enumerate every priority-eligible owner assignment. Retain those with the necessary private-bearing path bounds (73)–(75) consistent. If none exists, the profile cannot be an actual normalized graph, since every actual priority assignment has just been proved to satisfy these bounds. There are 303 surviving orbit representatives. For each one fix the first eligible assignment in lexicographic order as a baseline for candidate generation. Each actual graph admits this choice; no search for the unknown original ownership is needed.

(4) Enumerate integer private counts satisfying (78) and

$$68 - s_{\mathrm{tot}} \le \sum_{i=0}^{3} k_i \le 155 - s_{\mathrm{tot}}, \qquad s_{\mathrm{tot}} = \#\{\text{shared vertices}\}.$$

For the fixed baseline assignment put $q_i = k_i + s_i$, and form its full individual path interval from (73)–(77), $p_i \le q_i$, and $p_i \le 7 - t$. An empty interval disproves realization for this assignment and hence for the profile. Otherwise compute

$$g_i(k_i) = E(q_i, l_i, u_i) - 2(q_i + 1)S_2.$$

If $\sum_i g_i(k_i) \ge 0$, pinching alone pays the target. The candidate list is defined to contain *every* remaining tuple with $\sum_i g_i(k_i) < 0$ and the displayed order domain.

To reconstruct it independently, make the two lists (k_0, k_1, g_0+g_1) and (k_2, k_3, g_2+g_3). Sort the second by its charge. For each first-list entry use binary search for all second-list charges strictly below the negative first charge and filter their combined private count by the displayed interval. Sort the resulting four-tuples, reject duplicates, and require entry-by-entry equality to the stored candidate array. This is a complete meet-in-the-middle enumeration of the defining inequalities. It does not assume the producer's recursive pruning was correct.

(5) For each candidate the certificate specifies a witness among the following alternatives. The chosen ownership is also stored.

- A baseline block payment, a selected positive semidefinite core of size two, three, or four, or one of the three pair matchings, using (82).
- An exact-p fallback. Enumerate every four-tuple in the individual path intervals with $\sum p_i \le 10 - t$. For each use that same tuple in energy, vector, variance, and cross-contact quantities. Maximize over the valid cores, pair matchings, and block payment, then take the minimum over *all* admissible p tuples. Require that minimum to be at least $2nS_2$. If no tuple exists, the selected ownership cannot occur, which alone precludes an actual profile.
- A whole-hub payment (84) with actual selected hub degree $d_i = k_i + a_i + \nu_i$.

- An explicit impossibility certificate: for every eligible assignment the exact-p domain is empty. This last alternative is allowed by the reader, but no unverified "unrealizable" flag is accepted in place of its enumeration.

For interval witnesses, an inconsistent path interval for the chosen ownership also disproves the profile; every actual graph would admit that ownership. Every numerically claimed lower bound must be at least $2nS_2$ and must be rederived, rather than accepted from its stored value.

The exact catalogue cardinalities are as follows. Edge lists use the fixed labels $0, 1, 2, 3$; they identify rows independently of shorthand names for their isomorphism types.

	$E(H)$	raw	orbits	admitted	candidates
0	$\varnothing$	552	47	47	819
1	01	301	99	99	12293
2	01, 02	121	69	64	59804
3	01, 23	146	31	31	16895
4	01, 02, 03	44	13	11	120956
5	01, 02, 12	104	26	17	1699
6	01, 02, 13	36	23	19	58578
7	01, 02, 03, 12	32	20	12	15091
8	01, 02, 03, 12, 13	16	7	3	634
	total	1352	335	303	286769

Here "raw" already imposes the necessary partial-core conditions; it is not a count of all actual graphs. The identity rule for the orbit reduction is relabeling of *parameter profiles* by a selected-core automorphism. It makes no assertion that the actual MOPs realizing a profile are mutually isomorphic.

The complete bridge is carried by `coupled-bridge-244-certificate.json` and `coupled-bridge-244-witnesses.npz`. The independent implementation is `bridge_structure_reader_244.py` together with `verify_coupled_bridge_244.py`; the latter rebuilds every energy/vector table using rational fractions and integer square roots and compares its exact shape, type, and entries to the archive. It reconstructs the nine cores, all shared-type multisets, all relabeling maps, and every complete candidate list, then verifies each witness as just specified. There are 249629 energy/coupling payments and 37140 whole-hub payments. The exact-p fallbacks require 1874 valid tuple checks. Their sum accounts for every one of the 286769 candidates; no unpaid candidate is discarded. All archive keys must be exactly those generated by the reader; extra or missing arrays are rejected. Array dimensions and integer types, candidate ordering, and absence of duplicates are checked.

All principal minors are evaluated by integer Laplace expansion. The signed 64-bit operations are guarded by $|C_{ij}| < 4096$ and $24 \cdot 4096^4 < 2^{63}$; scalar fractions and Gram arithmetic use unbounded integers. Thus machine overflow does not silently replace the stated arithmetic. The reader additionally rejects a missing, duplicated, or changed candidate and a zeroed variance table through the same equality checks used on the genuine objects. Algebraic controls reject a nonpositive-semidefinite core and an interface which is negative or exceeds the real contact. These checks are controls on the implementation, while Lemma C.4 is the all-order mathematical justification.

A further standalone script `verify_removed_runs_244.py` calibrates (76) against every triangulation of orders 4 through 9 and every dominating marked four-set and actual owner assignment. This finite check is not substituted for the neighbor-path proof. It also includes a genuine adjacent-pair diamond control, which rejects the erroneous rule that would force its opposite vertices to be adjacent.

Run the finite readers with Python 3 and assertions enabled, in a fresh writable copy of the companion files, using NumPy for the `.npz` archives. No SciPy, NetworkX, numerical

eigensolver, producer import, or original workspace module is required for the coupled bridge reader. The commands, in dependency order, are

```
python -B -X utf8 verify_tighter_charge_small_190.py
python -B -X utf8 verify_tighter_split_fan_charge_190.py
python -B -X utf8 verify_nonlinear_small_190.py
python -B -X utf8 verify_nonlinear_split_fan_190.py
python -B -X utf8 extraction_gamma3_verify.py
python -B -X utf8 verify_coupled_bridge_244.py --manifest
python -B -X utf8 verify_removed_runs_244.py
```

The aggregate reader must also enforce the exceptional-shape set equalities stated after (65); individual success words alone are insufficient. File hashes in the companion manifest bind the reviewed versions to these statements, while the mathematical content is the complete arithmetic readback, coverage equality, and the proofs above.

C.8. **The unbounded four-hub range and assembly.** For $n \geq 160$, use (61), (70), and the square-root concentration inequality already proved. With $h = 4$ and $P \leq 10 - t$,

$$\begin{aligned}\mathcal{F}(G) - 2n &> 2\sqrt{n-7} - 173/7 + 11t/7 \\ &\geq 2\sqrt{n-7} - 173/7 > 0.\end{aligned}$$

The last inequality holds because $7 + (173/14)^2 = 31301/196$ lies strictly between 159 and 160. This covers every order beyond the finite bridge.

Proof of Theorem 4.3. For orders $8 \leq n \leq 71$ apply Theorem 4.1. For $n \geq 72$ choose a normalized minimum dominating set. Its size is one, two, three, or four. The first three sizes were paid in the corresponding subsection without any four-hub assumption. For size four and $72 \leq n \leq 159$, the actual graph induces one of the catalogued profiles by the structural proofs above. Its fixed priority assignment either has a nonnegative block charge, or its private tuple appears in the complete candidate list. Every such candidate is paid by a verified block/core, exact-p, or whole-hub certificate. An inconsistency certificate cannot hold for an actual graph, since each of its eligible ownership choices has an actual path forest satisfying all derived bounds. At $n \geq 160$ the strict analytic bound just proved applies. Thus in every case $\mathcal{F}(G) \geq 2n$, proving the claimed domination subclass. In particular the argument does not assert that an arbitrary MOP has domination number at most four. □

Appendix D. Exact vertex-retirement cost

Let $\mathscr{C}$ be the class of finite simple chordal outerplanar graphs, allowing disconnected graphs and isolated vertices. Its nontrivial blocks are bridges or maximal outerplanar graphs. Indeed every two-connected outerplanar block has a convex polygon embedding; a bounded face of length at least four would be an induced cycle, since any chord between its vertices would lie inside that face. Conversely maximal outerplanar graphs and bridges are chordal, and gluing blocks at cut vertices does not introduce an induced cycle. The class $\mathscr{C}$ is hereditary under induced vertex deletion, since both chordality and outerplanarity are hereditary.

D.1. Sectors after actual induced deletions.

Lemma D.1 (Hereditary root-clique sector)**.** *Let $X \in \mathscr{C}$ and let Q consist of one vertex or two adjacent vertices. With $B = A_X[V(X) \setminus Q]$ and the actual root contacts T, put $F_Q(t) = T(tI + iB)^{-1}T^T$. For every $t > 0$,*

$$2\operatorname{Re} F_Q(t) \pm t \operatorname{Im} F_Q(t) \succeq 0.$$

For a single nonisolated root both scalar inequalities are strict.

Proof. Let S be the set of nonroot neighbors of Q. If Q is a single vertex, its neighbors in each incident maximal outerplanar block induce a vertex-link path, and an incident bridge contributes a singleton. Different incident blocks give disjoint paths. If Q is an edge, its own block gives a path on each nonempty chord side, as in Lemma 3.1; for an internal edge the two sides are disjoint and have no cross-edges. Other blocks incident with either root give additional disjoint link paths. A bridge root edge gives no central path. Thus the actual adjacency matrix $P = A_X[S]$ is a disjoint union of paths, including singleton components.

For any connected component W outside $Q \cup S$, its neighborhood $N(W) \cap S$ is a clique. Otherwise choose nonadjacent x, y in that neighborhood. A shortest x–y path with its internal vertices in W is induced. A shortest path from x to y through the root clique has length two or three and is also induced. No root vertex has a neighbor in W, and shortestness excludes the remaining cross-chords at x, y. The two paths therefore form an induced cycle of length at least four, contrary to chordality. A clique in the path forest on S has size one or two. A component with no contact in S contributes zero to the root response and may be omitted from this response computation.

Apply simultaneous induction on the order of X for both signs and both root-clique sizes. The induced graph on $W \cup N(W)$ belongs to $\mathscr{C}$, is rooted at its one- or two-vertex clique $N(W)$, and is smaller because it omits Q. Let F_W be its actual correction, and let L_W insert its root coordinates into S. Elimination of the disjoint far components gives exactly

$$Z = tI + iP + \sum_W L_W F_W L_W^T, \qquad F_Q = T_0 Z^{-1} T_0^T,$$

where T_0 consists of the actual root-to-S contacts. Every edge in P is counted once. Its real part satisfies $\operatorname{Re} Z \succeq tI$. The induction hypotheses give

$$\operatorname{Re}((2+it)Z) = t(2I - P) + \sum_W L_W (2 \operatorname{Re} F_W - t \operatorname{Im} F_W) L_W^T \succ 0,$$

$$\operatorname{Re}((2-it)Z) = t(2I + P) + \sum_W L_W (2 \operatorname{Re} F_W + t \operatorname{Im} F_W) L_W^T \succ 0.$$

The path quadratic-form identities in the proof of Lemma 3.1 apply to each path component. Inverse congruence and the real congruence by T_0 give the two claimed signs. Empty interiors give zero corrections and supply the induction bases. For a single nonisolated root, T_0 is a nonzero row, so the scalar congruences of these positive definite matrices are strictly positive. For two roots no rank-two assertion is needed; internal-edge contacts can have rank one. □

D.2. **The scalar envelope over every positive frequency.** Let v have actual degree $D \geq 1$ in $X \in \mathscr{C}$, and let a be its contact row with $X - v$, so $\|a\|^2 = D$. Put

$$f(t) = a(tI + iA_{X-v})^{-1} a^T = u(t) + iw(t).$$

The spectral theorem and Lemma D.1 give

$$0 < u(t) \leq D/t, \qquad |w(t)| < 2u(t)/t.$$

The scalar Schur determinant $t + f(t)$ has positive real part. Its continuous phase, normalized at infinity, is therefore the ordinary argument

$$\delta(t) = \arctan \frac{w(t)}{t + u(t)}.$$

Since $u \mapsto u/(t+u)$ increases on $u \geq 0$,

$$|\delta(t)| < \arctan \frac{2D}{t(t^2 + D)}. \tag{85}$$

The strictness follows from the strict one-root sector, including graphs with bridges or disconnected components. The equal-trace energy identity (3), now with one zero-diagonal private vertex, gives

$$s^+(X) - s^+(X - v) = D - \frac{2}{\pi}\int_0^\infty t\delta(t)\,dt.$$

The majorant in (85), multiplied by t, is integrable at both endpoints. Set

$$I(D) = \int_0^\infty t \arctan\frac{2D}{t(t^2 + D)}\,dt.$$

Then $s^+(X) - s^+(X - v) < D + 2I(D)/\pi$. Principal interlacing for both positive and negative square energies independently gives $0 \le s^+(X) - s^+(X - v) \le 2D$, since the total trace-square increase is $2D$. An isolated vertex has zero retirement cost exactly.

D.3. **Exact evaluation of the complete integral.**

Lemma D.2 (Exact all-degree retirement integral)**.** *For $D > 0$, let α be the unique real root greater than $\sqrt{D}$ of $\alpha^3 - D\alpha - 2D = 0$, and put $R(D) = \alpha^2$. Then*

$$I(D) = \frac{\pi}{2}(R(D) - D), \qquad R(D) > D, \qquad R(D)(R(D) - D)^2 = 4D^2.$$

Consequently, for every actual induced degree-D deletion in $\mathscr{C}$,

$$s^+(X) - s^+(X - v) < R(D).$$

Proof. Integration by parts has zero boundary term: at zero it is bounded by $\pi t^2/4$, and at infinity it is $O(1/t)$. Differentiating the arctangent gives

$$I(D) = \int_0^\infty \frac{Dt^2(3t^2 + D)}{(t^3 + Dt)^2 + 4D^2}\,dt.$$

The polynomial $x^3 - Dx - 2D$ is negative at $\sqrt{D}$, has positive derivative for $x \ge \sqrt{D}$, and tends to infinity. Thus the required α exists uniquely. Let $r = \alpha^2 > D$ and define

$$q_+(z) = z^3 + Dz + 2Di, \qquad q_-(z) = z^3 + Dz - 2Di.$$

Their factorization is

$$q_+(z) = (z - i\alpha)\{z^2 + i\alpha z + (D - r)\}.$$

The quadratic roots are $(-i\alpha \pm \sqrt{3r - 4D})/2$. If $3r \ge 4D$, both have imaginary part $-\alpha/2 < 0$. If $3r < 4D$, their imaginary parts are $(-\alpha \pm \sqrt{4D - 3r})/2 < 0$, since $r > D$ implies $\sqrt{4D - 3r} < \alpha$. Hence q_+ has exactly one upper-half-plane root, $i\alpha$. The roots of q_- are its conjugates, so q_- has two upper roots counting multiplicity and the lower root $-i\alpha$. There are no real roots of either polynomial.

Let $W(z)$ be the displayed rational integrand. The exact logarithmic- derivative identity

$$W(z) = \frac{i}{4}z^2\left\{\frac{q_+'(z)}{q_+(z)} - \frac{q_-'(z)}{q_-(z)}\right\}$$

shows that a q_+ root z_0 of multiplicity m contributes residue $imz_0^2/4$, and a q_- root contributes $-imz_0^2/4$. This also covers repeated roots: a polynomial logarithmic derivative has a simple pole whose residue is the root multiplicity. In particular the repeated-pole case $D = 27$ is not omitted.

The sum of the squares of all roots of q_- is $-2D$ by Vieta. Its lower root has square $-r$, so its two upper squared roots sum to $r - 2D$. The COMPLETE upper-half-plane residue sum of W is therefore

$$\frac{i}{4}(-r) - \frac{i}{4}(r - 2D) = -\frac{i}{2}(r - D).$$

The integrand is even and real on the real axis, and is $O(|z|^{-2})$ on sufficiently large semicircles. The semicircle contribution tends to zero. The residue theorem gives $2I(D) =$

$2\pi i[-i(r-D)/2] = \pi(r-D)$. Finally $\alpha(r-D) = 2D$ gives $r(r-D)^2 = 4D^2$. The strict retirement inequality follows from (85) and its continuous, strictly smaller integrand on every positive compact interval. □

D.4. **The constants actually used by the parent payments.** For fixed $D > 0$, $x(x-D)^2$ is strictly increasing on $x > D$ because its derivative is $(x-D)(3x-D) > 0$. Exact rational substitutions therefore give

$$\frac{111}{20}\left(\frac{111}{20} - 3\right)^2 - 36 = \frac{711}{8000} > 0, \qquad R(3) < \frac{111}{20},$$

and

$$\frac{351}{50}\left(\frac{351}{50} - 4\right)^2 - 64 = \frac{3151}{125000} > 0, \qquad R(4) < \frac{351}{50}.$$

The first bound is the degree-three retirement cost used in the long inward-fan payment; the second is the degree-four cost used in the large double-parent and sibling partitions. These are actual degrees in the specific retained graph, not the degrees before a different deletion. The lemma applies after earlier induced deletions because $\mathscr{C}$ is hereditary.

For completeness an admissible monotone cost is $U(0) = 0$, $U(D) = \min\{2D, R(D)\}$ for $D > 0$. The relation $D = \alpha^3/(\alpha+2)$ gives

$$R'(D) = \frac{(\alpha+2)^2}{\alpha(\alpha+3)} > 0, \qquad \frac{d}{d\alpha}\log R'(D) = -\frac{\alpha+6}{\alpha(\alpha+2)(\alpha+3)} < 0.$$

Thus R is increasing and strictly concave. The two parts of U meet at $D = 2$, with slopes 2 on the left and 8/5 on the right; U is concave on $[0,\infty)$. Also $R(D) - D = 2D/\alpha < 2\sqrt{D}$. If two adjacent old roots have degrees $d_a \geq d_b$, one actual deletion chain therefore has cost at most $U(d_a) + U(d_b - 1)$. Concavity shows this is no larger than $U(d_b) + U(d_a - 1)$. The common edge is counted once; this does not add independently optimized overlapping graph witnesses.

Appendix E. A uniform bound for the bare external branch

E.1. **The normalized imaginary sign and the low-frequency phase.** The scalar weighted-trace inequalities alone do not supply the sharp low-frequency estimate. We prove the additional matrix inequality that is needed. All lengths in this subsection are positive integers. Use the actual path $B = P_k$, $k = p+q+1$, and the two contact rows $T_a = \mathbf{1}^T_{\{0,\dots,p\}}$ and $T_b = \mathbf{1}^T_{\{p,\dots,p+q\}}$. Put

$$E = \operatorname{diag}(p-1, q-1), \qquad F_* = T(tI + iB)^{-1}T^T - \frac{E}{t+2i}.$$

This is the same signed normalization as in (88); it is not assigned a graph energy. To fix the auxiliary matrices used below, let D be the $k \times (k+1)$ incidence matrix with row $e_j^{\mathsf{T}} - e_{j+1}^{\mathsf{T}}$, let $C = 2I - \mathsf{D}^{\mathsf{T}}\mathsf{D}$, and put $V = (e_{p+1} - e_0, e_k - e_p)/\sqrt{2}$. Then $V^{\mathsf{T}}V = I$ and $F_* = (K + 2iV^{\mathsf{T}}(tI + iC)^{-1}V)/(t+2i)$, as is verified by the incidence calculation in (88). The matrix C has unit path off-diagonals and unit diagonal entries at its two endpoints.

Lemma E.1 (All-length normalized imaginary sign)**.** *For every $p, q \geq 1$ and $t > 0$,* $\operatorname{Im} F_* \preceq 0$.

Proof. Let $D_{-1} = 0$, $D_0 = 1$, $D_1 = t$ and $D_j = tD_{j-1} + D_{j-2}$. These are the real positive determinants $\det(tI + iP_j)$; explicitly $D_j = \sum_{r=0}^{\lfloor j/2 \rfloor} \binom{j-r}{r} t^{j-2r}$. For $0 \leq i \leq j < k$, tridiagonal cofactors give

$$[(tI + iB)^{-1}]_{ij} = \frac{(-i)^{j-i} D_i D_{k-j-1}}{D_k}.$$

Consequently, if A_{ab} denotes an entry of the real symmetric polynomial matrix $A = -D_k \operatorname{Im}\{T(tI + iB)^{-1}T^T\}$, it is obtained by summing the displayed cofactors over the

two specified contact sets. Only odd distances contribute: a pair at distance $2r+1$ contributes $(-1)^r D_i D_{k-j-1}$ to A. For a diagonal contact entry both ordered orientations are counted. Define

$$H_{p,q}(t) = (t^2+4)A(t) - 2ED_k(t), \qquad \nu = k \bmod 2, \quad x = t^2,$$
$$R_{p,q}(x) = H_{p,q}(t)/t^\nu, \qquad d_k(x) = D_k(t)/t^\nu.$$

The parity of each cofactor shows that these are integer polynomials in x, and $d_k(x) > 0$ for $x > 0$. The exact identity to be proved positive is

$$-\operatorname{Im} F_* = \frac{R_{p,q}(x)}{(x+4)d_k(x)}. \tag{86}$$

Division by the common factor t^ν cancels polynomials for $t > 0$; it does not remove a private zero mode.

We next prove the recurrences for every index, before applying a finite certificate. Let $h(z) = 1 - tz - z^2$ and $U_p = \sum_{r=0}^{p} (-i)^r D_{p-r}$, so

$$\sum_{p\ge 0} D_p z^p = \frac{1}{h(z)}, \qquad \sum_{p\ge 0} U_p z^p = \frac{1}{(1+iz)h(z)}.$$

At fixed q, an odd-distance pair in A_{aa} can be written $j = i + 2r + 1$ and $m = p - j$, with $i + 2r + m = p - 1$. The three sums have generating functions $1/h(z)$, $1/(1+z^2)$ and $(D_q + D_{q-1}z)/h(z)$, respectively. Hence

$$A_{aa}(p,q) = 2[z^{p-1}]\frac{D_q + D_{q-1}z}{(1+z^2)h(z)^2}.$$

Set $A_0 = D_q$, $B_0 = D_{q-1}$. The shifted continuants satisfy

$$\sum_{p\ge 0} D_{p+q+1} z^p = \frac{tA_0 + B_0 + A_0 z}{h(z)}.$$

Applying $z\partial_z - 1$ to this last expression computes the $(p-1)D_{p+q+1}$ subtraction. Multiplication by the denominators and cancellation gives the exact formal identity

$$\sum_{p\ge 0} H_{aa}(p,q) z^p = \frac{2\{A_0(t + 4z + 2tz^2 + 2z^3) + B_0(1 - tz + 3z^2)\}}{(1+z^2)h(z)}. \tag{87}$$

Here the formal $p = 0$ extension has $A_{aa} = 0$ and $E_{aa} = -1$; only $p \ge 1$ is used for the actual response. This formal extension explains the constant term in the identity. The bb entry is its reflected expression. At fixed q its finite cofactor sums are linear combinations of D_p, D_{p-1}, and its generating denominator divides $h(z)$. For the cross entry, the sum over $i \le p \le j$ factors into the two sums U_p, U_q, giving $A_{ab} = -\operatorname{Im}(U_p U_q)$. Its poles are therefore the path poles and $\pm i$.

For $t > 0$ all these generating functions are proper, with distinct poles among the two path poles and $\pm i$. If $\lambda_\pm$ solve $\lambda^2 = t\lambda + 1$, their fourth powers have product one and sum $a(x) = x^2 + 4x + 2$. The fourth powers of $\pm i$ are one. It follows on every residue-four subsequence in p that the characteristic polynomial divides

$$(\mu - 1)(\mu^2 - a(x)\mu + 1).$$

There is no initial transient: proper rational functions are sums of these pole sequences from index zero. Reflection gives the same assertion in q. The parity ν is unchanged on either subsequence, so dividing by t^ν preserves the recurrences for R. Equality at every $t > 0$ proves the corresponding polynomial identities.

We use the cone of symmetric polynomial matrices whose every coefficient matrix is positive semidefinite. It is closed under addition and multiplication by a scalar polynomial with nonnegative coefficients. For a sequence Z_j with the displayed characteristic, $Z_{j+2} -$

$aZ_{j+1} + Z_j = L$ is constant. Put $b = a - 2 = x^2 + 4x$, $\Delta_0 = Z_1 - Z_0$ and $K_0 = Z_2 - aZ_1 + (a-1)Z_0 = L + bZ_0$. A direct subtraction gives

$$\Delta_{j+1} = \Delta_j + K_0 + b(Z_{j+1} - Z_0), \qquad \Delta_j = Z_{j+1} - Z_j.$$

If Z_0, Δ_0, K_0 lie in the cone, induction proves that all Z_j and all Δ_j do. In this argument $Z_{j+1} - Z_0 = \sum_{s=0}^{j} \Delta_s$ is already in the cone. Positivity of L is neither assumed nor generally true.

Define the commuting residue-four operators

$$I_p R = R(p,q), \quad \Delta_p R = R(p+4,q) - R(p,q),$$
$$K_p R = R(p+8,q) - aR(p+4,q) + (a-1)R(p,q).$$

and define I_q, Δ_q, K_q by reflection. The finite certificate `normalized-sign-jets.json` contains exactly

$$O_q O_p R(r,s), \qquad r,s \in \{1,2,3,4\}, \qquad O_p, O_q \in \{I, \Delta, K\}.$$

There are $16 \cdot 9 = 144$ polynomial matrices and 924 coefficient matrices, including zero coefficients up to each declared degree. Each coefficient $\left(\begin{smallmatrix} u & v \\ v & w \end{smallmatrix}\right)$ is checked by the integer comparisons $u \geq 0$, $w \geq 0$, $uw - v^2 \geq 0$. All pass. The reader reconstructs the complete index set and every coefficient from the actual path adjacency adjugate recurrence, rather than accepting the supplied coefficients as a premise. Missing, duplicated or altered entries reject the certificate; the formal identity (87) is checked separately by polynomial multiplication.

For each residue pair first apply the cone induction in q to the three initial p objects $I_p R, \Delta_p R, K_p R$. Their q recurrences follow by commutation of the polynomial operators. They belong to the cone for every q. Apply the induction in p next. Thus every $R_{p,q}$ has positive semidefinite coefficients for every positive pair, and is positive semidefinite at $x > 0$. Equation (86) proves the lemma. □

Lemma E.2 (The same-parent low phase)**.** *At every actual old boundary of a maximal outerplanar graph of order at least three, let F_0 be its complete correction and put $Z_0 = tI + iJ + F_0$, $Z_1 = Z_0 + F_*$. With the accretive trace-log phase relative to the same old graph,*

$$\delta_*(t) = \operatorname{Im} \operatorname{Tr} \operatorname{Log} Z_1 - \operatorname{Im} \operatorname{Tr} \operatorname{Log} Z_0 < 2 \arctan(2/t).$$

The physical cap phase is at most δ_* by lossless monotonicity.

Proof. We verify all the normalized sectors needed here. The finite auxiliary path has $-2I \prec C \preceq 2I$. The first inequality follows because $2I + C$ has the quadratic form consisting of consecutive squared sums and twice the two endpoint squares; the second follows from the incidence factorization $2I - C = D^T D$. Write

$$Q = V^T (t^2 I + C^2)^{-1} V \succ 0, \qquad M = V^T C (t^2 I + C^2)^{-1} V, \qquad S = tQ - iM.$$

Since $M + 2Q \succeq 0$, direct substitution into $F_* = (K + 2iS)/(t + 2i)$ gives

$$\operatorname{Re} F_* = \frac{t(K + 2M + 4Q)}{t^2 + 4} \succeq \frac{tK}{t^2 + 4} \succ 0,$$
$$2 \operatorname{Re} F_* + t \operatorname{Im} F_* = 2tQ \succ 0,$$
$$2 \operatorname{Re} F_* - t \operatorname{Im} F_* = \frac{2t}{t^2 + 4} \left\{ 2K - I + V^T (C + 2I)^2 (t^2 I + C^2)^{-1} V \right\} \succ 0.$$

The last compressed spectral function is positive semidefinite and $2K - I$ has eigenvalues one and five. The two old-boundary sector inequalities now imply $2 \operatorname{Re} Z_j \pm t \operatorname{Im} Z_j \succ 0$ for $j = 0, 1$; both matrices are strictly accretive.

Write $R_j = \operatorname{Re} Z_j \succ 0$, $H_j = \operatorname{Im} Z_j$ and $B_j = R_j^{-1/2} H_j R_j^{-1/2}$. Every eigenvalue of this Hermitian B_j lies in $(-2/t, 2/t)$. Congruence gives $\det Z_j = \det R_j \det(I + iB_j)$ with $\det R_j > 0$. Its accretive trace phase is exactly $\sum_\lambda \arctan \lambda(B_j)$: the phases agree modulo

2π and continuously on the connected positive-real-part domain, and agree when $H_j = 0$. No scalar determinant is rewrapped.

By the preceding lemma $H_1 \preceq H_0$. Let m be the negative inertia of H_0 and r the positive inertia of H_1. Their negative and positive subspaces have zero intersection: a nonzero vector in both would have $x^*H_1x > 0 > x^*H_0x$, contradicting the order. Hence $m + r \le 2$. Congruence preserves inertia. With $\theta = \arctan(2/t) > 0$, the phase of Z_0 is at least $-m\theta$, strictly so if $m > 0$, and the phase of Z_1 is at most $r\theta$, strictly so if $r > 0$. Their difference is therefore less than 2θ if $m + r > 0$. If both inertias vanish, the difference is at most zero, which is still strictly less than 2θ. This proves the bound on the entire interval $t > 0$ with the same physical old response. □

The following bound is used only in the large-middle partitions. It does not assert a nonnegative increment for every bare external branch.

Lemma E.3. *At every ordered boundary of any old maximal outerplanar graph H, $|H| \ge 3$, a bare $E_{p,q}$, $p, q \ge 1$, with $k = p + q + 1$ private vertices satisfies*

$$\mathcal{F}(G) - \mathcal{F}(H) > 2k - 184/157.$$

Proof. The private matrix is the path $B = P_k$. Its root contacts are the prefix $0, \dots, p$ and suffix $p, \dots, k-1$. Let D be the $k \times (k+1)$ oriented incidence matrix with row $e_j^{\mathsf{T}} - e_{j+1}^{\mathsf{T}}$, and put $C = 2I - D^{\mathsf{T}}D$. Thus C is the path on $k+1$ vertices, with unit off-diagonal entries and unit diagonal entries at its two endpoints. Also $B = 2I - DD^{\mathsf{T}}$. Define

$$W = (e_{p+1} - e_0,\ e_k - e_p), \qquad V = W/\sqrt{2}, \qquad S = V^{\mathsf{T}}(tI + iC)^{-1}V.$$

The columns of V are orthonormal and $TD = -W^{\mathsf{T}}$. The intertwining identity for these two incidence products gives

$$T(tI + iB)^{-1}T^{\mathsf{T}} = \frac{TT^{\mathsf{T}} + 2iS}{t + 2i}. \tag{88}$$

It can also be checked by multiplying by $tI + iB$ before applying T: $(tI + iB)^{-1} = (I + iD(tI + iC)^{-1}D^{\mathsf{T}})/(t + 2i)$. The same finite Hermitian matrix C is used at every frequency.

Subtract the two nonnegative rays $(p-1)/(t+2i)$ and $(q-1)/(t+2i)$ on the respective diagonals. Then $F_* = (K + 2iS)/(t + 2i)$. This is a phase normalization, with no graph energy assigned to F_*. Put $x = t^2$, $K_i = K^{-1}$ and

$$a = \operatorname{Im}\operatorname{tr}(K_iS), \qquad Q = V^{\mathsf{T}}(t^2I + C^2)^{-1}V.$$

We will use the two all-length estimates

$$-a \ge g(x) := \frac{x}{3(x+1)^2}, \qquad \operatorname{tr}(K_i^2Q) \le h(x) := \frac{1}{x} + \frac{1}{9(x+2)}. \tag{89}$$

Here is an explicit finite-coefficient proof with an all-index recurrence; checking lengths up to a cutoff alone would not suffice.

Let D_j be the continuants in (5.4), also $D_{-1} = 0$, and define $U_p = \sum_{j=0}^{p}(-i)^j D_{p-j}$, $U_{-1} = 0$. Path cofactors, or multiplication of each resolvent column by $tI + iC$, give $S_{ij} = N_{ij}/(2D_k)$ with

$$\begin{aligned} N_{aa} &= 2D_qU_p + D_{q-1}(U_{p-1} + iU_p), \\ N_{bb} &= 2D_pU_q + D_{p-1}(U_{q-1} + iU_q), \\ N_{ab} &= [(-i)^q + D_q]U_p - D_{q+1}U_{p-1} + iD_{p-1}D_q. \end{aligned}$$

For $\nu = k \bmod 2$, parity makes the following polynomials in $x = t^2$:

$$d = D_k/t^\nu, \quad N = -\operatorname{Im}(N_{aa} + N_{bb} - N_{ab})/t^\nu, \quad R = \operatorname{Re}(5N_{aa} + 5N_{bb} - 8N_{ab})/t^{1-\nu}.$$

Since $d > 0$, the required inequalities are equivalent to nonnegativity of

$$A_{p,q} = (x+1)^2N - xd, \qquad B_{p,q} = 2(10x + 18)d - x^{1-\nu}(x+2)R. \tag{90}$$

Indeed $-a = N/(3d)$ and $\operatorname{tr}(K_i^2 Q) = R/(18x^\nu d)$, by taking the stated real and imaginary parts of the full symmetric matrix.

Both families in (90) have nonnegative coefficients for all $p, q \geq 1$. To prove the all-index claim, at fixed q the proper generating function of the four objects $D_p, D_{p-1}, U_p, U_{p-1}$ has denominator $(1+iz)(1-tz-z^2)$ and numerator degree at most two. Taking real and imaginary parts adds its conjugate poles. On each residue-four subsequence the characteristic therefore divides

$$(\mu - 1)(\mu^2 - (x^2 + 4x + 2)\mu + 1).$$

The two path poles have fourth powers with sum x^2+4x+2 and product one; the additional poles $\pm i$ have fourth power one. This proves the recurrence at every index, not merely after a transient. Reflection gives the same recurrence in q. Parity and the factors in (90) stay fixed on each subsequence.

For a sequence Z_j obeying that third-order recurrence, put $z_0 = x^2+4x+2$, $b_0 = z_0-2 = x^2 + 4x$ and form the initial objects Z_0, $\Delta_0 = Z_1 - Z_0$, $K_0 = Z_2 - z_0 Z_1 + (z_0 - 1)Z_0$. The recurrence implies

$$\Delta_{j+1} = \Delta_j + K_0 + b_0(Z_{j+1} - Z_0).$$

If the three initial objects have nonnegative coefficients, this identity inductively gives nonnegative Z_j and differences. For each of the sixteen residue pairs $p, q \in \{1, 2, 3, 4\}$, check the nine initial operators $O_q O_p$, $O \in \{I, \Delta_4, K_4\}$ on each of A, B. There are exactly 288 polynomial coefficient lists. Constructing them from the displayed cofactor formulas gives nonnegative coefficients throughout. The supplied integer polynomial reader also reconstructs the initial $1 \leq p, q \leq 12$ grid by adjacency traces, Newton identities and contact moments, and compares every coefficient. Applying the cone induction first in q to each of the three p objects, then in p, proves the full all-length claim. The finite coefficient lists are complete supplementary proof objects, with missing or negative coefficients rejected. This proves (89).

Let $L = K_i^{1/2} S K_i^{1/2}$, $\beta = \|L\|_F^2$. In the real eigenbasis of K_i, with weights $1, 1/3$, the symmetry of S and $2uv \leq u^2 + v^2$ give $\beta \leq \operatorname{tr}(K_i^2 S^* S)$. Compression of the same resolvent gives $S^* S \preceq Q$, by inserting $VV^{\mathsf{T}} \preceq I$ between its two full factors. Thus $\beta \leq h(x)$. The exact linear ball support of F_* is

$$tE(F_*, t) = a - 1 + \tfrac{1}{2} \|I + 2iL\|_* .$$

The sum of its two singular values is at most $2\sqrt{1 - 2a + 2\beta}$, by the arithmetic–quadratic mean inequality and expansion of the squared Frobenius norm. The function $a - 1 + \sqrt{1 - 2a + 2\beta}$ is nondecreasing for $a \leq 0$, and increases with β. Hence

$$t\delta(t) \leq -g(x) - 1 + \sqrt{1 + 2g(x) + 2h(x)}. \tag{91}$$

This bounds the physical phase by the normalized support; it does not assert simultaneously realizable extremizers for a, β.

For the whole low interval $0 < t < \varepsilon = 5/8$, use lemma E.2. Its proof requires the additional all-length normalized matrix sign, not merely the two sector bounds. It bounds the physical phase by the same-parent normalized phase $\delta_* < 2 \arctan(2/t)$. Its exact weighted integral is

$$\pi\varepsilon^2/2 + 2\varepsilon - (\varepsilon^2 + 4) \arctan(\varepsilon/2).$$

For $t \geq \varepsilon$ set $u = 1/t$, and write $g(t^2) = u^2 G(u)$, $h(t^2) = u^2 H(u)$ with

$$G(u) = 1/[3(1 + u^2)^2], \qquad H(u) = 1 + 1/[9(1 + 2u^2)].$$

Rationalization in (91) gives the full tail integral on $0 \leq u \leq 8/5$ of

$$W(u) = \frac{2(G + H)}{\sqrt{1 + 2u^2(G + H)} + 1} - G.$$

At fixed u this decreases with G and increases with H; at fixed G, H it decreases with u. Since G, H decrease, a whole-cell upper at $[l, r]$ is

$$\frac{2(G(r) + H(l))}{\sqrt{1 + 2l^2(G(r) + H(l))} + 1} - G(r).$$

The complete 256 equal cells of $[0, 8/5]$ have checked 80-bit outward square-root lower brackets and upper areas. Together with an eight-term alternating lower sum for $\arctan(5/16)$ and $\pi < 22/7$, their exact rational sum plus the low integral is less than $46/25$. The private path has energy $k - 1$ and contact mass $k + 1$. Consequently (3) and $\pi > 157/50$ give $\Delta\mathcal{F} > 2k - (100/157)(46/25) = 2k - 184/157$. □

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